ELECTRONICS ENGINEERING

BOARD EXAMINATION COMPREHENSIVE REVIEWER

PB EXAM 20 · 100 Complete Solved Examination Questions, Formulas, & Cram Sheet

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**GENERAL ENGINEERING AND APPLIED SCIENCES (GEAS),
ENGINEERING MANAGEMENT, TECHNOPRENEURSHIP,
RA 9292 / ECE LAWS, CODE OF ETHICS, MATERIALS SCIENCE,
APPLIED PHYSICS, CHEMISTRY, ENVIRONMENTAL ENGINEERING, & OOP**

Document Specifications & Review Scope:

- **Comprehensive Coverage:** Questions 1 through 100 completely solved without omission.
- **Rigorous Numerical Verification:** Step-by-step arithmetic and formula derivations for all calculations.
- **Structured Analysis:** Every question features Given, Find, Formula, Step-by-Step Solution, Answer, Explanation, and Board Exam Tips.
- **Synthesized Summaries:** Part II Terms & Concepts, Part III Complete Formula Sheet, Part IV Quick Answer Key Table, and Part V Final Board Exam Cram Sheet.

PRC Electronics Engineering Licensure Examination Standard Edition

Professional Licensing Review Series

Contents

1 Part I — Complete Solutions to Examination Questions **2**

2 Part II — Important Terms & Concepts Summary **22**
2.1 Engineering Management & Technopreneurship 22
2.2 RA 9292 & Professional Ethics 22
2.3 Materials Science & Solid-State Physics 22
2.4 General Engineering Sciences & Environmental Engineering 22

3 Part III — Comprehensive Formula Sheet **23**

4 Part IV — Quick Answer Key Table **24**

5 Part V — Final Board Exam Cram Sheet **25**
5.1 Must-Remember Formulas 25
5.2 Must-Remember One-Line Concepts 25
5.3 Common Traps & Quick Rules 25

1 Part I — Complete Solutions to Examination Questions

Question 1

In project management, the Critical Path Method (CPM) identifies the longest sequence of dependent tasks in a project plan. What is the fundamental characteristic of any task strictly located on the critical path?
ANSWER: It has zero slack or float time (Slack $=$ LS $-$ ES $=$ LF $-$ EF $= 0$).

- **Concept:** The critical path represents the minimum duration required to complete the entire project. Tasks located along this sequence have equal early start (ES) and late start (LS) times.
- **Explanation:** Any delay in an activity located on the critical path directly postpones the final project completion date. Hence, total float (TF) and free float (FF) are identically zero.
- **Board Exam Tip:** Critical Path = **Longest path** in duration through the network, but represents the **shortest time** to complete the project. Every activity on it has **Zero Float**.

Question 2

In organizational control systems, which specific type of control focuses on monitoring resources and preventing problems before the actual production or operational process begins?
ANSWER: Feedforward control (Preliminary / Preventive control)

- **Explanation:**
  - **Feedforward Control:** Anticipates problems before they occur by screening incoming inputs (raw materials, employee qualifications, capital equipment).
  - **Concurrent Control:** Monitors ongoing activities during the transformation process to ensure standards are met in real time.
  - **Feedback Control:** Evaluates finished outputs after an operation is completed to correct future processes.
- **Board Exam Tip:** Before process = **Feedforward**. During process = **Concurrent**. After process = **Feedback**.

Question 3

In ferromagnetic materials, the critical temperature above which the material's magnetic domains are randomized by thermal agitation, causing it to transition into a paramagnetic state, is strictly defined as the:
ANSWER: Curie temperature (T_c)

- **Explanation:** Above the Curie point, thermal kinetic energy exceeds the quantum exchange coupling that aligns neighboring electron magnetic dipole moments. Spontaneous magnetization vanishes, and magnetic susceptibility drops to follow the Curie-Weiss law: $\chi = \frac{C}{T - T_c}$. For pure iron (α-Fe), $T_c \approx 770°$C (1043 K).
- **Board Exam Tip:** Ferromagnetic $\rightarrow$ Paramagnetic transition occurs at the **Curie Temperature**. Antiferromagnetic $\rightarrow$ Paramagnetic occurs at the **Néel Temperature**.

Question 4

What type of physical system exhibits an amplitude of oscillation that gradually decreases over time due to non-conservative forces like friction or air resistance?
ANSWER: Damped harmonic oscillator

- **Formula:** Equation of motion: $m\frac{d^2x}{dt^2} + b\frac{dx}{dt} + kx = 0$, where b is the damping coefficient.
- **Explanation:** Mechanical dissipation dissipates total mechanical energy ($E = \frac{1}{2}kA^2$) into thermal waste heat, causing oscillation amplitude to decay exponentially as $A(t) = A_0 e^{-\gamma t}$.
- **Board Exam Tip:** Oscillators: $\zeta = 0$ is undamped; $0 < \zeta < 1$ is underdamped (oscillates with decaying amplitude); $\zeta = 1$ is critically damped; $\zeta > 1$ is overdamped.

Question 5

In UML Class Diagrams, numerical notations such as `1..*` or `0..1` placed strictly at the ends of an association line define the specific constraints on the number of object instances that can participate in the relationship. This concept is termed:
ANSWER: Multiplicity (Cardinality)

- **Explanation:** Multiplicity specifies the allowable range of instances of one class linked to a single instance of another class:
  - `0..1`: Zero or one (optional single instance).
  - `1`: Exactly one.
  - `0..*`: Zero or more (unbounded).
  - `1..*`: One or more (at least one required).
- **Board Exam Tip:** UML notations: Multiplicity defines *how many* objects participate in an association.

Question 6

For a simple harmonic mass-spring system, what analytically happens to the period of oscillation if the stiffness of the spring (the spring constant, k) is quadrupled while the mass remains constant?
ANSWER: The period is halved ($T_{\text{new}} = \frac{1}{2}T_{\text{old}}$).
- **Given:** $k_2 = 4k_1$, $m_1 = m_2$.
- **Find:** New period of oscillation T_2.
- **Formula:** Simple harmonic oscillator period: $T = 2\pi\sqrt{\frac{m}{k}}$.
- **Solution:**

$$T_2 = 2\pi\sqrt{\frac{m}{4k_1}} = \frac{1}{\sqrt{4}}\left(2\pi\sqrt{\frac{m}{k_1}}\right) = \frac{1}{2}T_1$$

- **Explanation:** The period of oscillation is inversely proportional to the square root of the spring constant ($T \propto 1/\sqrt{k}$). Quadrupling k doubles the natural frequency ($\omega = \sqrt{k/m}$) and halves the period.
- **Board Exam Tip:** Mass-spring period: $T = 2\pi\sqrt{m/k}$. Pendulum period: $T = 2\pi\sqrt{L/g}$.

Question 7

In exception handling architectures, what is the explicit structural purpose of the `finally` block when placed after a `try-catch` sequence?
ANSWER: To execute critical cleanup code, such as closing database connections or file streams, regardless of whether an exception is thrown or caught.
- **Explanation:** The `finally` block guarantees execution of cleanup routines even if an unhandled exception occurs or a `return`, `break`, or `continue` statement is encountered inside the `try` or `catch` blocks. It is only bypassed if the process terminates abruptly (e.g., `System.exit(0)`).
- **Board Exam Tip:** Code inside a `finally` block **always** executes, making it ideal for resource deallocation.

Question 8

An engineering firm fails to deliver a completed telecommunications tower on the date explicitly stipulated in the contract, despite no natural disasters occurring. The client suffers financial losses due to the delay. Under Philippine Civil Code obligations, the specific legal status that applies to the ECE firm is default or delay on the part of the debtor, which is also known as:
ANSWER: *Mora Solvendi*
- **Explanation:** Under Philippine Law on Obligations and Contracts (Civil Code):
  - ***Mora Solvendi*:** Delay on the part of the debtor (obligor) to fulfill their obligation.
  - ***Mora Accipiendi*:** Delay on the part of the creditor (obligee) to accept the performance of the obligation.
  - ***Compensatio Morae*:** Mutual delay of both parties in reciprocal obligations (delays cancel out).
- **Board Exam Tip:** Debtor delay = *Mora Solvendi*. Creditor delay = *Mora Accipiendi*.

Question 9

Analytically, materials designed for use as permanent magnets (like neodymium magnets) must possess which specific magnetic characteristic to resist demagnetization from external fields?
ANSWER: High coercivity (H_c)
- **Explanation:** Coercivity (H_c) is the intensity of the reverse magnetic field required to reduce the magnetic induction B of a magnetized material to zero. "Hard" magnetic materials possess broad hysteresis loops with high coercivity and high remanence, resisting thermal, mechanical, and external demagnetizing influences.

- **Board Exam Tip:** Permanent magnets (hard magnetics) require **High Coercivity** and **High Retentivity**. Transformer cores (soft magnetics) require **Low Coercivity** and **Narrow Hysteresis Area**.

Question 10

An engineering manager is evaluating a project with a 60% chance of yielding a P$100,000$ profit and a 40% chance of resulting in a P$20,000$ loss. What is the Expected Monetary Value (EMV) of this project?
ANSWER: P$52,000$

- **Given:** Outcome 1: $X_1 = +$P$100,000$, $P(X_1) = 0.60$; Outcome 2: $X_2 = -$P$20,000$, $P(X_2) = 0.40$.
- **Find:** Expected Monetary Value (EMV).
- **Formula:** EMV $= E[X] = \sum_{i=1}^{n} P(X_i) \cdot X_i$.
- **Solution:**

$$\text{EMV} = (0.60 \times 100,000) + (0.40 \times -20,000) = 60,000 - 8,000 = \text{P}52,000$$

- **Board Exam Tip:** In EMV calculations, remember to assign a negative sign to losses and costs!

Question 11

According to Frederick Herzberg's Two-Factor Theory of motivation, factors such as company policy, supervision, and salary that can prevent dissatisfaction but do not inherently motivate employees are classified as:
ANSWER: Hygiene factors (Extrinsic factors / Maintenance factors)

- **Explanation:** Herzberg's dual-factor theory states:
  - **Hygiene Factors:** Extrinsic to the work itself (salary, job security, working conditions, company policy, interpersonal relations). If inadequate, they cause dissatisfaction; if adequate, they establish a neutral baseline but do not motivate.
  - **Motivator Factors:** Intrinsic to the job (achievement, recognition, the work itself, responsibility, advancement). These directly drive job satisfaction and high performance.
- **Board Exam Tip:** Herzberg: Hygiene factors prevent dissatisfaction; Motivators produce satisfaction.

Question 12

For total internal reflection to occur in optical fibers, light must strike the core boundary at an angle strictly greater than the critical angle, and it must initially be traveling within the medium that possesses the:
ANSWER: Higher index of refraction ($n_{\text{core}} > n_{\text{cladding}}$)

- **Formula:** Snell's Law at critical angle ($\theta_2 = 90°$): $\sin(\theta_c) = \frac{n_2}{n_1}$. For $\sin(\theta_c) \leq 1$, we must have $n_1 > n_2$.
- **Explanation:** Total internal reflection (TIR) requires two simultaneous physical conditions:
  1. Light must originate in the denser medium ($n_1 > n_2$).
  2. The incident angle must exceed the critical angle ($\theta_i > \theta_c$).
- **Board Exam Tip:** Fiber optic light confinement: Core index is ALWAYS greater than cladding index ($n_{\text{core}} > n_{\text{clad}}$).

Question 13

Why is modern photolithography strictly classified as a "top-down" approach to nanomaterial fabrication?
ANSWER: Because it involves carving or etching nanoscale patterns out of a larger, pre-existing bulk substrate.

- **Explanation:**
  - **Top-Down Fabrication:** Removes, carves, or etches material from bulk wafers to form micro/nanostructures (e.g., photolithography, chemical etching, mechanical milling).
  - **Bottom-Up Synthesis:** Assembles atoms or molecules from precursors into nanostructures (e.g., chemical vapor deposition, self-assembly, sol-gel synthesis).
- **Board Exam Tip:** Top-Down = Bulk to nanoscale (etching/carving). Bottom-Up = Atoms/molecules to nanoscale (assembly).

Question 14

The detrimental process by which the internal concentration of a persistent toxic substance, such as DDT, increases progressively in the tissues of organisms at successively higher trophic levels is called:
ANSWER: Biomagnification (Biological magnification)

- **Explanation:** Persistent, fat-soluble pollutants (like DDT, heavy metals, or PCBs) resist metabolic degradation. As biomass transfers up the food chain, predators consume contaminated prey, concentrating the toxin in apex organisms.
- **Board Exam Tip:** **Bioaccumulation** = Accumulation within a single organism over its lifespan. **Biomagnification** = Concentration increase across successive trophic levels of a food chain.

Question 15

What is the primary regulatory role of the Professional Regulation Commission (PRC) in the Electronics Engineering profession in the Philippines?
ANSWER: To administer the registration, licensing, and regulation of Electronics Engineering professionals in accordance with applicable laws and regulations.
- **Explanation:** Under RA 8981 (PRC Modernization Act) and RA 9292, the PRC and the Professional Regulatory Board of Electronics Engineering (PRBECE) oversee licensure examinations, issue Certificates of Registration and Professional Identification Cards, and enforce the code of ethics.
- **Board Exam Tip:** PRC administers licensing and regulation of professions in the Philippines.

Question 16

Which programming paradigm primarily avoids the use of mutable state and mutable data by emphasizing immutable data and stateless operations?
ANSWER: Functional Programming
- **Explanation:** Functional programming treats computation as the evaluation of mathematical functions, avoiding side effects and mutable shared state (e.g., Haskell, Lisp, Erlang). In contrast, Object-Oriented and Imperative paradigms rely on mutating internal object states.
- **Board Exam Tip:** Immutable state + pure functions without side effects = **Functional Programming**.

Question 17

Based on the Doppler effect, how is the frequency of a sound wave perceived by a stationary observer when the source is accelerating directly towards them?
ANSWER: The perceived frequency is higher than the emitted frequency and continuously increases.
- **Formula:** Doppler equation for moving source: $f' = f\left(\frac{v}{v - v_s(t)}\right)$.
- **Explanation:** Since the source is approaching, the wavefronts are compressed, so $f' > f$. Because the source is accelerating, its velocity $v_s(t)$ increases over time, making the denominator $(v - v_s)$ smaller, which causes perceived frequency f' to continuously rise.
- **Board Exam Tip:** Source approaching at constant speed $\implies$ Constant elevated frequency. Source accelerating toward observer $\implies$ Continuously increasing frequency.

Question 18

Which renewable energy process explicitly converts organic biological materials into a combustible gas (primarily methane and carbon dioxide) through anaerobic digestion?
ANSWER: Biomass gasification / Anaerobic digestion (Biogas generation)
- **Explanation:** In the absence of oxygen, anaerobic microorganisms decompose organic animal manure or biomass in a digester, producing biogas (≈ 55–70% CH_4 and 30–45% CO_2).
- **Board Exam Tip:** Anaerobic digestion of biomass produces **Biogas** (predominantly Methane, CH_4).

Question 19

For a pure metallic element exhibiting a simple cubic (SC) crystal structure, how many total equivalent atoms are contained strictly within the boundaries of a single unit cell?
ANSWER: 1 atom
- **Formula:** In a simple cubic (SC) cell, an atom sits at each of the 8 corners. Each corner atom is shared among 8 adjacent unit cells:

$$N_{\text{atoms}} = 8 \text{ corners} \times \frac{1}{8} \text{ atom/corner} = 1 \text{ equivalent atom}$$

- **Board Exam Tip:** Unit cell atom counts:

- Simple Cubic (SC) = 1 atom (Coordination number = 6, APF = 0.52).
- Body-Centered Cubic (BCC) = 2 atoms (Coordination number = 8, APF = 0.68).
- Face-Centered Cubic (FCC) = 4 atoms (Coordination number = 12, APF = 0.74).

Question 20

A zinc metal plate with a work function of 4.31 eV is illuminated by an ultraviolet laser with a wavelength of 250 nm. Using the approximation $hc \approx 1240$ eV · nm, what is the maximum kinetic energy of the emitted photoelectrons?
ANSWER: 0.65 eV

- **Given:** $\Phi = 4.31$ eV, $\lambda = 250$ nm, $hc \approx 1240$ eV · nm.
- **Find:** Maximum kinetic energy K_{max}.
- **Formula:** Einstein's Photoelectric Equation: $K_{\text{max}} = E_{\text{photon}} - \Phi = \frac{hc}{\lambda} - \Phi$.
- **Solution:**

$$E_{\text{photon}} = \frac{1240 \text{ eV} \cdot \text{nm}}{250 \text{ nm}} = 4.96 \text{ eV}$$

$$K_{\text{max}} = 4.96 \text{ eV} - 4.31 \text{ eV} = 0.65 \text{ eV}$$

- **Board Exam Tip:** Photon energy shortcut: $E(\text{eV}) = \frac{1240}{\lambda(\text{nm})}$.

Question 21

To tune an RF circuit, an ECE places a $10\,\mu$F capacitor and a $40\,\mu$F capacitor strictly in series. What is the equivalent total capacitance of this configuration?
ANSWER: $8\,\mu$F

- **Given:** $C_1 = 10\,\mu$F, $C_2 = 40\,\mu$F in series.
- **Formula:** Series capacitance: $C_{eq} = \frac{C_1 C_2}{C_1 + C_2}$.
- **Solution:**

$$C_{eq} = \frac{10 \times 40}{10 + 40} = \frac{400}{50} = 8\,\mu\text{F}$$

- **Board Exam Tip:** Capacitors in series combine like resistors in parallel; capacitors in parallel add directly.

Question 22

Which classical management theorist is widely credited with developing the principles of organizational bureaucracy, emphasizing clear hierarchical structures, formal rules, and impersonal relationships?
ANSWER: Max Weber

- **Explanation:** German sociologist Max Weber formulated the Bureaucratic Management Theory, advocating for clear division of labor, formal rules and procedures, a well-defined hierarchy of authority, and merit-based advancement.
- **Board Exam Tip:** Bureaucracy = **Max Weber**. Scientific Management = **Frederick Taylor**. Administrative Management = **Henri Fayol** (14 principles).

Question 23

What is the greenhouse effect?
ANSWER: The process by which the Earth's atmosphere traps heat, primarily through greenhouse gases that absorb and re-emit infrared radiation.

- **Explanation:** Solar shortwave ultraviolet and visible light pass through the atmosphere and warm the surface. The Earth then radiates this energy back as longwave infrared (thermal radiation). Greenhouse gases (CO_2, H_2O, CH_4) absorb and re-radiate this infrared energy downward, maintaining a habitable planetary temperature.
- **Board Exam Tip:** The primary greenhouse gas by abundance is **Water Vapor**; the primary anthropogenic driver is **Carbon Dioxide**.

Question 24

For a vehicle negotiating a banked circular curve without relying on friction, how must the ideal banking angle conceptually change if the design velocity of the curve is doubled?
ANSWER: The tangent of the angle must quadruple ($\tan\theta_{\text{new}} = 4\tan\theta_{\text{old}}$).
- **Formula:** Frictionless curve banking: $\tan\theta = \frac{v^2}{Rg}$.
- **Solution:** If velocity v is doubled ($v' = 2v$):

$$\tan\theta' = \frac{(2v)^2}{Rg} = \frac{4v^2}{Rg} = 4\tan\theta$$

- **Board Exam Tip:** The tangent of the banking angle is proportional to the square of velocity: $\tan\theta \propto v^2$.

Question 25

In the microstructural analysis of crystalline materials, a "grain boundary"—the interface where two single crystals of different orientations meet—is structurally classified as what specific type of defect?
ANSWER: Interfacial defect (Two-dimensional / Planar defect)
- **Explanation:**
  - **0D (Point defects):** Vacancies, interstitials, substitutional impurities.
  - **1D (Linear defects):** Edge dislocations, screw dislocations.
  - **2D (Planar / Interfacial defects):** Grain boundaries, phase boundaries, twin boundaries, free surfaces.
  - **3D (Volume defects):** Voids, cracks, precipitate inclusions.
- **Board Exam Tip:** Grain boundaries are **2D (Planar/Interfacial) defects**.

Question 26

The specific amount of thermal energy required to change 1 kilogram of a substance from a solid to a liquid state without any change in temperature is strictly defined as the:
ANSWER: Latent heat of fusion (L_f)
- **Formula:** $Q = mL_f$. For water at $0°$C, $L_f \approx 334$ kJ/kg (79.7 cal/g).
- **Explanation:** During a phase change, heat input breaks intermolecular crystalline bonds without increasing the average kinetic energy of the molecules, meaning temperature remains constant.
- **Board Exam Tip:** Solid $\leftrightarrow$ Liquid = **Latent Heat of Fusion**. Liquid $\leftrightarrow$ Gas = **Latent Heat of Vaporization**.

Question 27

A pure, ideal inductor is connected in series with a resistor, a switch, and a DC battery. At the exact instant the switch is closed ($t = 0^+$), what is the instantaneous state of the electrical current in the circuit and the voltage across the inductor?
ANSWER: The current is zero ($i(0^+) = 0$), and the voltage drop across the inductor is equal to the battery voltage ($v_L(0^+) = V_B$).
- **Explanation:** Current through an inductor cannot change instantaneously because infinite power would be required: $v_L(t) = L\frac{di}{dt}$. Since $i(0^-) = 0$, $i(0^+) = 0$. By KVL: $V_B = i(0^+)R + v_L(0^+) = 0 + v_L(0^+) \implies v_L(0^+) = V_B$.
- **Board Exam Tip:** At $t = 0^+$: Inductors act as **Open Circuits** ($i = 0, v = V_s$); Capacitors act as **Short Circuits** ($v = 0, i = V_s/R$).

Question 28

In Graphical User Interface (GUI) programming, what specific programmatic component is explicitly registered to a button or widget to wait for, detect, and respond to user actions like a mouse click?
ANSWER: The event listener (Event Handler / Callback function)
- **Explanation:** GUI architectures follow the Observer Design Pattern. When a user interacts with a widget, the operating system generates an event object that is dispatched to registered listeners (e.g., `ActionListener` in Java or `onClick` in JavaScript).
- **Board Exam Tip:** GUI inputs trigger events that are captured and handled by **Event Listeners**.

Question 29

A hardware startup has P$600,000$ in cash reserves. Their monthly gross revenue is P$50,000$, and their total monthly operating expenses are P$150,000$. What is the startup's current "runway" before they run out of cash?

ANSWER: 6 months

- **Given:** Cash reserves $=$ P$600,000$, monthly revenue $=$ P$50,000$, monthly expenses $=$ P$150,000$.
- **Find:** Cash runway in months.
- **Formula:**

$$\text{Net Burn Rate} = \text{Monthly Expenses} - \text{Monthly Revenue}$$

$$\text{Runway} = \frac{\text{Cash Reserves}}{\text{Net Burn Rate}}$$

- **Solution:**

$$\text{Net Burn Rate} = 150,000 - 50,000 = \text{P}100,000/\text{month}$$

$$\text{Runway} = \frac{\text{P}600,000}{\text{P}100,000/\text{month}} = 6 \text{ months}$$

- **Board Exam Tip:** Always subtract monthly revenue from expenses to find the **Net Burn Rate** before dividing into cash reserves.

Question 30

Within the global nitrogen cycle, which specific biological process involves bacteria converting usable soil nitrates back into inert nitrogen gas, releasing it into the atmosphere?

ANSWER: Denitrification

- **Explanation:** In the nitrogen cycle:
  - **Nitrogen Fixation:** Converts atmospheric N_2 into ammonia (NH_3 or NH_4^+) by bacteria like *Rhizobium*.
  - **Nitrification:** Aerobic oxidation converting $NH_3 \rightarrow NO_2^- \rightarrow NO_3^-$.
  - **Denitrification:** Anaerobic bacteria (e.g., *Pseudomonas*) reduce nitrates (NO_3^-) back to nitrous oxide (N_2O) and nitrogen gas (N_2).
- **Board Exam Tip:** Nitrates $\rightarrow$ Nitrogen gas = **Denitrification**. Nitrogen gas $\rightarrow$ Ammonia = **Nitrogen Fixation**.

Question 31

A software startup enters a highly saturated CRM market. Instead of competing on features against entrenched giants, they target independent gig-economy workers who find enterprise CRMs too complex and expensive, offering a hyper-simplified, mobile-only tool. Which strategic framework are they applying?

ANSWER: Blue Ocean Strategy

- **Explanation:** Developed by W. Chan Kim and Renée Mauborgne, Blue Ocean Strategy advocates creating uncontested market space rather than competing in crowded, existing markets ("red oceans"). By focusing on non-consumers with a simplified value proposition, the startup makes the competition irrelevant.
- **Board Exam Tip:** Creating uncontested market space to make competition irrelevant = **Blue Ocean Strategy**.

Question 32

Which fundamental equation explicitly models non-steady-state diffusion, where the concentration of the diffusing species changes with respect to both time and position?

ANSWER: Fick's Second Law ($\frac{\partial C}{\partial t} = D\frac{\partial^2 C}{\partial x^2}$)

- **Explanation:**
  - **Fick's First Law (Steady-State):** Flux is proportional to concentration gradient: $J = -D\frac{dC}{dx}$ (no time dependence).
  - **Fick's Second Law (Non-Steady-State):** Rate of concentration change over time equals the divergence of diffusion flux: $\frac{\partial C}{\partial t} = D\frac{\partial^2 C}{\partial x^2}$.
- **Board Exam Tip:** Fick's 1st Law = Steady state ($J = -D\frac{dC}{dx}$). Fick's 2nd Law = Non-steady state / time-dependent ($\frac{\partial C}{\partial t} = D\frac{\partial^2 C}{\partial x^2}$).

Question 33

Under Section 24 of RA 9292, how many years after the revocation of a Certificate of Registration may the Board reinstate the validity of the certificate upon proper application?
ANSWER: Two (2) years

- **Statutory Basis:** RA 9292, Section 24 (*Reinstatement, Re-issuance or Replacement of Certificate of Registration and Professional Identification Card*): The Board may, after the expiration of two (2) years from the date of revocation, reinstate the validity of a revoked Certificate of Registration.
- **Board Exam Tip:** RA 9292 license reinstatement after revocation = **2 years**.

Question 34

In manufacturing and engineering economics, the additional financial cost incurred by producing exactly one more unit of a product is strictly known as the:
ANSWER: Marginal cost ($MC = \frac{\Delta TC}{\Delta Q} = \frac{d(TC)}{dQ}$)

- **Explanation:** Marginal cost is the derivative of total production cost with respect to output quantity. Profit is maximized when marginal revenue equals marginal cost ($MR = MC$).
- **Board Exam Tip:** Additional cost of one extra unit = **Marginal Cost**.

Question 35

When two identical waves traveling in opposite directions along the same medium superimpose, they can form a wave pattern that appears completely stationary. This phenomenon is known as a:
ANSWER: Standing wave (Stationary wave)

- **Formula:** $y(x,t) = A\sin(kx - \omega t) + A\sin(kx + \omega t) = 2A\sin(kx)\cos(\omega t)$.
- **Explanation:** Superposition produces spatial nodes (points of zero amplitude where $\sin(kx) = 0$) and antinodes (points of maximum displacement where $|\sin(kx)| = 1$) that remain fixed in space.
- **Board Exam Tip:** Identical counter-propagating waves create **Standing Waves** with stationary nodes and antinodes.

Question 36

An ECE startup's founders originally own 100% of their company, comprising $1,000,000$ shares. During a Series A funding round, they issue an additional $250,000$ new shares to a Venture Capital firm in exchange for P$2,000,000$. What is the exact percentage of equity dilution experienced by the original founders?
ANSWER: 20%

- **Given:** Original founder shares $= 1,000,000$; Newly issued shares $= 250,000$.
- **Find:** Percentage equity dilution.
- **Formula:**

$$\text{Total Post-Money Shares} = 1,000,000 + 250,000 = 1,250,000$$

$$\text{Founder Post-Money Ownership} = \frac{1,000,000}{1,250,000} = 0.80 = 80\%$$

$$\text{Dilution} = 100\% - 80\% = 20\%$$

- **Board Exam Tip:** Dilution percentage equals the incoming investor's post-money ownership share: $\frac{250,000}{1,250,000} = 20\%$.

Question 37

A startup is developing drone software for agricultural mapping. Early farmers complain that processing raw data takes hours on their old laptops. Using the Value Proposition Canvas, how should the technopreneur actively pivot their offering?
ANSWER: Shift software processing entirely to a cloud-based server and deliver finished PDF reports directly to the farmers' phones as a Pain Reliever.

- **Explanation:** In the Value Proposition Canvas, customer pains (slow processing, hardware limitations) are addressed by specific **Pain Relievers**. Offloading computation to the cloud removes the hardware bottleneck for the user.
- **Board Exam Tip:** Value Proposition Canvas matches customer pains/gains to product pain relievers and gain creators.

Question 38

When a falling object reaches a constant downward velocity where the accelerating force of gravity is balanced by the opposing drag force of air resistance, this velocity is called:
ANSWER: Terminal velocity (v_t)

- **Formula:** $\sum F_y = mg - F_{\text{drag}} = 0 \implies mg = \frac{1}{2}\rho v_t^2 C_d A \implies v_t = \sqrt{\frac{2mg}{\rho C_d A}}$.
- **Explanation:** When aerodynamic drag equals gravitational pull, net acceleration becomes zero ($a = 0$), and the object continues falling at a constant terminal velocity.
- **Board Exam Tip:** At terminal velocity: Net force $= 0$, Acceleration $= 0$, Velocity is constant and maximum.

Question 39

When performing a present economy study for a public works project (like a bridge) that is assumed to have an infinite useful life, which specific economic evaluation method calculates the present worth of its perpetual maintenance costs?
ANSWER: Capitalized Cost Analysis

- **Formula:** $CC = FC + \frac{A}{i}$, where FC is initial capital investment, A is annual perpetual operating cost, and i is interest rate.
- **Board Exam Tip:** Infinite project lifespan = **Capitalized Cost** ($CC = FC + A/i$).

Question 40

Under Section 8 of RA 9292, what is the minimum number of years of active professional practice required for an appointee to qualify as a member of the Board of Electronics Engineering?
ANSWER: Ten (10) years

- **Statutory Basis:** RA 9292, Section 8 (*Qualifications of Board Members*): Appointees must be certified PECEs with at least ten (10) years of continuous active professional practice prior to appointment.
- **Board Exam Tip:** PRBECE Board Member qualification requirement = **10 years** of continuous active practice as a PECE.

Question 41

A technopreneur wishes to synthesize a purely "Circular Economy" business model for a new line of smartwatches. Which comprehensive structural strategy fulfills this goal?
ANSWER: Implementing a product-as-a-service model where the company retains ownership, modularly upgrades components, and recycles 100% of the materials at end-of-life.

- **Explanation:** A circular economy replaces the traditional linear "take-make-dispose" model with restorative industrial loops: design for durability, reuse, refurbishment, and closed-loop material recycling.
- **Board Exam Tip:** Circular economy = Closed-loop lifecycles that eliminate waste via reuse, remanufacturing, and recycling.

Question 42

A heat engine operates in a Carnot cycle between a high-temperature reservoir at 800 K and a low-temperature reservoir at 300 K. What is the theoretical maximum thermodynamic efficiency of this engine?
ANSWER: 62.5%

- **Given:** $T_H = 800$ K, $T_C = 300$ K.
- **Find:** Carnot efficiency η_{Carnot}.
- **Formula:** $\eta = 1 - \frac{T_C}{T_H} = \frac{T_H - T_C}{T_H}$.
- **Solution:**

$$\eta = \frac{800 - 300}{800} = \frac{500}{800} = \frac{5}{8} = 0.625 = 62.5\%$$

- **Board Exam Tip:** In thermodynamic efficiency calculations, temperatures MUST always be in absolute units (Kelvin, not Celsius).

Question 43

According to the First Law of Thermodynamics, energy transferred as heat into a closed, stationary system must equal the sum of the work done by the system and the change in its:
ANSWER: Internal energy (ΔU)
- **Formula:** First Law: $Q = \Delta U + W \implies \Delta U = Q - W$.
- **Board Exam Tip:** First Law of Thermodynamics is the law of **conservation of energy** applied to thermal systems.

Question 44

A startup is developing a wearable biometric ring using the Lean Canvas model. They identify that competitive smartwatches are too bulky for sleep tracking. In the Lean Canvas framework, this observation is best categorized under which specific block?
ANSWER: Existing Alternatives (sub-block under Problem)
- **Explanation:** In Ash Maurya's Lean Canvas, the Problem box includes an "Existing Alternatives" subsection identifying current products used by target customers and their shortcomings.
- **Board Exam Tip:** Identifying competitor limitations in a Lean Canvas belongs under **Existing Alternatives / Problem**.

Question 45

Which organizational structure assigns technical specialists from functional departments to work under a specific project manager, resulting in a dual-reporting relationship that violates the classical "unity of command" principle?
ANSWER: Matrix structure
- **Explanation:** In a matrix organization, an engineer reports to both a functional department head (e.g., VP of Engineering) and a project manager, creating a dual-chain of command.
- **Board Exam Tip:** Dual reporting that violates unity of command = **Matrix Organization**.

Question 46

In fluid mechanics and solid elasticity theory, the Bulk Modulus is a fundamental parameter that measures a material's resistance to uniform:
ANSWER: Volume compression ($K = -V\frac{\Delta P}{\Delta V}$)
- **Explanation:**
  - **Young's Modulus (E):** Resistance to uniaxial tensile/compressive deformation.
  - **Shear Modulus (G):** Resistance to planar shearing deformation.
  - **Bulk Modulus (K):** Resistance to hydrostatic/uniform volume change.
- **Board Exam Tip:** Linear elasticity = **Young's Modulus**. Shear = **Shear Modulus**. Volume change = **Bulk Modulus**.

Question 47

When evaluating multiple engineering alternatives under strict uncertainty (probabilities are completely unknown), the "Minimax Regret" criterion requires the decision-maker to choose the alternative that:
ANSWER: Minimizes the maximum possible opportunity loss (regret) across all states of nature (Savage criterion).
- **Explanation:** The decision-maker builds a regret matrix by subtracting each payoff from the maximum payoff in that state, and selects the strategy that minimizes the worst-case regret.
- **Board Exam Tip:** Savage criterion = **Minimax Regret**. Wald criterion = **Maximin** (pessimistic). Maximax = Optimistic.

Question 48

In fundamental programming control structures, which iterative statement is explicitly designed to evaluate its boolean condition strictly before executing the code block, potentially executing zero times?
ANSWER: A `while` loop (Pre-test loop)
- **Explanation:** A `while` loop checks its condition at the entry point; if initially false, the loop body never executes. A `do-while` loop evaluates its condition at the exit point, executing at least once.

- **Board Exam Tip:** Pre-test loop (0 or more iterations) = `while`. Post-test loop (at least 1 iteration) = `do-while`.

Question 49

By convention, the direction of magnetic field lines exterior to a permanent bar magnet is defined as pointing from:
ANSWER: North pole to South pole
- **Explanation:** Outside a magnet, field lines emerge from the North pole and loop into the South pole. Inside the magnet, lines return from South to North to form continuous closed loops ($\nabla \cdot \vec{B} = 0$).
- **Board Exam Tip:** Magnetic field lines outside: **North to South**. Inside: **South to North**.

Question 50

Which project evaluation method explicitly calculates the discount rate that makes the present worth of a project's cash inflows exactly equal to the present worth of its cash outflows?
ANSWER: Internal Rate of Return (IRR)
- **Formula:** NPV $= \sum_{t=0}^{n} \frac{CF_t}{(1+\text{IRR})^t} = 0$.
- **Board Exam Tip:** Discount rate where NPV $= 0$ is the **Internal Rate of Return (IRR)**. If IRR $>$ MARR, the project is accepted.

Question 51

A radioactive isotope used in medical imaging has a half-life of 6 hours. If a hospital initially receives a 100-gram sample, exactly how many grams of the active isotope will remain after 24 hours?
ANSWER: 6.25 grams
- **Given:** $N_0 = 100$ g, $T_{1/2} = 6$ hrs, total elapsed time $t = 24$ hrs.
- **Formula:** Remaining mass: $N(t) = N_0 \left(\frac{1}{2}\right)^{t/T_{1/2}}$.
- **Solution:**

$$n = \frac{24 \text{ hrs}}{6 \text{ hrs}} = 4 \text{ half-lives}$$

$$N(24) = 100 \times \left(\frac{1}{2}\right)^4 = 100 \times \frac{1}{16} = 6.25 \text{ grams}$$

- **Board Exam Tip:** Exponential decay by half-lives: $100\text{g} \xrightarrow{1} 50\text{g} \xrightarrow{2} 25\text{g} \xrightarrow{3} 12.5\text{g} \xrightarrow{4} 6.25\text{g}$.

Question 52

A 2.0-meter length of straight wire carries a direct current of 15 A. It is placed in a uniform magnetic field of 0.4 T oriented at a $30°$ angle to the direction of the current. What is the magnitude of the magnetic force experienced by this section of the wire?
ANSWER: 6.0 N
- **Given:** $L = 2.0$ m, $I = 15$ A, $B = 0.4$ T, $\theta = 30°$.
- **Formula:** Lorentz force on current-carrying conductor: $F = ILB \sin\theta$.
- **Solution:**

$$F = (15 \text{ A}) \times (2.0 \text{ m}) \times (0.4 \text{ T}) \times \sin(30°) = 30 \times 0.4 \times 0.5 = 6.0 \text{ N}$$

- **Board Exam Tip:** Magnetic force on a wire is $F = ILB \sin\theta$.

Question 53

According to Bernoulli's equation, as an incompressible, non-viscous fluid flows through a horizontal pipe that narrows in cross-sectional area, what must occur to maintain the conservation of energy?
ANSWER: The fluid velocity increases, causing the static fluid pressure to decrease (Venturi effect).
- **Formula:** Continuity: $A_1 v_1 = A_2 v_2$. Bernoulli: $P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2$.
- **Explanation:** When cross-sectional area narrows ($A_2 < A_1$), continuity dictates velocity must increase ($v_2 > v_1$). Because potential energy is constant along a horizontal streamline, dynamic pressure ($\frac{1}{2}\rho v^2$) rises, forcing static pressure (P) to drop.
- **Board Exam Tip:** Constriction in a pipe: **Velocity increases $\implies$ Static Pressure decreases**.

Question 54

In engineering project scheduling, a Gantt chart is a specialized visual tool primarily utilized to explicitly illustrate the:
ANSWER: Start and finish dates of the project's terminal elements over a calendar timeline.
- **Explanation:** Developed by Henry Gantt, a Gantt chart uses horizontal bars along a calendar axis to display task start dates, durations, and completion milestones.
- **Board Exam Tip:** Gantt chart displays **project schedules and task durations over time**.

Question 55

An electronics technician notices that a circuit breaker repeatedly trips whenever a specific circuit is in use. To ensure electrical safety and proper system operation, what should he do FIRST?
ANSWER: Inspect the circuit for possible overloads, short circuits, or other faults to determine the root cause before making repairs.
- **Explanation:** Repeated breaker trips indicate an active fault (overload or short circuit). Resetting the breaker repeatedly without diagnosing the cause risks electrical fires and equipment damage.
- **Board Exam Tip:** Safety first: Always troubleshoot to identify the root cause before resetting or replacing protection devices.

Question 56

Analytically, what is the fundamental structural difference between an Abstract Class and an Interface in Java architecture?
ANSWER: An interface traditionally contains only constant declarations and abstract method signatures, whereas an abstract class can maintain state via instance variables and provide fully implemented methods.
- **Explanation:** A class can implement multiple interfaces (multiple inheritance of type), but can extend only a single abstract class (single inheritance of state).
- **Board Exam Tip:** Abstract classes support state and code reuse; interfaces define behavioral contracts.

Question 57

Under Republic Act No. 9292, what is the appropriate action when an individual offers or performs electronics engineering services without a valid PRC license?
ANSWER: The individual should immediately cease practicing electronics engineering and be reported to the appropriate authorities for unauthorized practice.
- **Statutory Basis:** RA 9292, Section 35 (*Penal Provisions*): Illegal practice of electronics engineering is a criminal misdemeanor punishable by fines of P$100,000$ to P$1,000,000$, imprisonment of 6 months to 6 years, or both.
- **Board Exam Tip:** Unauthorized practice of ECE is a criminal violation under RA 9292 Section 35.

Question 58

In standard CAD drafting software, which specific command is utilized to connect two non-parallel lines by trimming or extending them to meet at a sharp, angled corner, or by creating a flat, beveled edge between them?
ANSWER: `CHAMFER`
- **Explanation:** The `CHAMFER` command creates a beveled or angled transition between two lines. In contrast, the `FILLET` command connects lines with a rounded tangent arc of defined radius.
- **Board Exam Tip:** Flat/beveled angled corner = **CHAMFER**. Rounded corner arc = **FILLET**.

Question 59

According to the Hume-Rothery rules for forming a continuous substitutional solid solution, the atomic radii of the solute and solvent atoms must not differ by more than approximately what percentage?
ANSWER: 15%
- **Hume-Rothery Rules for Complete Miscibility:**
  1. **Size difference:** Atomic radius difference $\Delta r \leq 15\%$.
  2. **Crystal structure:** Must have the same crystal structure.
  3. **Electronegativity:** Must have similar electronegativity.

4. **Valence:** Solvent and solute should have the same valence.

- **Board Exam Tip:** Hume-Rothery atomic radius limit is **15%**.

Question 60

Analytically, Carbon-12 and Carbon-14 exhibit identical chemical behavior but possess different atomic masses. Structurally, what atomic variation defines these two isotopes?
ANSWER: They have a different number of neutrons (Carbon-12 has 6 neutrons, while Carbon-14 has 8 neutrons).

- **Explanation:** Isotopes share the same atomic number $Z = 6$ (same number of protons and electrons, dictating identical chemical properties) but differ in neutron count N, yielding mass numbers $A = 12$ and $A = 14$.
- **Board Exam Tip:** Isotopes have the **same number of protons**, but **different numbers of neutrons**.

Question 61

Which atmospheric trace gas, primarily released through the combustion of fossil fuels, is the most abundant anthropogenic contributor to the enhanced greenhouse effect?
ANSWER: Carbon dioxide (CO_2)

- **Explanation:** While methane (CH_4) has a higher Global Warming Potential per molecule, carbon dioxide accounts for roughly 76% of all human-generated greenhouse gas emissions due to fossil fuel combustion.
- **Board Exam Tip:** Most abundant human-generated greenhouse gas = **Carbon Dioxide** (CO_2).

Question 62

According to Section 6 of RA 9292, how is the Professional Regulatory Board of Electronics Engineering composed, and by whom are its members officially appointed?
ANSWER: A Chairman and two (2) members appointed by the President of the Philippines from a list of recommendees submitted by the PRC.

- **Statutory Basis:** RA 9292, Section 6: The Board consists of a Chairman and two members appointed by the President of the Philippines.
- **Board Exam Tip:** PRBECE composition: **1 Chairman and 2 Members** appointed by the **President of the Philippines**.

Question 63

The specific ecological process where life gradually establishes itself on newly formed, barren substrates entirely devoid of soil, such as freshly cooled volcanic lava, is strictly called:
ANSWER: Primary succession

- **Explanation:**
  - **Primary Succession:** Starts on lifeless, soil-free substrate (cooled lava, retreating glaciers); pioneer species like lichens build initial soil.
  - **Secondary Succession:** Follows a disturbance that removes vegetation but leaves existing soil intact (abandoned farmland, burnt forest).
- **Board Exam Tip:** Succession starting with NO pre-existing soil = **Primary Succession**.

Question 64

In ecology, the zones of Earth's land, water, and atmosphere that are inhabited by living organisms are collectively and strictly defined as the:
ANSWER: Biosphere

- **Explanation:** The biosphere encompasses all global ecosystems supporting life, spanning parts of the lithosphere (crust), hydrosphere (water), and atmosphere (air).
- **Board Exam Tip:** Global sum of all ecosystems supporting life = **Biosphere**.

Question 65

When navigating a complex CAD drawing, a user needs to slide the view vertically or horizontally across the screen without altering the zoom scale or the physical coordinates of the objects. Which specific viewing command executes this action?
ANSWER: `PAN`

- **Explanation:** The `PAN` command shifts the viewport display window parallel to the drawing plane without altering object coordinates or magnification scale.
- **Board Exam Tip:** Move viewport without changing scale = **PAN**. Change magnification = **ZOOM**.

Question 66

Fick's First Law of Diffusion states that the steady-state diffusion flux of atoms in a solid material is mathematically proportional to the:
ANSWER: Concentration gradient ($\frac{dC}{dx}$)

- **Formula:** $J = -D\frac{dC}{dx}$, where J is net atomic flux, D is the diffusion coefficient, and $\frac{dC}{dx}$ is the spatial concentration gradient.
- **Board Exam Tip:** Fick's First Law: Flux is directly proportional to the **Concentration Gradient**.

Question 67

A force is strictly classified as conservative if the work it does on a particle moving between two points is:
ANSWER: Independent of the path taken between the two points.

- **Explanation:** A force field $\vec{F}$ is conservative if the work done around any closed loop is zero ($\oint \vec{F} \cdot d\vec{r} = 0$), which implies the force can be expressed as the gradient of a potential function ($\vec{F} = -\nabla U$). Examples include gravity and electrostatics.
- **Board Exam Tip:** Conservative force: Work is **path-independent**; net work around a closed loop is **zero**.

Question 68

In behavioral economics under uncertainty, an investor who strictly prefers to accept a guaranteed P50 payoff over taking a 50% chance of winning P100 is analytically described as:
ANSWER: Risk-averse

- **Explanation:** The expected value of the gamble is $(0.50 \times 100) + (0.50 \times 0) =$ P50. A risk-neutral person is indifferent between the two. A risk-averse person exhibits a concave utility function ($U''(W) < 0$), strictly preferring a sure payout over a risky bet of equal expected value.
- **Board Exam Tip:** Prefers guaranteed payoff over gamble of equal expected value = **Risk-Averse**.

Question 69

In nuclear chemistry, the "mass defect" of a nucleus is directly converted into which of the following quantities that holds the protons and neutrons together?
ANSWER: Nuclear binding energy ($\Delta E = \Delta m \cdot c^2$)

- **Explanation:** The total mass of a stable nucleus is always less than the sum of its individual constituent free protons and neutrons. By mass-energy equivalence, this missing mass defect (Δm) is released when the nucleus forms and represents the energy required to disassemble it into free nucleons.
- **Board Exam Tip:** Mass defect converts directly into **Nuclear Binding Energy** via $E = mc^2$.

Question 70

In the Technology Adoption Life Cycle model, the critical market gap or "chasm" that a tech startup must successfully cross to achieve mainstream success lies specifically between the "Early Adopters" and the:
ANSWER: Early Majority

- **Explanation:** Geoffrey Moore's *Crossing the Chasm* model identifies a market adoption gap between visionary Early Adopters and pragmatist Early Majority buyers who require proven references before purchasing.
- **Board Exam Tip:** Technology Adoption Chasm lies between **Early Adopters** and **Early Majority**.

Question 71

A circular wire loop with a radius of 0.1 meters is placed in a uniform magnetic field that is perpendicular to the plane of the loop. If the magnetic field strength decreases at a constant rate of 0.5 T/s, what is the magnitude of the induced electromotive force (EMF) in the loop?
ANSWER: 0.016 V (15.7 mV)
- **Given:** $r = 0.1$ m, $\frac{dB}{dt} = 0.5$ T/s, $N = 1$.
- **Formula:** Faraday's Law: $|\mathcal{E}| = \frac{d\Phi}{dt} = A\frac{dB}{dt} = \pi r^2 \frac{dB}{dt}$.
- **Solution:**

$$A = \pi(0.1)^2 = 0.01\pi \approx 0.031416 \text{ m}^2$$

$$|\mathcal{E}| = 0.01\pi \times 0.5 = 0.005\pi \approx 0.015708 \text{ V} \approx 0.016 \text{ V}$$

- **Board Exam Tip:** Induced EMF magnitude is $|\mathcal{E}| = A\frac{dB}{dt}$.

Question 72

In engineering economics, what specific financial term describes the present worth of a project or asset that has an infinite assumed lifespan, commonly used for evaluating public works like bridges or dams?
ANSWER: Capitalized cost
- **Formula:** $CC = \text{Initial Cost} + \frac{\text{Annual Operating Cost}}{i}$.
- **Board Exam Tip:** Present worth of perpetual service = **Capitalized Cost**.

Question 73

Which Windows API function is used in C++ to create and register a graphical application window?
ANSWER: `CreateWindow()` / `CreateWindowEx()`
- **Explanation:** In the Win32 API, `CreateWindow()` creates an overlapped, pop-up, or child window, specifying its class name, window title, style, position coordinates, parent handle, and menu handle.
- **Board Exam Tip:** Win32 window creation function = `CreateWindow()`.

Question 74

The excessive runoff of agricultural fertilizers containing high concentrations of nitrogen and phosphorus into bodies of water directly causes which severe environmental degradation process?
ANSWER: Eutrophication
- **Explanation:** Nutrient enrichment spurs dense algal blooms. When these algae die, decomposing aerobic bacteria consume dissolved oxygen, creating hypoxic "dead zones" that suffocate aquatic life.
- **Board Exam Tip:** Excess nitrogen/phosphorus causing algal blooms and oxygen depletion = **Eutrophication**.

Question 75

Under the Philippine Environmental Impact Statement (EIS) System, an EIS is a comprehensive regulatory document prepared specifically to:
ANSWER: Assess the environmental impacts of a proposed project and specify appropriate mitigation measures before project implementation.
- **Explanation:** Established under Presidential Decree No. 1586, the EIS system evaluates potential environmental consequences of proposed developments to secure an Environmental Compliance Certificate (ECC) prior to construction.
- **Board Exam Tip:** EIS is an advance environmental assessment tool used to obtain an **ECC** under PD 1586.

Question 76

Which Java package provides the standard Abstract Window Toolkit (AWT) classes commonly used to develop graphical user interface (GUI) applications?
ANSWER: `java.awt`
- **Explanation:** `java.awt` contains the base GUI components (`Button`, `TextField`, `Frame`) and layout managers.
- **Board Exam Tip:** Standard AWT package in Java = `java.awt`.

Question 77

A startup develops an advanced LIDAR sensor that is vastly superior in accuracy to competitors but extremely expensive to produce. Using Porter's Generic Strategies, if they target strictly high-end autonomous aerospace applications rather than the general automotive market, what strategy are they applying?
ANSWER: Focused Differentiation (Niche Strategy)

- **Explanation:** Michael Porter's Generic Strategies:
  - **Cost Leadership:** Lowest industry production costs across a broad market.
  - **Differentiation:** Premium, unique product across a broad market.
  - **Cost Focus:** Cost advantage within a narrow, specialized niche.
  - **Focused Differentiation:** Unique, high-end product tailored to a narrow target niche.
- **Board Exam Tip:** Premium, specialized product serving a narrow market segment = **Focused Differentiation**.

Question 78

In a perfectly elastic one-dimensional collision between two isolated masses, what is the theoretical value of the coefficient of restitution?
ANSWER: 1.0 ($e = 1$)

- **Formula:** Coefficient of restitution: $e = \frac{v_2' - v_1'}{v_1 - v_2}$.
- **Explanation:** In a perfectly elastic collision ($e = 1.0$), both momentum and total kinetic energy are conserved. In a completely inelastic collision ($e = 0$), colliding bodies stick together and kinetic energy loss is maximized.
- **Board Exam Tip:** Perfectly Elastic: $e = 1$. Completely Inelastic: $e = 0$. Real-world Inelastic: $0 < e < 1$.

Question 79

Within how many days following the administration of the licensure examination are the Board and the Commission mandated to release the official results under Section 17 of RA 9292?
ANSWER: Fifteen (15) days

- **Statutory Basis:** RA 9292, Section 17 (*Release of Results of Examination*): The Board and the Commission shall correct and release the results within fifteen (15) days after the examination.
- **Board Exam Tip:** RA 9292 exam results release mandate = **15 days**.

Question 80

In Object-Oriented Analysis, which of the following BEST describes the problem domain of a software system?
ANSWER: The real-world entities, concepts, and relationships that the software system is designed to model.

- **Explanation:** The problem domain defines the real-world operational environment and business logic that software models before technical implementation details are introduced.
- **Board Exam Tip:** Problem domain encompasses **real-world business entities and requirements**.

Question 81

On a standard engineering stress-strain curve for a ductile metal, the specific stress point at which the material transitions from elastic, recoverable deformation to permanent, plastic deformation is known as the:
ANSWER: Yield strength (σ_y)

- **Explanation:** Below yield strength, atomic bonds stretch elastically and recover when unloaded. Above yield strength, slip occurs along crystalline planes, causing permanent plastic deformation. For materials without a sharp yield point, yield strength is conventionally identified using the **0.2% offset method** ($\varepsilon = 0.002$).
- **Board Exam Tip:** Transition from elastic to plastic deformation = **Yield Strength**. Maximum stress before necking = **Ultimate Tensile Strength (UTS)**.

Question 82

In UML Use Case diagrams, a stick figure is conventionally drawn to formally represent an entity that interacts directly with the system from the outside. What is this entity called?
ANSWER: The actor
- **Explanation:** An actor represents an external user, piece of hardware, or external software system that interacts with the system to initiate use cases or receive outputs.
- **Board Exam Tip:** External entity interacting with a system in UML = **Actor**.

Question 83

Which unique electromechanical property allows certain advanced crystalline ceramics, such as barium titanate, to generate a physical, mechanical strain when subjected to an external electric field?
ANSWER: Inverse piezoelectric effect
- **Explanation:**
  - **Direct Piezoelectric Effect:** Applying mechanical stress produces an internal electric charge/voltage.
  - **Inverse Piezoelectric Effect:** Applying an external electric field causes mechanical strain or dimensional change.
- **Board Exam Tip:** Electric field producing mechanical strain = **Inverse Piezoelectric Effect**.

Question 84

An electronics firm sources cobalt for its batteries. A human rights audit reveals their primary mining supplier utilizes child labor. To strictly apply global corporate social responsibility (CSR) standards, what immediate action must the firm's leadership take?
ANSWER: Cease procurement from the violating supplier immediately and transition to certified, ethically sourced cobalt vendors, despite higher raw material costs.
- **Explanation:** Corporate Social Responsibility (CSR) and international labor standards (such as UN Guiding Principles on Business and Human Rights) prohibit participating in child labor or human rights violations anywhere along supply chains.
- **Board Exam Tip:** CSR ethical standards require terminating relationships with supply-chain human rights violators.

Question 85

Engr. Jannatul discovers that a colleague is practicing the electronics engineering profession without a valid PRC license. What is the most appropriate action under professional ethical standards?
ANSWER: Report the unauthorized practice to the appropriate PRC authorities and inform the employer of the violation.
- **Explanation:** The ECE Code of Ethics requires practitioners to uphold the dignity of the profession and report illegal or unauthorized engineering practice to protect public safety and regulatory compliance.
- **Board Exam Tip:** Unlicensed practice must be formally reported to the **PRC / Professional Regulatory Board**.

Question 86

What is the standard SI derived unit for linear momentum?
ANSWER: Kilogram-meter per second (kg · m/s) or Newton-second (N · s)
- **Formula:** Linear momentum $p = mv \implies [\text{kg}] \times [\text{m/s}] = \text{kg} \cdot \text{m/s}$. Impulse: $J = \Delta p = F\Delta t \implies \text{N} \cdot \text{s}$.
- **Board Exam Tip:** Momentum unit: $\text{kg} \cdot \text{m/s} = \text{N} \cdot \text{s}$.

Question 87

An ideal electrical transformer has 100 turns on its primary coil and 500 turns on its secondary coil. If an alternating voltage of 120 V is applied to the primary, what is the output voltage on the secondary coil?
ANSWER: 600 V
- **Given:** $N_p = 100$, $N_s = 500$, $V_p = 120$ V.
- **Formula:** Ideal transformer turns ratio: $\frac{V_s}{V_p} = \frac{N_s}{N_p}$.

- **Solution:**

$$V_s = V_p \left(\frac{N_s}{N_p}\right) = 120\ \text{V} \times \left(\frac{500}{100}\right) = 120 \times 5 = 600\ \text{V}$$

- **Board Exam Tip:** Transformer ratio: $\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$.

Question 88

In a chemical reaction, if the total enthalpy of the products is strictly less than the total enthalpy of the reactants, what specific type of reaction has occurred?
ANSWER: Exothermic reaction ($\Delta H < 0$)
- **Formula:** Enthalpy change $\Delta H = H_{\text{products}} - H_{\text{reactants}}$.
- **Explanation:** Since $H_{\text{products}} < H_{\text{reactants}}$, ΔH is negative, meaning net chemical potential energy was released into the surroundings as heat.
- **Board Exam Tip:** Releases heat ($\Delta H < 0$) = **Exothermic**. Absorbs heat ($\Delta H > 0$) = **Endothermic**.

Question 89

In decision analysis recognizing uncertainty, a decision tree utilizes "chance nodes" (represented by circles). What specific mathematical process is performed at these nodes to evaluate the viability of the path?
ANSWER: Calculating the Expected Monetary Value (EMV) by multiplying potential payoffs by their respective probabilities and summing the results.
- **Formula:** EMV $= \sum_{j=1}^{m} p_j \cdot X_j$.
- **Board Exam Tip:** Decision nodes (squares) = Decision maker chooses best path. Chance nodes (circles) = Calculate weighted average (**EMV**).

Question 90

In ecology, the non-living physical and chemical elements of an environment that influence living organisms—such as sunlight, ambient temperature, water, and soil minerals—are collectively defined as:
ANSWER: Abiotic factors
- **Explanation:** Ecosystem components are divided into:
  - **Biotic Factors:** Living organisms (plants, animals, fungi, bacteria).
  - **Abiotic Factors:** Non-living physical and chemical variables (temperature, sunlight, pH, water, mineral nutrients).
- **Board Exam Tip:** Living = **Biotic**. Non-living environmental conditions = **Abiotic**.

Question 91

Photochemical smog is a hazardous atmospheric condition triggered primarily by the interaction of sunlight with volatile organic compounds (VOCs) and which other specific class of pollutants?
ANSWER: Nitrogen oxides (NO_x, such as NO and NO_2)
- **Explanation:** Solar ultraviolet radiation drives photochemical reactions between automotive NO_x and VOCs, producing secondary pollutants including tropospheric ground-level ozone (O_3) and peroxyacetyl nitrates (PAN).
- **Board Exam Tip:** Photochemical smog reactants = **NO_x + VOCs + Sunlight** $\implies$ Ground-level Ozone.

Question 92

In environmental management, the "Mitigation Hierarchy" is a sequence of actions designed to limit negative impacts. Analytically, which of the following represents the most preferred, highest-priority step in this hierarchy?
ANSWER: Avoid the impact entirely by altering the project site or canceling harmful activities.
- **Explanation:** The environmental Mitigation Hierarchy priority order is:
  1. **Avoid:** Prevent the impact from happening in the first place (highest priority).
  2. **Minimize:** Reduce the duration, intensity, or extent of impacts.
  3. **Rehabilitate / Restore:** Repair damaged habitats on-site.
  4. **Offset:** Compensate for residual unavoidable damage elsewhere (last resort).
- **Board Exam Tip:** Highest priority in the environmental mitigation hierarchy is ALWAYS **Avoidance**.

Question 93

A consumer electronics company sells its proprietary smart home hub at a steep discount, nearly at a loss, but requires users to purchase high-margin, proprietary smart-plugs and sensors to make the hub functional. This is a direct application of which business model?
ANSWER: Razor and Blade Model (Bait-and-Hook model)

- **Explanation:** The company sells the primary durable platform at or below cost ("the razor") to drive adoption, while generating steady high-margin profits from recurring, proprietary consumables or accessories ("the blades").
- **Board Exam Tip:** Low-cost base product dependent on high-margin proprietary accessories = **Razor and Blade Model**.

Question 94

On the periodic table of elements, the property of "electronegativity"—the fundamental tendency of an atom to attract shared electrons in a bond—generally exhibits which spatial trend?
ANSWER: Increases from left to right across a period and increases from bottom to top within a group.

- **Explanation:** Across a period, increasing nuclear charge pulls electrons closer, increasing electron attraction. Up a group, fewer electron shells reduce shielding, increasing core attraction. Fluorine (F) sits at the top right with the highest electronegativity (3.98 on the Pauling scale).
- **Board Exam Tip:** Electronegativity increases **Left $\rightarrow$ Right** and **Bottom $\rightarrow$ Top**. Highest electronegativity element is **Fluorine**.

Question 95

A leadership style where the engineering manager adopts a "hands-off" approach, granting team members total freedom to make decisions and complete their work without intervention, is known as:
ANSWER: Laissez-faire leadership (Free-rein leadership)

- **Explanation:**
  - **Autocratic:** Centralized authority, unilateral decision-making.
  - **Democratic / Participative:** Involves team members in decision-making.
  - **Laissez-faire:** Delegative approach with minimal supervisory oversight, most effective with highly autonomous, experienced experts.
- **Board Exam Tip:** Complete hands-off management delegation = **Laissez-faire leadership**.

Question 96

Which classification of polymer material undergoes a permanent, non-reversible chemical cross-linking reaction upon initial heating, meaning it will char and degrade rather than melt if reheated?
ANSWER: Thermosetting polymer (Thermoset)

- **Explanation:**
  - **Thermoplastics:** Linear or branched polymer chains held by weak van der Waals forces; melt when heated and resolidify when cooled (e.g., polyethylene, PVC).
  - **Thermosets:** Polymer chains bonded into a rigid 3D covalent cross-linked network; heat cures them permanently, and reheating causes decomposition/charring (e.g., epoxy, bakelite).
- **Board Exam Tip:** Covalently cross-linked, chars without melting = **Thermosetting polymer**. Melts and can be reshaped = **Thermoplastic**.

Question 97

If the physical length of a simple pendulum is quadrupled, how does its theoretical period of oscillation change?
ANSWER: It is doubled ($T_{\text{new}} = 2T_{\text{old}}$).

- **Formula:** Period of a simple pendulum: $T = 2\pi\sqrt{\frac{L}{g}}$.
- **Solution:** If $L' = 4L$:

$$T' = 2\pi\sqrt{\frac{4L}{g}} = \sqrt{4}\left(2\pi\sqrt{\frac{L}{g}}\right) = 2T$$

- **Board Exam Tip:** Pendulum period is proportional to $\sqrt{L}$. Quadrupling length doubles the period.

Question 98

An ECE establishes a new consulting firm and places a full-page advertisement in a national newspaper. The ad claims their firm is "The Absolute Best and Only Competent Electronics Engineering Firm in the Philippines, Guaranteed." Why does this violate the Code of Ethics for Electronics Practitioners?
ANSWER: Because it utilizes self-laudatory, exaggerated, and unprofessional language that degrades the dignity of the profession and maligns other practitioners.

- **Ethical Basis:** The ECE Code of Ethics requires advertising to be factual, dignified, and modest. Self-laudatory claims of superiority or statements that discredit other licensed practitioners violate ethical guidelines.
- **Board Exam Tip:** Engineers must advertise in a professional, factual manner and avoid self-laudatory claims.

Question 99

When white light passes through a triangular glass prism, it separates into its component colors and forms a spectrum on a screen. What physical phenomenon primarily causes this separation?
ANSWER: Dispersion (Different wavelengths refract at different angles due to wavelength-dependent refractive index $n(\lambda)$).

- **Explanation:** In optical media, the refractive index varies with wavelength (optical dispersion, described by Cauchy's equation: $n(\lambda) = A + B/\lambda^2$). Shorter wavelengths (violet, $\lambda \approx 400$ nm) encounter a higher refractive index and bend the most; longer wavelengths (red, $\lambda \approx 700$ nm) experience a lower refractive index and bend the least.
- **Board Exam Tip:** Separation of light into component colors by wavelength-dependent refraction = **Dispersion**.

Question 100

For ferromagnetic materials like iron, the specific critical temperature threshold above which the material loses its permanent magnetic properties and becomes strictly paramagnetic is known as the:
ANSWER: Curie temperature (T_c)

- **Explanation:** At and above the Curie temperature, thermal energy breaks the aligned spin coupling within magnetic domains, destroying spontaneous macroscopic magnetization and transforming the material into a paramagnetic substance.
- **Board Exam Tip:** Ferromagnetic to paramagnetic transition threshold = **Curie Temperature**.

2 Part II — Important Terms & Concepts Summary

2.1 Engineering Management & Technopreneurship

- **Critical Path Method (CPM):** Finds the longest sequence of dependent tasks. Critical tasks have zero float (Slack $= 0$); delays to these tasks directly delay project completion.
- **Managerial Control Types:**
  - *Feedforward:* Anticipates and prevents problems before work begins.
  - *Concurrent:* Real-time monitoring during task execution.
  - *Feedback:* Evaluates finished deliverables after completion.
- **Herzberg's Two-Factor Theory:** Hygiene factors (salary, policies) prevent dissatisfaction but do not motivate; Motivator factors (advancement, responsibility) drive satisfaction and performance.
- **Blue Ocean Strategy:** Creates uncontested market space to make the competition irrelevant, contrasting with competing in saturated "red oceans."
- **Razor and Blade Model:** Sells a base item at low margin to generate recurring high-margin sales from proprietary consumables or accessories.
- **Startup Runway:** Runway (months) $= \frac{\text{Cash Reserves}}{\text{Monthly Burn Rate}}$.

2.2 RA 9292 & Professional Ethics

- **RA 9292 (Electronics Engineering Law of 2004):** Governs the registration, licensing, and regulation of PECEs, ECEs, and ECTs.
- **Board Composition:** 1 Chairman and 2 Members, appointed by the President of the Philippines (Section 6).
- **Board Qualification:** Appointees must be PECEs with at least ten (10) years of continuous active professional practice (Section 8).
- **Licensure Results Release:** Must be released within fifteen (15) days after the exam (Section 17).
- **License Reinstatement:** Revoked licenses may be considered for reinstatement after two (2) years (Section 24).
- **Unlicensed Practice Penalties:** Fines of P$100,000$ to P$1,000,000$ and/or imprisonment of 6 months to 6 years (Section 35).
- **Legal Delay (*Mora*):** *Mora Solvendi* is debtor default; *Mora Accipiendi* is creditor delay.

2.3 Materials Science & Solid-State Physics

- **Magnetic Transitions:** Ferromagnetic to paramagnetic transition occurs at the Curie Temperature (T_c).
- **Magnetic Materials:** Permanent magnets require high coercivity; transformer cores require low coercivity.
- **Crystal Imperfections:** 0D = Point defects (vacancies); 1D = Line defects (dislocations); 2D = Planar defects (grain boundaries); 3D = Volume defects (voids).
- **Unit Cell Atom Counts:** Simple Cubic (SC) = 1; Body-Centered Cubic (BCC) = 2; Face-Centered Cubic (FCC) = 4.
- **Hume-Rothery Rules:** Substitutional solid solutions require atomic radius difference $\leq 15\%$, similar crystal structures, and close electronegativities.
- **Polymers:** Thermoplastics melt and can be reshaped; Thermosets cure into cross-linked networks that char when heated.

2.4 General Engineering Sciences & Environmental Engineering

- **Doppler Effect:** Approaching source compresses waves to higher frequency; accelerating source causes perceived frequency to rise continuously.
- **Fluid Dynamics:** In horizontal fluid flow, a narrowing cross-section increases velocity and decreases static pressure (Bernoulli's principle / Venturi effect).
- **Capitalized Cost:** Present worth of perpetual service: $CC = FC + A/i$.
- **Environmental Impacts:** Excess nitrogen and phosphorus runoff causes eutrophication; photochemical smog is produced by reactions between NO_x, VOCs, and sunlight.

3 Part III — Comprehensive Formula Sheet

Subject / Law	Mathematical Formula	Application / Description
Expected Monetary Value	$\text{EMV} = \sum p_i \cdot X_i$	Decision tree chance node evaluation
Startup Cash Runway	$\text{Runway} = \frac{\text{Cash Reserves}}{\text{Operating Expenses} - \text{Revenues}}$	Time before cash exhaustion (months)
Equity Dilution Ratio	$\text{Dilution} = \frac{\text{New Shares}}{\text{Existing Shares} + \text{New Shares}}$	Equity percentage issued to investors
Capitalized Cost (Perpetual)	$CC = FC + \frac{A}{i}$	Infinite-life asset present worth
Marginal Cost	$MC = \frac{\Delta TC}{\Delta Q} = \frac{d(TC)}{dQ}$	Incremental cost to produce one more unit
Mass-Spring System Period	$T = 2\pi\sqrt{\frac{m}{k}}$	Simple harmonic oscillator period
Simple Pendulum Period	$T = 2\pi\sqrt{\frac{L}{g}}$	Small-angle pendulum period
Photoelectric Effect	$K_{\max} = \frac{hc}{\lambda} - \Phi \approx \frac{1240\ \text{eV·nm}}{\lambda(\text{nm})} - \Phi$	Max photoelectron kinetic energy
Capacitors in Series	$\frac{1}{C_{eq}} = \sum \frac{1}{C_i} \implies C_{eq} = \frac{C_1 C_2}{C_1 + C_2}$	Total equivalent series capacitance
Frictionless Banked Curve	$\tan\theta = \frac{v^2}{Rg}$	Ideal roadway/track banking angle
Fick's First Law (Diffusion)	$J = -D\frac{dC}{dx}$	Steady-state mass diffusion flux
Fick's Second Law	$\frac{\partial C}{\partial t} = D\frac{\partial^2 C}{\partial x^2}$	Time-dependent non-steady diffusion
Carnot Engine Efficiency	$\eta = 1 - \frac{T_C}{T_H} = \frac{T_H - T_C}{T_H}$	Maximum thermodynamic efficiency
First Law of Thermodynamics	$Q = \Delta U + W$	Closed-system conservation of energy
Radioactive Decay Law	$N(t) = N_0\left(\frac{1}{2}\right)^{t/T_{1/2}} = N_0 e^{-\lambda t}$	Half-life decay of active isotopes
Magnetic Force on Conductor	$F = ILB\sin\theta$	Lorentz force on current-carrying wire
Faraday's Law of Induction	$\lvert\mathcal{E}\rvert = N\frac{d\Phi}{dt} = NA\frac{dB}{dt}$	Induced electromotive force (EMF)
Coefficient of Restitution	$e = \frac{v_2' - v_1'}{v_1 - v_2}$	Elasticity of collision ($e = 1$ is elastic)
Ideal Transformer Voltage	$\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$	Voltage and current turns ratio
Terminal Velocity	$v_t = \sqrt{\frac{2mg}{\rho C_d A}}$	Equilibrium falling velocity in a fluid
Doppler Shift (Approaching)	$f' = f\left(\frac{v}{v - v_s}\right)$	Perceived frequency from moving source
Bulk Modulus	$K = -V\frac{\Delta P}{\Delta V}$	Resistance to volumetric compression

4 Part IV — Quick Answer Key Table

5 Part V — Final Board Exam Cram Sheet

5.1 Must-Remember Formulas

1. **Carnot Cycle Efficiency:** $\eta = \frac{T_H - T_C}{T_H} \times 100\%$ (Temperatures must be in Kelvin).
2. **Photoelectric Equation:** $K_{\max} = \frac{1240}{\lambda(\text{nm})} - \Phi(\text{eV})$.
3. **Capitalized Cost Equation:** $CC = FC + \frac{A}{i}$.
4. **Expected Monetary Value (EMV):** $\text{EMV} = \sum p_i X_i$.
5. **Harmonic Periods:** Mass-Spring: $T = 2\pi\sqrt{\frac{m}{k}}$; Pendulum: $T = 2\pi\sqrt{\frac{L}{g}}$.
6. **Frictionless Banked Curve:** $\tan\theta = \frac{v^2}{Rg}$.
7. **Faraday's Induced EMF:** $|\mathcal{E}| = NA\frac{dB}{dt}$.
8. **Transformer Turns Ratio:** $\frac{V_s}{V_p} = \frac{N_s}{N_p} = \frac{I_p}{I_s}$.
9. **Lorentz Magnetic Force on Wire:** $F = ILB\sin\theta$.
10. **Startup Runway Formula:** $\text{Runway} = \frac{\text{Cash Reserves}}{\text{Monthly Burn Rate}}$.
11. **Equity Dilution Ratio:** $\text{Dilution} = \frac{\text{New Shares}}{\text{Total Shares After Round}}$.
12. **Fick's Laws:** 1st (Steady): $J = -D\frac{dC}{dx}$; 2nd (Unsteady): $\frac{\partial C}{\partial t} = D\frac{\partial^2 C}{\partial x^2}$.

5.2 Must-Remember One-Line Concepts

- Tasks on the **Critical Path** have **Zero Slack (Float)**.
- **Feedforward** control acts before the process; **Concurrent** control acts during; **Feedback** control acts after.
- Ferromagnetic materials lose spontaneous magnetization above the **Curie Temperature**.
- Permanent magnets require **High Coercivity**; transformer cores require **Low Coercivity**.
- Total Internal Reflection requires $n_{\text{core}} > n_{\text{clad}}$ and incident angle greater than the critical angle.
- **Primary Succession** occurs on barren, soil-free ground; **Secondary Succession** occurs where soil remains.
- **Biomagnification** concentrates persistent toxins up successive trophic levels of a food chain.
- **Hume-Rothery Rule:** Complete solid solubility requires atomic size difference $\leq 15\%$.
- Unit cell atom contents: Simple Cubic = 1, BCC = 2, FCC = 4.
- Under RA 9292, the PRBECE consists of **1 Chairman and 2 Members**, appointed by the **President of the Philippines**.
- PRBECE appointees must have at least **10 years** of active practice as a registered PECE.
- Licensure examination results must be officially released within **15 days**.
- Revoked licenses may be considered for reinstatement after **2 years**.
- Debtor delay is termed ***Mora Solvendi***; creditor delay is ***Mora Accipiendi***.
- **Venturi Effect:** Fluid entering a constriction speeds up, causing static pressure to drop.
- The highest priority in the environmental mitigation hierarchy is always **Avoidance**.
- Ground-level photochemical smog is driven by reactions of **NO_x + VOCs + Sunlight**.

5.3 Common Traps & Quick Rules

- **Temperature Units Trap:** In Carnot engine and thermodynamic equations, always convert °C to Kelvin ($K = {}^\circ\text{C} + 273.15$).
- **Series/Parallel Capacitance vs Resistance:** Capacitors in series combine like resistors in parallel: $C_{eq} = \frac{C_1 C_2}{C_1 + C_2}$. Capacitors in parallel add directly: $C_{eq} = C_1 + C_2$.
- **Initial Conditions ($t = 0^+$):** Inductors act as open circuits ($i = 0$, full voltage drop); capacitors act as short circuits ($v = 0$, maximum current).
- **Net Burn Rate Trap:** When calculating startup runway, remember to subtract incoming monthly revenue from gross monthly expenses before dividing into cash reserves.
- **Dilution Base Trap:** Dilution is calculated over **post-money shares** (original shares + new shares), not original shares alone.
- **Bioaccumulation vs Biomagnification:** Bioaccumulation occurs within a single organism over time; Biomagnification multiplies up the food chain across trophic levels.
- **Pre-test vs Post-test Loops:** A `while` loop can execute 0 times; a `do-while` loop executes at least once.
- **Electronegativity Trend:** Increases toward the top-right of the periodic table (Fluorine is highest); decreases toward the bottom-left.

No.	Key Answer	Topic	No.	Key Answer	Topic
1	Zero slack or float time	Project Management / CPM	51	6.25 grams	Physics / Radioactive Half-Life
2	Feedforward control	Management / Control Systems	52	6.0 N	Physics / Magnetic Force (BIL)
3	Curie temperature	Materials / Ferromagnetism	53	Velocity up, static pressure down	Fluid Dynamics / Bernoulli
4	Damped harmonic oscillator	Physics / Oscillations	54	Start/finish dates on timeline	Project Management / Gantt
5	Multiplicity	Software Engineering / UML	55	Inspect circuit for root cause	Electrical Safety / Breakers
6	The period is halved	Physics / Mass-Spring Period	56	Interface constants vs class state	Programming / Java OOP
7	Execute critical cleanup code	Programming / Exceptions	57	Cease and report practice	RA 9292 / Penal Sanctions
8	Mora Solvendi	Civil Code / Obligations	58	Chamfer	CAD Drafting / Commands
9	High coercivity	Materials / Permanent Magnets	59	15% atomic radius difference	Metallurgy / Hume-Rothery
10	P52,000	Engineering Economics / EMV	60	Different number of neutrons	Chemistry / Isotopes
11	Hygiene factors	Motivation / Herzberg	61	Carbon dioxide (CO_2)	Environmental / Greenhouse
12	Higher index of refraction	Optics / Total Internal Refl	62	Chairman + 2, by President	RA 9292 / Board Composition
13	Etching out of bulk substrate	Nanotechnology / Lithography	63	Primary succession	Ecology / Ecological Succession
14	Biomagnification	Environmental / Toxicology	64	Biosphere	Ecology / Earth Systems
15	Register, license, regulate	ECE Laws / PRC Mandate	65	Pan	CAD Drafting / Navigation
16	Functional Programming	Programming Paradigms	66	Concentration gradient	Materials / Fick's 1st Law
17	Higher and continuously rising	Physics / Doppler Effect	67	Independent of path taken	Classical Mechanics / Forces
18	Biomass gasification / Biogas	Renewable Energy / Digestion	68	Risk-averse	Behavioral Economics / Utility
19	1 atom	Materials / Simple Cubic	69	Nuclear binding energy	Nuclear Physics / Mass Defect
20	0.65 eV	Modern Physics / Photoelectric	70	Early Majority	Technopreneurship / Chasm
21	8 μF	Circuit Theory / Capacitors	71	0.016 V (15.7 mV)	Electromagnetics / Faraday
22	Max Weber	Management / Bureaucracy	72	Capitalized cost	Engineering Economics / Perpetuity
23	Traps heat via IR absorption	Environmental / Greenhouse	73	CreateWindow()	Programming / Win32 API
24	Tangent of angle quadruples	Mechanics / Banked Curves	74	Eutrophication	Environmental / Nutrient Runoff
25	Interfacial (2D planar) defect	Materials / Crystal Defects	75	Assess impacts and mitigation	Environmental / EIS System
26	Latent heat of fusion	Thermodynamics / Phase Change	76	java.awt	Programming / Java GUI
27	$i = 0, v_L = V_B$	Transients / Inductor at $t = 0^+$	77	Focused Differentiation	Strategic Management / Porter
28	The event listener	GUI Programming / Event Models	78	1.0 (e = 1)	Mechanics / Restitution
29	6 months	Technopreneurship / Runway	79	Fifteen (15) days	RA 9292 / Exam Results
30	Denitrification	Environmental / Nitrogen Cycle	80	Real-world problem entities	Software Engineering / OOAD
31	Blue Ocean Strategy	Business Strategy / Blue Ocean	81	Yield strength	Materials / Stress-Strain
32	Fick's Second Law	Materials / Non-Steady Diffusion	82	The actor	Software Engineering / UML
33	Two (2) years	RA 9292 / Reinstatement	83	Inverse piezoelectric effect	Smart Materials / Transducers
34	Marginal cost	Engineering Economics / Cost	84	Cease violating supplier	Business Ethics / CSR
35	Standing wave	Physics / Wave Interference	85	Report to PRC / Board	ECE Code of Ethics / Reporting
36	20% dilution	Technopreneurship / Equity	86	kg·m/s (or N·s)	Mechanics / Momentum Units
37	Cloud processing to phones	Technopreneurship / VPC	87	600 V	Electromagnetics / Transformer
38	Terminal velocity	Fluid Mechanics / Aerodynamics	88	Exothermic reaction	Chemistry / Thermochemistry
39	Capitalized Cost Analysis	Engineering Economics / Perpetuity	89	Calculating EMV	Decision Analysis / Chance Nodes
40	Ten (10) years	RA 9292 / Board Qualifications	90	Abiotic factors	Ecology / Ecosystem Elements
41	Product-as-a-service circular	Technopreneurship / Circular	91	Nitrogen oxides (NO_x)	Environmental / Smog
42	62.5%	Thermodynamics / Carnot Engine	92	Avoid the impact entirely	Environmental / Mitigation
43	Internal energy	Thermodynamics / First Law	93	Razor and Blade Model	Business Models / Monetization
44	Existing Alternatives	Technopreneurship / Lean Canvas	94	Increases Left→Right, Bottom→Top	Chemistry / Electronegativity
45	Matrix structure	Management / Organizational	95	Laissez-faire leadership	Management / Leadership Styles
46	Volume compression	Mechanics / Bulk Modulus	96	Thermosetting polymer	Materials / Polymers
47	Minimizes maximum regret	Decision Theory / Savage	97	It is doubled	Physics / Pendulum Length
48	A while loop	Programming / Loop Constructs	98	Unprofessional self-laudation	ECE Code of Ethics / Ads
49	North pole to South pole	Electromagnetics / Field Lines	99	Dispersion (refractive index)	Optics / Prisms
50	Internal Rate of Return (IRR)	Engineering Economics / IRR	100	Curie temperature	Materials / Ferromagnetism

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PHILIPPINE ELECTRONICS ENGINEERING LICENSURE EXAMINATION | COMPREHENSIVE TECHNICAL REVIEW
PRACTICE BOARD EXAMINATION SERIES (PB-11) - 100 ITEMS (GEAS, LAWS, MGMT & ICT)
COMPLETE FORMULA COMPILATION, TERMINOLOGY GUIDE & DETAILED SOLUTIONS

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[FORMAT: Monospace ASCII Plain-Text Reference Card Layout | Letter / A4 Paper: Exactly 22 Pages]

PART I: COMPLETE MATHEMATICAL FORMULAS & GOVERNING LAWS COMPILATION

This compilation aggregates every governing formula, physical law, mathematical equation, statutory provision, and economic factor required to solve the 100 examination problems in Series PB-11 (Set 2) covering Engineering Mechanics, Strength of Materials, Thermodynamics, Chemistry, Materials Science, Engineering Economics, Management, OOP, and RA 9292 / RA 7925.

Subject Domain	Mathematical Formula / Governing Expression	Definition of Variables & Physical Constants
Rotational Mechanics	$I = \int r^2\, dm = \Sigma\, m_i * r_i^2$ $\tau_{net} = I * \alpha = dL / dt$	I = Moment of inertia (kg·m²), r = radial distance from axis, τ = torque (N·m), L = angular momentum (kg·m²/s).
Photoelectric Effect	$E = h * f = h * c / \lambda$ $K_{max} = h * f - \Phi$; $f_0 = \Phi / h$	$h = 6.626 \times 10^{-34}$ J·s (Planck's constant), Φ = work function, f_0 = threshold frequency. Emission occurs only if $f \geq f_0$.
Fluid Hydrostatics	$F_{buoyant} = \rho_{fluid} * V_{displaced} * g = W_{displaced}$	Archimedes' Principle: ρ = fluid density, V = submerged volume, g = 9.81 m/s². Upward force equals displaced fluid weight.
Electrostatics (Coulomb)	$F = k_e * (\lvert q1 * q2 \rvert) / r^2$ $E = F / q$ (N/C or V/m)	$k_e = 8.98755 \times 10^9$ N·m²/C², r = distance between charges. Halving distance (r/2) quadruples the electrostatic force (4F).
Centripetal Dynamics	$a_c = v^2 / r = \omega^2 * r$ $F_c = (m * v^2) / r$	m = mass (kg), v = tangential speed (m/s), r = curve radius (m). For 1000 kg, 10 m/s, 50 m: $F_c = 1000*(100)/50 = 2000$ N.
Crystallography (BCC)	$4R = a * \sqrt{3}$ APF = (N * V_atom) / V_cell = $(\pi * \sqrt{3}) / 8 \approx 0.68$	R = atomic radius, a = lattice parameter, N = 2 net atoms/cell. BCC packing fraction is exactly 68% (0.68).
Harmonic Oscillations	$T = 2 * \pi * \sqrt{(m / k)}$ $f = 1 / T = (1 / 2\pi) * \sqrt{(k / m)}$	T = oscillation period, m = oscillating mass, k = spring constant. Quadrupling mass (4m) exactly doubles the period (2T).
Engineering Economics (Annuity)	(P/A, i, n) = [(1 + i)^n - 1] / [i * (1 + i)^n] F = P * (1 + i)^n; AW = PW * (A/P, i, n)	P = present worth, A = uniform annual payment, i = interest rate per period, n = number of periods.
Optics & Critical Angle	$n1 * \sin(\theta 1) = n2 * \sin(\theta 2)$ $\sin(\theta_c) = n2 / n1$ (where n1 > n2)	Total Internal Reflection occurs only when transitioning from denser (higher n1) to less dense (lower n2) medium.
Thermodynamics (Carnot)	$\eta_{Carnot} = 1 - (T_{cold} / T_{hot})$ Adiabatic: Q = 0 => ΔU = -W	Temperatures strictly in Kelvin. For T_{hot} = 400 K, T_{cold} = 300 K: η = 1 - 300/400 = 25%. Rapid expansion cools gas.
Hooke's Law & Elasticity	$E = \sigma / \varepsilon = (F / A) / (\Delta L / L_0)$	E = Young's Modulus (Pa), σ = tensile stress, ε = tensile strain. Quantifies linear axial stiffness within proportional limit.
Electromagnetic Induction	$EMF = - N * (d\Phi_B / dt)$ $\Phi_B = \int B \bullet dA$; Lenz's Law = Opposing sign	Faraday's Law: Induced voltage equals rate of change of magnetic flux. Lenz's Law: Induced current opposes change in flux.
Steady-State Diffusion	$J = - D * (dC / dx)$ (Fick's First Law)	J = diffusion flux (atoms/m²·s), D = diffusion coefficient, dC/dx = concentration gradient. Negative sign denotes flow down gradient.
Financial Liquidity & EMV	Quick Ratio = (Cash + Marketables + A/R) / Current Liab. EMV = Σ (P_i * Payoff_i)	Quick ratio excludes inventory. Expected Monetary Value weights each possible branch payoff by its state probability.
Ground Fault Impedance (PEC)	$Z_{fault} = R_{cond} + j * 2\pi f * L_{loop}$	Routing ground outside metallic conduit enlarges loop area, spiking AC inductive reactance and preventing OCPD clearing.
RA 9292 Board & Practice	Board: 1 Chairman + 2 Members = 3 Total PECE Sole Authority: Sign/Seal Plans, Consulting	Republic Act 9292 Sections 5, 6, 27, and 35 define regulatory board composition, exclusive PECE stamping rights, and penalties.

PART II: COMPREHENSIVE TERMINOLOGY & TECHNICAL GLOSSARY GUIDE

Nanomaterial: Materials possessing at least one external dimension in the nanoscale regime of 1 to 100 nanometers, characterized by quantum confinement and high specific surface area.

Moment of Inertia: The rotational analog of linear mass, quantifying a body's resistance to angular acceleration based on its total mass and squared radial distance distribution relative to the rotational axis.

Bootstrapping: An entrepreneurial financing strategy where founders build and scale a business relying solely on personal savings and reinvested customer revenues without taking external venture capital.

Brittle Fracture in Ceramics: Catastrophic cleavage failure occurring with little to no prior plastic deformation, driven by the absence of independent dislocation slip systems under high bond directional stiffness.

Mora / Legal Delay (Civil Code): A condition where a debtor fails to perform an obligation on time. Under Article 1165, an obligor in delay remains strictly liable for damages even if destroyed by a fortuitous event.

Encapsulation & Information Hiding: An OOP principle where an object's internal fields are made `private`, forcing all external interactions to pass through `public` accessor and mutator methods.

Ortho Mode (CAD): A drafting constraint toggle (F8) that restricts mouse movement and coordinate entry strictly to horizontal and vertical Cartesian grid axes.

Quick Ratio (Acid-Test): A stringent liquidity metric that evaluates immediate debt-paying ability by dividing quick liquid assets (cash, marketable securities, receivables) by current liabilities, excluding illiquid inventory.

Polymorphism: The OOP capability that allows subclasses to provide specific overridden implementations of inherited parent methods, dispatched dynamically at runtime.

Greenhouse Effect: The warming of Earth's surface caused by atmospheric triatomic gases (CO_2, H_2O, CH_4) absorbing outgoing longwave terrestrial infrared radiation and re-radiating it isotropically.

Atomic Packing Factor (APF): The fraction of solid unit cell volume occupied by hard-sphere atoms. For Body-Centered Cubic (BCC) metallic crystals, the APF is exactly 0.68 (68%).

Nuclear Reactor Moderator: A light-element medium (water, heavy water, graphite) that slows down energetic fast neutrons (~2 MeV) via elastic scattering into thermal neutrons (~0.025 eV) to sustain U-235 fission.

Annual Worth Method: An economic equivalence evaluation technique that converts capital cash flows into equivalent uniform annual amounts over one life cycle, enabling unbiased comparison of unequal-life alternatives without LCM expansion.

Thermoplastic vs. Thermoset: Thermoplastics feature linear/branched chains held by weak secondary bonds that can be repeatedly melted and reshaped; thermosets possess permanent cross-links that degrade upon heating.

Scientific Management (Taylorism): A classical management school developed by Frederick Taylor focusing on standardizing job workflows and maximizing labor output via empirical time-and-motion studies.

PECE Stamping Authority (RA 9292): The exclusive statutory legal prerogative granted solely to duly registered Professional Electronics Engineers to sign, seal, and issue official electronics plans, specifications, and reports.

Intrinsically Safe Systems (PEC / NEC): Equipment designed such that operating electrical and thermal energy levels remain strictly below the Minimum Ignition Energy (MIE) of explosive atmospheric mixtures under normal and fault conditions.

Blue Ocean Strategy: A business growth framework where a venture escapes cutthroat competition ('Red Oceans') by pioneering an entirely new, uncontested market space through value innovation.

Triple Bottom Line (TBL): A holistic sustainability accounting framework measuring organizational performance across three interconnected dimensions: Economic (Profit), Social (People), and Environmental (Planet).

Minimum Viable Product (MVP): The core artifact in Lean Startup methodology built with minimum resources to begin the Build-Measure-Learn feedback loop and validate fundamental business hypotheses with early adopters.

Model Space vs. Paper Space: Model Space is the infinite 1:1 scale geometric authoring environment in CAD; Paper Space (Layout) arranges scaled viewports and annotations for standardized physical printing.

Lenz's Law: An induction principle stating that the polarity of an induced EMF generates an electric current whose magnetic field resists the change in magnetic flux that produced it.

PART III: STEP-BY-STEP PROBLEM CARDS & DETAILED SOLUTIONS (QUESTIONS 001 - 100)

QUESTION 001 [MATERIALS SCIENCE & NANOTECHNOLOGY]

Problem: By standard scientific definition, a nanomaterial must have at least one external dimension in which of the following size ranges?

VERIFIED ANSWER: 1 to 100 nanometers

Governing Formula / Principle: $1\text{ nm} \le d_{nano} \le 100\text{ nm}$ (10^{-9} m to 10^{-7} m)

Step-by-Step Technical Derivation:

- ISO/TS 80004 and IUPAC standard definitions define a nanomaterial as an object with at least one external dimension, internal structure, or surface structure in the nanoscale.
- The nanoscale is explicitly established as the range spanning approximately 1 nm to 100 nm.
- Within this specific regime, quantum mechanical effects and extreme surface-area-to-volume ratios dominate material behavior.

QUESTION 002 [ENGINEERING MECHANICS / ROTATIONAL DYNAMICS]

Problem: What physical property describes the resistance of an object to changes in its rotational motion, depending on both its mass and the distribution of that mass relative to the axis?

VERIFIED ANSWER: Moment of inertia

Governing Formula / Principle: $I = \int r^2\, dm = \Sigma\, m_i * r_i^2$; Torque $\tau = I * \alpha$

Step-by-Step Technical Derivation:

- In translational dynamics, inertial mass m quantifies resistance to linear acceleration (F = m*a).
- In rotational mechanics, the rotational inertia depends not merely on total mass, but on how that mass is distributed relative to the axis of rotation.
- This second moment of mass with respect to the rotational axis is termed the Moment of Inertia (I).

QUESTION 003 [ENGINEERING MANAGEMENT & ENTREPRENEURSHIP]

Problem: An entrepreneur funds their startup entirely through personal savings and the reinvestment of early revenues, deliberately avoiding external equity investors. What is this financial execution strategy called?

VERIFIED ANSWER: Bootstrapping

Governing Formula / Principle: Self-Sustaining Financial Model: Working Capital = Owner Equity + Retained Earnings

Step-by-Step Technical Derivation:

- Bootstrapping refers to building and scaling a company solely utilizing existing personal resources, operating cash flows, and customer revenues.
- It deliberately foregoes venture capital, angel investment, or third-party equity financing.
- This ensures the founders retain 100% equity ownership and governance control.

QUESTION 004 [MATERIALS SCIENCE & CERAMICS]

Problem: Which of the following is a characteristic mechanical property of traditional structural ceramics?

VERIFIED ANSWER: Brittle fracture

Governing Formula / Principle: Fracture Toughness $K_{Ic} = Y * \sigma * \sqrt{\pi * a}$; Lack of dislocation slip systems

Step-by-Step Technical Derivation:

- Ceramics feature strong ionic and covalent bonding with high bond directional stiffness.
- The absence of independent slip systems severely restricts dislocation motion and plastic yield.
- Consequently, ceramics undergo catastrophic brittle fracture along microcrack stress concentrations without significant preceding plastic deformation.

QUESTION 005 [MODERN PHYSICS / PHOTOELECTRIC EFFECT]

Problem: In the photoelectric effect, if the frequency of the incident light is strictly below the threshold frequency of the target metal, what occurs?

VERIFIED ANSWER: No electrons are emitted, regardless of the light's intensity.

Governing Formula / Principle: $E_{photon} = h * f$; $K_{max} = h * f - \Phi$; Emission condition: $f \ge f_0 = \Phi / h$

Step-by-Step Technical Derivation:

- Einstein's photoelectric law establishes that energy transfer occurs in discrete quanta (photons) of energy E = h*f.
- The minimum energy required to liberate an electron from the metal surface is the work function $\Phi = h*f_0$.
- If $f < f_0$, an individual photon lacks sufficient energy to overcome Φ; increasing intensity merely increases the number of sub-threshold photons, yielding zero emission.

QUESTION 006 [LAWS & ETHICS / OBLIGATIONS & CONTRACTS]

Problem: An ECE firm delays a network installation project, and subsequently, a massive typhoon destroys the partially completed work.

VERIFIED ANSWER: The firm is strictly liable for all damages because they were already in legal delay when the act of God occurred.

Governing Formula / Principle: Civil Code of the Philippines Art. 1165, 1170, 1174: Fortuitous Event Exception in Mora

Step-by-Step Technical Derivation:

• Under Philippine contract law, an obligor is generally exempt from liability for failure to perform due to a fortuitous event (force majeure / act of God).

• However, Article 1165 explicitly establishes an exception: if the obligor is already in default (mora) or delay prior to the event, they remain fully liable.

• Because the firm incurred default prior to the typhoon, fortuitous event defenses are legally extinguished.

QUESTION 007 [COMPUTER PROGRAMMING & OOP (JAVA / C++)]

Problem: In Java and C++, which block of code is explicitly designed to contain the statements that might generate a runtime error, allowing it to be caught without crashing the application?

VERIFIED ANSWER: Try block

Governing Formula / Principle: Exception Handling Architecture: try { ... } catch (Exception e) { ... } finally { ... }

Step-by-Step Technical Derivation:

• A `try` block encloses code statements that have the potential to throw monitored checked or unchecked exceptions at runtime.

• When an abnormal condition occurs inside the try block, program execution halts inside that block and transfers control to the matching `catch` handler.

• This structured exception handling prevents abnormal program abortion.

QUESTION 008 [FLUID MECHANICS & HYDROSTATICS]

Problem: Which fundamental fluid mechanics principle states that the buoyant force exerted on a body fully or partially immersed in a fluid is equal to the weight of the fluid that the body displaces?

VERIFIED ANSWER: Archimedes' Principle

Governing Formula / Principle: $F_{buoyant} = \rho_{fluid} * V_{displaced} * g = W_{fluid}$

Step-by-Step Technical Derivation:

• Fluid hydrostatic pressure increases linearly with depth ($P = \rho*g*h$), creating an upward net vertical pressure gradient across any submerged body.

• Integrating net vertical pressure forces over the entire surface area equals the weight of the displaced fluid.

• This law of buoyancy was formulated by Archimedes of Syracuse.

QUESTION 009 [RA 9292 (ELECTRONICS ENGINEERING LAW OF 2004)]

Problem: According to Republic Act No. 9292, the science dealing with the development and application of devices involving the flow of electrons in a vacuum, gaseous media, plasma, or semiconductors is defined as:

VERIFIED ANSWER: Electronics

Governing Formula / Principle: RA 9292 Section 3(a): Statutory Definition of Electronics

Step-by-Step Technical Derivation:

• Section 3, Article I of Republic Act No. 9292 establishes statutory definitions for the profession.

• Section 3(a) defines 'Electronics' verbatim as the science dealing with electron flow in vacuum, gases, plasma, or semiconductors.

• It serves as the legal foundation distinguishing electronics practice from general electrical engineering.

QUESTION 010 [OBJECT-ORIENTED PROGRAMMING (OOP)]

Problem: Which fundamental Object-Oriented Programming concept is achieved by strictly hiding the internal state of an object and requiring all interaction to occur through its public methods?

VERIFIED ANSWER: Encapsulation

Governing Formula / Principle: Information Hiding: private fields + public getters/setters

Step-by-Step Technical Derivation:

• Encapsulation bundles internal data attributes and the functions manipulating them into a single class unit.

• By declaring instance variables `private` and exposing `public` accessor and mutator methods, internal representation is protected from external corruption.

• This guarantees data integrity and modular class abstraction.

QUESTION 011 [COMPUTER PROGRAMMING & DATA TYPES]

Problem: Which of the following is universally classified as a primitive data type in most OOP languages, used specifically to store a single 16-bit Unicode character?

VERIFIED ANSWER: Char

Governing Formula / Principle: Type Primitive: char (16 bits / 2 bytes, UTF-16 code unit, range '' to ' ')

Step-by-Step Technical Derivation:

• In Java and C#, the `char` primitive type is defined as an unsigned 16-bit numeric type representing a UTF-16 Unicode code point.

• Its values range numerically from 0 to 65,535.

• Unlike reference types (such as `String`), `char` stores raw scalar character data directly in execution stack frames.

QUESTION 012 [SAFETY & OCCUPATIONAL HEALTH MANAGEMENT]

Problem: To synthesize the principles of 'Value-Based Safety' with Occupational Safety and Health Standards in a high-risk semiconductor plant, which management approach must an ECE implement?

VERIFIED ANSWER: A proactive framework embedding continuous hazard identification and mandatory safety training into daily metrics.

Governing Formula / Principle: OSH Integration: Proactive Leading Indicators + Hierarchy of Hazard Controls

Step-by-Step Technical Derivation:

• Reactive compliance relies strictly on lagging indicators (accident counts, worker injury logs).

• Value-Based Safety treats worker safety as an uncompromised core ethical value rather than a variable priority.

• Synthesizing this with DOLE OSH standards requires continuous proactive daily inspections, job hazard analyses, and embedded safety KPIs.

QUESTION 013 [COMPUTER-AIDED DESIGN (CAD)]

Problem: In CAD software, which drafting feature explicitly restricts the movement of the cursor to strictly horizontal and vertical directions relative to the current coordinate system?

VERIFIED ANSWER: Ortho mode

Governing Formula / Principle: Drafting Constraint: F8 Key Toggle; Cursor constrained to $\theta \in \{0°, 90°, 180°, 270°\}$

Step-by-Step Technical Derivation:

• In Autodesk AutoCAD and standard engineering CAD suites, Orthogonal (Ortho) mode constrains pointer vector input to the Cartesian X and Y axes.

• Toggled via function key F8 or status bar command `ORTHOMODE` = 1.

• This guarantees exact perpendicularity and alignment during geometric modeling.

QUESTION 014 [NUCLEAR PHYSICS & RADIOACTIVE DECAY]

Problem: During the radioactive decay of a heavy nucleus, which specific emission process does not alter the mass number of the parent atom?

VERIFIED ANSWER: Gamma decay

Governing Formula / Principle: $^{A}_{Z}X^{*} \rightarrow {}^{A}_{Z}X + \gamma$ ($\Delta A = 0$, $\Delta Z = 0$)

Step-by-Step Technical Derivation:

• Alpha decay emits a helium nucleus ($^{4}_{2}He$), reducing mass number A by 4 and atomic number Z by 2.

• Beta decay converts a nucleon, preserving mass number A ($\Delta A = 0$) but altering atomic number Z by ±1.

• Gamma decay emits a high-energy photon (γ) as an excited nuclear isomer drops to a lower ground state; neither mass number A nor atomic number Z changes.

QUESTION 015 [ENGINEERING ECONOMICS / FINANCIAL RATIOS]

Problem: When analyzing a technology firm's financial health, why is the quick ratio (acid-test ratio) considered a more stringent measure of short-term liquidity than the current ratio?

VERIFIED ANSWER: It excludes inventory from current assets, recognizing that inventory cannot always be rapidly converted into cash.

Governing Formula / Principle: Current Ratio = Current Assets / Current Liabilities; Quick Ratio = (Cash + Marketable Securities + Receivables) / Current Liabilities

Step-by-Step Technical Derivation:

• The Current Ratio includes all current assets, which encompasses physical inventories and pre-paid items.

• In technology and hardware manufacturing, inventories may suffer rapid obsolescence or require deep markdowns to liquidate.

• The Quick (Acid-Test) Ratio subtracts inventory, testing whether immediate cash, short-term marketable instruments, and receivables can settle all short-term liabilities.

QUESTION 016 [ELECTROMAGNETIC THEORY & DIELECTRICS]

Problem: The dielectric constant of an insulating material is a fundamental measure of the material's ability to:

VERIFIED ANSWER: Store electrical potential energy in an electric field

Governing Formula / Principle: $C = \varepsilon_r * \varepsilon_0 * (A / d)$; Energy Stored $U = 0.5 * C * V^2$

Step-by-Step Technical Derivation:

- The relative permittivity (dielectric constant ε_r) indicates how effectively an applied external electric field polarizes internal atomic bound dipoles.
- The resulting bound surface charges create an opposing internal field, allowing more charge to accumulate for a given potential difference.
- Hence, it quantifies the material's electrostatic capacitive energy storage capacity.

QUESTION 017 [ENVIRONMENTAL ENGINEERING & EIA]

Problem: What is the primary analytical goal of conducting an Environmental Impact Assessment for a proposed engineering project?

VERIFIED ANSWER: To evaluate and predict the potential environmental consequences of the project before it commences.

Governing Formula / Principle: EIA Cycle: Screening -> Scoping -> Baseline Assessment -> Impact Prediction -> Mitigation Plan

Step-by-Step Technical Derivation:

- An Environmental Impact Assessment (EIA) is an anticipatory planning tool mandated prior to obtaining an Environmental Compliance Certificate (ECC).
- Its fundamental purpose is to identify, model, and quantify potential bio-physical and socio-economic damages of an infrastructure development beforehand.
- This enables engineering mitigation and alternative site selection prior to construction.

QUESTION 018 [SOFTWARE ENGINEERING & UML]

Problem: In the Unified Modeling Language (UML), which diagram is primarily used to represent the static view of a system by showing its attributes, operations, and relationships?

VERIFIED ANSWER: Class Diagram

Governing Formula / Principle: UML Structure: Class { Name, Attributes, Operations } + Structural Connectors (Association, Inheritance)

Step-by-Step Technical Derivation:

- UML divides diagrams into behavioral diagrams (Use Case, Sequence, State Machine) and structural diagrams.
- The Class Diagram is the foundational structural diagram in object-oriented design.
- It depicts system classes, interfaces, attributes, methods, visibility specifiers, and static coupling relationships.

QUESTION 019 [ELECTROSTATICS / COULOMB'S LAW]

Problem: According to Coulomb's Law, if the distance between two point charges is halved, the electrostatic force between them is:

VERIFIED ANSWER: Quadrupled

Governing Formula / Principle: $F = k_e * (|q1 * q2|) / r^2$; $F_{new} = k_e * (|q1 * q2|) / (r/2)^2 = 4 * F_{orig}$

Step-by-Step Technical Derivation:

- Coulomb's Law dictates that electrostatic interaction follows an inverse-square relationship with distance: $F \propto 1/r^2$.
- If separation distance r is scaled by factor 1/2: $r_{new} = 0.5 * r$.
- Force scales by $(1 / (0.5)^2) = 1 / 0.25 = 4$. The electrostatic force quadruples.

QUESTION 020 [OBJECT-ORIENTED PROGRAMMING (OOP)]

Problem: The ability of different object classes to respond to the exact same method call in their own unique way is known as:

VERIFIED ANSWER: Polymorphism

Governing Formula / Principle: Dynamic Method Dispatch / Late Binding: base_ptr->executeMethod() invokes derived override

Step-by-Step Technical Derivation:

- Polymorphism (meaning 'many forms') allows subclasses to override inherited base class virtual method declarations.
- When invoking a method through an abstract superclass reference at runtime, the JVM/runtime executes the specific derived implementation.
- This decouples high-level interfaces from concrete object implementations.

QUESTION 021 [ENVIRONMENTAL SCIENCE & CLIMATOLOGY]

Problem: Which phenomenon describes the process by which specific atmospheric gases trap outgoing infrared radiation, thereby warming the Earth's surface?

VERIFIED ANSWER: The Greenhouse Effect

Governing Formula / Principle: Radiative Forcing: Earth reradiates in IR (λ ≈ 8-15 μm); absorbed by CO2, CH4, H2O vapor

Step-by-Step Technical Derivation:

- Solar shortwave radiation passes relatively unimpeded through atmospheric gases, heating the terrestrial surface.
- The Earth re-emits thermal energy outward as longer-wavelength terrestrial infrared radiation.
- Greenhouse gas molecules (CO2, H2O, CH4, N2O) absorb and re-emit this IR isotropically, trapping heat in the lower troposphere.

QUESTION 022 [ENGINEERING MECHANICS / CIRCULAR DYNAMICS]

Problem: A 1000 kg car negotiates a flat circular curve of radius 50 meters at a constant speed of 10 m/s. What is the magnitude of the centripetal force required to keep the car on the track?

VERIFIED ANSWER: 2000 N

Governing Formula / Principle: $F_c = (m * v^2) / r$

Step-by-Step Technical Derivation:

- Given: Mass m = 1000 kg; Speed v = 10 m/s; Curve radius r = 50 m.
- Centripetal acceleration $a_c = v^2 / r = (10)^2 / 50 = 100 / 50 = 2.0$ m/s².
- Required centripetal force $F_c = m * a_c = 1000$ kg * 2.0 m/s² = 2000 N (2 kN).

QUESTION 023 [MATERIALS SCIENCE & CRYSTALLOGRAPHY]

Problem: What is the standard theoretical atomic packing factor (APF) for a pure metallic element exhibiting a body-centered cubic (BCC) crystal structure?

VERIFIED ANSWER: 0.68

Governing Formula / Principle: APF = (N_atoms * V_atom) / V_unit_cell; For BCC: 4R = a * √3; APF = π * √3 / 8 ≈ 0.680

Step-by-Step Technical Derivation:

- A BCC unit cell contains net 2 whole atoms: (8 corners * 1/8) + 1 center atom = 2.
- Atoms touch along the cube diagonal: 4R = a * √3 => lattice parameter a = 4R / √3.
- Unit cell volume $V_{cell} = a^3 = (4R / \sqrt{3})^3 = 64R^3 / (3\sqrt{3})$.
- Total atomic volume = 2 * (4/3 * π * R³) = 8πR³ / 3.
- APF = [8πR³ / 3] / [64R³ / (3√3)] = π * √3 / 8 ≈ 0.68017 ≈ 0.68 (68% packing efficiency).

QUESTION 024 [PHYSICS / ELECTROSTATIC UNITS]

Problem: What is the standard SI unit of the electric field?

VERIFIED ANSWER: Newtons per Coulomb

Governing Formula / Principle: E = F / q => Unit: [Force] / [Charge] = N / C (equivalent to Volts per meter, V/m)

Step-by-Step Technical Derivation:

- The electric field intensity E at a point in space is defined as the electrostatic force experienced per unit positive test charge.
- In standard SI base and derived units: Force is measured in Newtons (N), and charge is measured in Coulombs (C).
- Hence, the fundamental unit is Newtons per Coulomb (N/C), which is dimensionally identical to Volts per meter (V/m).

QUESTION 025 [ENVIRONMENTAL SCIENCE & OZONE LAYER]

Problem: What is the primary protective function of the stratospheric ozone layer?

VERIFIED ANSWER: It absorbs and blocks a significant portion of harmful UV radiation from the sun.

Governing Formula / Principle: Chapman Cycle: O3 + hν (UV-B/UV-C, λ = 200-315 nm) -> O2 + O

Step-by-Step Technical Derivation:

- The stratospheric ozone layer (located ~15 to 35 km above Earth) contains elevated concentrations of triatomic oxygen (O3).
- Ozone molecules dissociate photochemically upon absorbing high-energy solar Ultraviolet-B and Ultraviolet-C photons.
- This screens the Earth's surface from destructive ionizing UV that causes cellular DNA mutation and skin cancers.

QUESTION 026 [STRATEGIC MANAGEMENT & VENTURE PLANNING]

Problem: When conducting a macro-environmental analysis for a new tech venture, which framework specifically evaluates the Political, Economic, Social, and Technological factors impacting the market?

VERIFIED ANSWER: PEST Analysis

Governing Formula / Principle: PEST Framework = Macro-environmental scanning across Political, Economic, Social, and Technological axes

Step-by-Step Technical Derivation:

• Venture planning distinguishes internal operational analysis from external macro-environmental scanning.
• PEST Analysis systematically surveys: Political stability and tax policies; Economic interest rates and consumer capital; Social demographics and cultural shifts; and Technological R&D disruptions.
• It identifies external threats and growth vectors prior to market entry.

QUESTION 027 [NUCLEAR ENGINEERING & REACTOR PHYSICS]

Problem: Which component of a nuclear reactor is primarily responsible for slowing down fast neutrons to sustain a controlled fission chain reaction?

VERIFIED ANSWER: Moderator

Governing Formula / Principle: Elastic Neutron Scattering: Fast neutrons (~2 MeV) thermalized to ~0.025 eV

Step-by-Step Technical Derivation:

• Uranium-235 fission releases energetic fast neutrons with kinetic energies averaging 2 MeV.
• Thermal neutrons (energies ~0.025 eV) possess a fission cross-section hundreds of times larger than fast neutrons.
• The moderator (light water, heavy water D2O, or graphite) undergoes elastic collisions with fast neutrons, thermalizing them without capturing them.

QUESTION 028 [ROTATIONAL DYNAMICS & CONSERVATION LAWS]

Problem: If the net external torque acting on a rigid rotating body is zero, which of the following must remain constant?

VERIFIED ANSWER: Angular momentum

Governing Formula / Principle: $\tau_{net} = dL / dt$; If $\tau_{net} = 0$, then $L = I * \omega = \text{Constant}$

Step-by-Step Technical Derivation:

• Newton's second law for rotation states that the net external torque equals the time rate of change of angular momentum.
• When the net torque acting on a mechanical system is zero ($\tau_{net} = 0$), $dL/dt = 0$.
• Therefore, total angular momentum L is conserved and remains constant in both magnitude and direction.

QUESTION 029 [GUI PROGRAMMING & EVENT HANDLING]

Problem: Which fundamental GUI Layout Manager arranges components in five specific geographic regions (North, South, East, West, Center)?

VERIFIED ANSWER: BorderLayout

Governing Formula / Principle: Java Swing/AWT: container.setLayout(new BorderLayout()); add(comp, BorderLayout.REGION)

Step-by-Step Technical Derivation:

• In Java Abstract Window Toolkit (AWT) and Swing, `BorderLayout` is the standard container layout manager.
• It divides space into five distinct topological zones: North, South, East, West, and Center.
• Components placed along edges maintain fixed preferred dimensions while the Center region expands to claim all remaining interior pixels.

QUESTION 030 [SUSTAINABLE DEVELOPMENT & ECOLOGY]

Problem: According to the 1987 Brundtland Report, 'sustainable development' is defined as development that meets the needs of the present without compromising:

VERIFIED ANSWER: The ability of future generations to meet their own needs.

Governing Formula / Principle: UN Brundtland Commission (Our Common Future, 1987) Core Principle

Step-by-Step Technical Derivation:

• The United Nations World Commission on Environment and Development codified the modern definition of sustainability.
• It explicitly mandates inter-generational equity.
• Present economic expansion must not degrade finite planetary resource stocks or ecological balance required by future generations.

QUESTION 031 [ENERGY STORAGE & ELECTROCHEMISTRY]

Problem: Why are lithium-ion batteries generally preferred over traditional lead-acid batteries in modern portable electronic devices?

VERIFIED ANSWER: They possess a significantly higher energy density.

Governing Formula / Principle: Specific Energy: Li-ion ≈ 150-250 Wh/kg vs. Lead-Acid ≈ 30-40 Wh/kg

Step-by-Step Technical Derivation:

• Lithium has the lowest atomic mass (M = 6.94 g/mol) of any solid element and the most negative standard electrochemical reduction potential (-3.04 V).

• This allows Li-ion cells to produce high cell voltages (~3.7 V) with minimal physical weight.

• Consequently, Li-ion delivers 4 to 6 times the gravimetric and volumetric energy density of heavy lead-acid chemistries.

QUESTION 032 [AC CIRCUIT THEORY & RESONANCE]

Problem: Because the reactive components cancel out at resonance in a series RLC circuit, what can be analytically stated about the relationship between the applied voltage and the resulting current?

VERIFIED ANSWER: The voltage and current are exactly in phase.

Governing Formula / Principle: $Z = R + j(X_L - X_C)$; At resonance $\omega_0 = 1/\sqrt{(LC)}$: $X_L = X_C \Rightarrow Z = R$; $\theta = \arctan(0/R) = 0°$

Step-by-Step Technical Derivation:

• Inductive reactance $X_L = \omega L$ leads current by 90°, while capacitive reactance $X_C = 1/(\omega C)$ lags current by 90°.

• At the resonant frequency f_0, $X_L = X_C$, causing net circuit reactance $X_{net} = 0$.

• The total impedance becomes purely resistive ($Z = R$), resulting in a unity power factor ($\cos(0°) = 1$) where voltage and current waveforms are perfectly in phase.

QUESTION 033 [ENGINEERING ECONOMICS / DECISION ANALYSIS]

Problem: Why is sensitivity analysis critical when evaluating an engineering economy project subject to parameter uncertainty?

VERIFIED ANSWER: It allows decision-makers to determine how variations in key estimates affect the project's economic viability.

Governing Formula / Principle: Elasticity / Sensitivity: $S_P = (\Delta NPV / NPV) / (\Delta P / P)$

Step-by-Step Technical Derivation:

• Engineering capital projects rely on estimated variables (initial cost, salvage value, annual revenues, interest rates, useful life).

• Sensitivity analysis varies one uncertain parameter at a time across a percentage range while keeping others constant.

• This identifies the 'critical parameters' that most dramatically alter Net Present Value (NPV) or Internal Rate of Return (IRR).

QUESTION 034 [BUSINESS MODELS & TECHNOPRENEURSHIP]

Problem: Which business model allows users to access basic software features for free while charging a premium for advanced features or virtual goods?

VERIFIED ANSWER: Freemium Model

Governing Formula / Principle: Business Architecture: Free Base Tier (User Acquisition) + Paid Premium Tier (Monetization)

Step-by-Step Technical Derivation:

• The portmanteau 'Freemium' combines 'Free' and 'Premium'.

• A software or SaaS vendor provides proprietary software with zero barrier to entry to rapidly expand active user footprint.

• Advanced functionality, enterprise security, cloud quotas, or specialized analytics are locked behind paid recurring subscriptions.

QUESTION 035 [MECHANICAL VIBRATIONS / HARMONIC MOTION]

Problem: For a mass-spring system undergoing simple harmonic motion, what happens to the period of oscillation if the mass is quadrupled?

VERIFIED ANSWER: It is doubled.

Governing Formula / Principle: $T = 2 * \pi * \sqrt{(m / k)}$; $T_{new} = 2 * \pi * \sqrt{(4m / k)} = 2 * T_{orig}$

Step-by-Step Technical Derivation:

• The fundamental natural period of oscillation for a mass-spring oscillator is given by $T = 2\pi\sqrt{(m/k)}$.

• Substituting $m_{new} = 4*m$: $T_{new} = 2\pi\sqrt{(4m/k)} = 2 * [2\pi\sqrt{(m/k)}]$.

• The square root of 4 is 2; thus, quadrupling the oscillating mass doubles the natural period.

QUESTION 036 [ENVIRONMENTAL BIOLOGY & ECOLOGY]

Problem: In ecology, what term defines the maximum population size of a biological species that a specific environment can sustainably support given its available resources?

VERIFIED ANSWER: Carrying capacity

Governing Formula / Principle: Logistic Growth Model: dN/dt = r * N * (1 - N / K); K = Carrying Capacity

Step-by-Step Technical Derivation:

- In ecological population dynamics, exponential growth (dN/dt = rN) is constrained by environmental resistance factors (food, water, territory, disease).
- The upper horizontal asymptote where birth rates equal death rates in a resource-limited ecosystem is designated K.
- K represents the biological carrying capacity of that habitat.

QUESTION 037 [ENGINEERING ECONOMICS & FINANCE]

Problem: What type of interest calculation applies the interest rate not only to the original principal but also to the accumulated interest of previous periods?

VERIFIED ANSWER: Compound interest

Governing Formula / Principle: F = P * (1 + i)^n; Compound Growth: I_t = i * F_{t-1}

Step-by-Step Technical Derivation:

- Simple interest computes interest strictly on the initial principal P (I = P*i*n).
- Compound interest credits accrued interest back into the principal balance at the close of every compounding period.
- Subsequent interest charges are calculated on the expanding cumulative total, yielding exponential rather than linear financial growth.

QUESTION 038 [POLLUTION MANAGEMENT & ENVIRONMENTAL LAWS]

Problem: In pollution management, an industrial discharge pipe directly releasing chemical waste into a river is analytically classified as what type of pollution source?

VERIFIED ANSWER: Point source

Governing Formula / Principle: Clean Water Act Definition: Discrete identifiable effluent conveyance = Point Source

Step-by-Step Technical Derivation:

- Under the Philippine Clean Water Act (RA 9275) and international environmental law, pollutants originating from a single, stationary, identifiable conduit (pipe, ditch, tunnel, well) are classified as Point Sources.
- Diffused, broad-area runoff (agricultural fields, urban street wash) is classified as Non-point Source pollution.

QUESTION 039 [ENGINEERING MANAGEMENT & CONTROL]

Problem: Which of the following is a primary objective of the controlling function in engineering management?

VERIFIED ANSWER: Tracking project progress against the baseline plan and applying corrective measures

Governing Formula / Principle: Management Control Cycle: Establish Standards -> Measure Performance -> Compare -> Take Corrective Action

Step-by-Step Technical Derivation:

- Controlling is the fourth fundamental managerial function alongside planning, organizing, and leading.
- It measures real-time performance against predetermined project baseline schedules and budgets.
- When variances exceed established engineering tolerances, the manager executes corrective interventions.

QUESTION 040 [ELECTROMAGNETISM & MAGNETIC MATERIALS]

Problem: Which type of magnetic material is characterized by a strong, permanent magnetic moment even in the complete absence of an external magnetic field?

VERIFIED ANSWER: Ferromagnetic

Governing Formula / Principle: Spontaneous Magnetization: Parallel electron spin alignment in Weiss domains; B-H Hysteresis

Step-by-Step Technical Derivation:

- Diamagnetic materials oppose magnetic fields weakly; paramagnetic materials align weakly only while an external field is present.
- Ferromagnetic materials (iron, nickel, cobalt) exhibit strong quantum-mechanical exchange coupling that aligns adjacent electron spins in microscopic magnetic domains.
- Even when the external excitation H drops to zero, a large remanent magnetization B_r persists.

QUESTION 041 [ENVIRONMENTAL SCIENCE & CARBON METRICS]

Problem: The total amount of greenhouse gases generated by our actions, commonly measured in equivalent tons of carbon dioxide, is referred to as a:

VERIFIED ANSWER: Carbon footprint

Governing Formula / Principle: Carbon Footprint = Σ (Activity Data * Emission Factor) in tCO2e

Step-by-Step Technical Derivation:

- A carbon footprint quantifies total direct (Scope 1) and indirect (Scope 2 and 3) greenhouse gas emissions associated with an individual, product, or enterprise.
- Because gases possess differing Global Warming Potentials (GWPs), non-CO2 gases (methane, nitrous oxide) are converted into equivalent metric tons of CO2 (tCO2e).

QUESTION 042 [ENGINEERING ECONOMICS & FINANCIAL ANALYSIS]

Problem: In engineering economics, what specific term refers to the potential financial benefit that is forfeited when one project alternative is chosen over another?

VERIFIED ANSWER: Opportunity cost

Governing Formula / Principle: Opportunity Cost = Return of Most Lucrative Foregone Option - Return of Chosen Option

Step-by-Step Technical Derivation:

• Capital resources in engineering firms are strictly finite, requiring capital budgeting rationing.

• When committing capital to Project A, the potential revenue or utility that would have been yielded by the second-best alternative (Project B) is forfeited.

• This foregone return is defined as the opportunity cost.

QUESTION 043 [MATERIALS SCIENCE & POLYMERS]

Problem: Why are elastomeric polymers capable of undergoing massive elastic deformations without suffering permanent structural damage?

VERIFIED ANSWER: Their highly kinked, long polymer chains uncoil under stress and return to shape due to sparse cross-linking.

Governing Formula / Principle: Entropic Elasticity: Thermodynamic conformational entropy driving chain recovery

Step-by-Step Technical Derivation:

• Elastomers possess long, amorphous polymer backbones that naturally exist in highly tangled, randomized coiled conformations.

• Applying mechanical tension straightens and aligns these convoluted chains along the stress axis.

• Covalent chemical cross-links sparsely distributed along the chains prevent individual macromolecules from sliding permanently past each other, pulling the network back when released.

QUESTION 044 [OPERATIONS & SUPPLY CHAIN MANAGEMENT]

Problem: Which operational management strategy focuses on minimizing inventory holding costs by strictly scheduling materials to arrive exactly when they are needed in the production cycle?

VERIFIED ANSWER: Just-In-Time (JIT) Inventory

Governing Formula / Principle: Lean Production / Toyota Production System (TPS): Pull-system driven by Kanban replenishment

Step-by-Step Technical Derivation:

• Traditional manufacturing relies on 'just-in-case' buffer inventories that tie up substantial working capital and introduce warehousing overhead.

• Just-In-Time (JIT) employs a pull-system where sub-assemblies and raw materials are delivered precisely as required on the assembly floor.

• This minimizes inventory holding costs and exposes production inefficiencies.

QUESTION 045 [RA 9292 PENAL PROVISIONS]

Problem: According to the penal provisions of RA 9292, allowing a non-licensed individual to use your Professional Identification Card or seal to secure an engineering contract is punishable by:

VERIFIED ANSWER: A fine, imprisonment, or both, at the discretion of the court.

Governing Formula / Principle: RA 9292 Section 35: Penal Provisions (Fine ₱50k-₱1M and/or 6 months to 6 years imprisonment)

Step-by-Step Technical Derivation:

• Section 35, Article V of RA 9292 outlines criminal liabilities for illegal electronics practice and fraudulent seal usage.

• Surrendering a professional seal, dry stamp, or license ID to an unregistered entity constitutes a statutory violation.

• The law mandates criminal prosecution resulting in a monetary fine, imprisonment, or both, alongside administrative license revocation.

QUESTION 046 [CLASSICAL MANAGEMENT PRINCIPLES]

Problem: According to Henri Fayol's 14 Principles of Management, what does the principle of 'Unity of Command' dictate?

VERIFIED ANSWER: Every employee should receive orders from and report to only one direct superior.

Governing Formula / Principle: Fayol's Principle: Single Reporting Hierarchy -> Minimizes Role Conflict & Confusion

Step-by-Step Technical Derivation:

• Henri Fayol formulated 14 general management principles for administrative efficiency.

• The Unity of Command principle states that an individual worker should be directly answerable to only one formal supervisor.

• Dual subordination creates conflicting directives, jurisdictional disputes, and compromised organizational accountability.

QUESTION 047 [THERMODYNAMICS & HEAT TRANSFER]

Problem: Which mode of heat transfer is characterized by the emission of electromagnetic waves and does not require a physical medium to propagate?

VERIFIED ANSWER: Radiation

Governing Formula / Principle: Stefan-Boltzmann Law: $q = \varepsilon * \sigma * A * (T_{hot}^4 - T_{cold}^4)$; $\sigma = 5.670374 \times 10^{-8}$ $W/(m^2 \cdot K^4)$

Step-by-Step Technical Derivation:

- Conduction requires physical microscopic atomic contact; convection requires bulk macroscopic fluid transport.
- Thermal radiation transfers energy through photons spanning infrared to visible electromagnetic bands.
- Because electromagnetic waves propagate through vacuum at the speed of light, radiative heat transfer requires no material medium.

QUESTION 048 [ENERGY RESOURCES & SUSTAINABILITY]

Problem: Which of the following energy sources is strictly classified as renewable because it is not depleted upon use and naturally replenishes?

VERIFIED ANSWER: Wind energy

Governing Formula / Principle: Renewable Power: $P_{wind} = 0.5 * \rho * A * v^3$; Inexhaustible atmospheric kinetic flux

Step-by-Step Technical Derivation:

- Fossil fuels (coal, oil, natural gas) and fissile uranium are non-renewable because their finite geologic reserves are depleted upon extraction.
- Wind energy harnesses planetary atmospheric air currents generated by solar heating and Earth rotation.
- Because atmospheric kinetic flux regenerates continuously without fuel consumption, it is strictly renewable.

QUESTION 049 [ORGANIZATIONAL DESIGN & MANAGEMENT]

Problem: In organizational structure design, what specific management term refers to the number of subordinates or direct reports a single manager can effectively supervise?

VERIFIED ANSWER: Span of control

Governing Formula / Principle: Organizational Span: N_direct_reports = Total Headcount / Number of Supervisors

Step-by-Step Technical Derivation:

- Span of control (or span of management) dictates the breadth of supervisory authority assigned to an engineering manager.
- A narrow span creates tall organizational hierarchies with many bureaucratic layers.
- A wide span creates flat organizational charts, fostering rapid communication but requiring highly autonomous engineers.

QUESTION 050 [ENGINEERING ECONOMICS / CAPITAL BUDGETING]

Problem: When evaluating two mutually exclusive engineering equipment alternatives with unequal useful lives, which evaluation method is typically preferred as it implicitly assumes reinvestment at the MARR over a common multiple of lives?

VERIFIED ANSWER: Annual Worth Method

Governing Formula / Principle: AW = PW * (A/P, i, n) = -Initial Cost * (A/P, i, n) + Annual Benefit + Salvage * (A/F, i, n)

Step-by-Step Technical Derivation:

- Comparing unequal-life alternatives via Net Present Value (NPV) requires finding the Least Common Multiple (LCM) of their useful lives, which can be computationally tedious.
- The Annual Worth (AW) method converts cash flows into an equivalent uniform annual series over one life cycle.
- Because the AW over one cycle equals the AW across repeated life cycles, it allows direct, unbiased comparison without LCM expansion.

QUESTION 051 [OPTICS & ELECTROMAGNETIC PROPAGATION]

Problem: The critical angle for total internal reflection can only be achieved when light travels from:

VERIFIED ANSWER: A denser medium to a less dense medium

Governing Formula / Principle: Snell's Law: $n1 * \sin(\theta_c) = n2 * \sin(90°) \Rightarrow \sin(\theta_c) = n2 / n1$ (Requires $n1 > n2$)

Step-by-Step Technical Derivation:

- Total internal reflection (TIR) occurs when the refracted ray bends along the interface ($\theta_r = 90°$), beyond which all energy reflects internally.
- By Snell's law, $\sin(\theta_c) = n2 / n1$. Since sine values cannot exceed 1.0, the quotient $n2/n1$ must be ≤ 1.
- Therefore, light must originate in a medium of higher refractive index (optically denser) and strike an interface with a lower index (less dense).

QUESTION 052 [SOLID STATE ELECTRONICS / SEMICONDUCTORS]

Problem: What is the typical range of the energy band gap for elemental semiconductors like silicon and germanium at room temperature?

VERIFIED ANSWER: 0.5-1.5 eV

Governing Formula / Principle: E_g(Ge, 300K) = 0.66 eV; E_g(Si, 300K) = 1.12 eV

Step-by-Step Technical Derivation:

• In solid-state physics, the bandgap E_g separates the valence band from the conduction band.
• Insulators have wide bandgaps (E_g > 3 to 5 eV), while metals have overlapping bands (E_g = 0 eV).
• Semiconductors have moderate gaps: Germanium is ~0.66 eV and Silicon is ~1.12 eV, squarely placing standard elemental semiconductors within 0.5 to 1.5 eV.

QUESTION 053 [MATERIALS SCIENCE & POLYMERS]

Problem: Which classification refers to a polymer that can be repeatedly melted and reformed upon heating without undergoing permanent chemical degradation?

VERIFIED ANSWER: Thermoplastic polymer

Governing Formula / Principle: Linear / Branched Macromolecular Architecture: Secondary van der Waals bonding between chains

Step-by-Step Technical Derivation:

• Thermoplastics consist of linear or branched polymer backbones held together by weak secondary bonds (van der Waals forces).
• Heating provides thermal energy to overcome these intermolecular bonds, allowing chains to flow as a viscous liquid.
• Unlike thermosets that crosslink permanently during initial curing, thermoplastics re-solidify upon cooling and can be remelted repeatedly.

QUESTION 054 [CIRCUIT THEORY / CAPACITOR NETWORKS]

Problem: When three identical capacitors, each with capacitance C, are connected in parallel across a voltage source, what is the equivalent total capacitance of the circuit?

VERIFIED ANSWER: 3C

Governing Formula / Principle: C_total = C1 + C2 + C3; For identical units: C_total = 3 * C

Step-by-Step Technical Derivation:

• When capacitors are placed in parallel, each capacitor experiences the exact same terminal voltage V.
• Total stored charge Q_total = Q1 + Q2 + Q3 = C*V + C*V + C*V = 3*C*V.
• By definition of equivalent capacitance C_eq = Q_total / V = 3C.

QUESTION 055 [OBJECT-ORIENTED PROGRAMMING (OOP)]

Problem: In Object-Oriented paradigm, what term describes the mechanism where a new class derives properties and behaviors from an existing parent class to promote code reusability?

VERIFIED ANSWER: Inheritance

Governing Formula / Principle: Subclassing: class Derived extends Base { ... }; 'is-a' relationship

Step-by-Step Technical Derivation:

• Inheritance allows a child class (subclass) to inherit fields, methods, and interface definitions from a superclass.
• The derived class can reuse existing parent code while introducing specialized attributes or overriding existing method behaviors.
• This eliminates duplicate code and establishes taxonomic object hierarchies.

QUESTION 056 [LEADERSHIP & ORGANIZATIONAL BEHAVIOR]

Problem: What specific leadership style is characterized by a manager who retains absolute decision-making authority and dictates tasks without soliciting input from subordinates?

VERIFIED ANSWER: Autocratic leadership

Governing Formula / Principle: Classical Leadership Taxonomy: Autocratic (Authoritarian) vs. Democratic vs. Laissez-faire

Step-by-Step Technical Derivation:

• Autocratic (authoritarian) leadership centralizes all decision-making authority within the manager.
• The leader dictates policies, assignment schedules, and operational steps without seeking employee feedback.
• While effective in high-stress tactical emergencies, it limits team innovation and autonomy in engineering environments.

QUESTION 057 [GEOMETRICAL OPTICS & REFRACTION]

Problem: When a light ray transitions from a medium with a lower refractive index to one with a higher refractive index, how does the ray bend relative to the normal line?

VERIFIED ANSWER: It bends towards the normal.

Governing Formula / Principle: n1 * sin(θ1) = n2 * sin(θ2); If n2 > n1, then sin(θ2) < sin(θ1) => θ2 < θ1

Step-by-Step Technical Derivation:

• Snell's Law dictates that n1*sin(θ1) = n2*sin(θ2), where θ is the angle relative to the perpendicular normal line.
• When entering an optically denser medium (n2 > n1), light slows down (v2 < v1).
• To satisfy the equality, sin(θ2) must decrease relative to sin(θ1), which means angle θ2 shrinks, causing the refracted beam to bend inward toward the normal.

QUESTION 058 [ACCOUNTING & FINANCIAL STATEMENTS]

Problem: How does depreciation structurally impact a company's cash flow statement?

VERIFIED ANSWER: It is a non-cash expense added back to net income in the operating activities section.

Governing Formula / Principle: Cash Flow from Operations = Net Income + Non-Cash Expenses (Depreciation) - ΔWorking Capital

Step-by-Step Technical Derivation:

• Depreciation allocates the capital cost of a physical asset across its useful accounting life, reducing reported Net Income on the Income Statement.
• However, depreciation does not involve an actual physical outflow of liquid currency during that accounting cycle.
• In the indirect Statement of Cash Flows, depreciation is added back to Net Income under Cash Flows from Operating Activities.

QUESTION 059 [ECOLOGICAL SYSTEMS & FOOD WEBS]

Problem: What is the primary role of autotrophs (producers) within an ecosystem's food web?

VERIFIED ANSWER: To synthesize their own food through processes.

Governing Formula / Principle: Primary Productivity: 6 CO2 + 6 H2O + hv -> C6H12O6 + 6 O2 (Photosynthesis)

Step-by-Step Technical Derivation:

• Autotrophs (green plants, algae, cyanobacteria) occupy the first trophic level in all ecosystems.
• They utilize solar energy (photosynthesis) or chemical oxidation (chemosynthesis) to convert inorganic carbon dioxide and water into energy-rich organic carbohydrates.
• They serve as the foundational energetic base supporting all heterotrophic consumers.

QUESTION 060 [MATERIALS SCIENCE / CRYSTAL DEFECTS]

Problem: How does the introduction of an interstitial impurity generally affect the mechanical properties of a crystalline metal lattice?

VERIFIED ANSWER: It creates lattice strains that restrict dislocation movement, increasing strength.

Governing Formula / Principle: Solid Solution Strengthening: Δτ = G * b * ε^(3/2) * √c

Step-by-Step Technical Derivation:

• Interstitial solute atoms (such as carbon in iron) lodge into void spaces between larger solvent lattice host atoms.
• This creates local compressive stress fields that distort surrounding lattice planes.
• These strain fields pin and obstruct dislocation motion, raising the required yield stress and increasing overall mechanical strength.

QUESTION 061 [CHEMISTRY & CHEMICAL BONDING]

Problem: Which primary interatomic bond is formed by the sharing of valence electron pairs between adjacent electronegative nonmetal atoms?

VERIFIED ANSWER: Covalent bond

Governing Formula / Principle: Molecular Orbital Overlap: Shared electron pairs achieve noble gas valence octets

Step-by-Step Technical Derivation:

• Ionic bonding transfers electrons between metals and nonmetals; metallic bonding involves delocalized electron gas.
• When two atoms with similar, relatively high electronegativities interact, electron transfer is energetically unfavorable.
• They overlap outer valence orbitals to mutually share electron pairs, forming strong directional covalent bonds.

QUESTION 062 [STRATEGIC PLANNING & ANALYSIS]

Problem: In the strategic planning process, which analytical framework is utilized specifically to evaluate a company's internal strengths and weaknesses alongside its external opportunities and threats?

VERIFIED ANSWER: SWOT Analysis

Governing Formula / Principle: 2x2 Matrix: Internal [Strengths, Weaknesses] × External [Opportunities, Threats]

Step-by-Step Technical Derivation:

• SWOT Analysis provides a structured diagnostic of organizational positioning.
• Internal audit identifies operational capabilities (Strengths) and structural limitations (Weaknesses).
• External environmental scanning identifies market vectors (Opportunities) and competitive risks (Threats).

QUESTION 063 [CLASSICAL MANAGEMENT THEORIES]

Problem: Which classical management theory, pioneered by Frederick W. Taylor, emphasizes the objective analysis of workflows and the optimization of labor productivity through time-and-motion studies?

VERIFIED ANSWER: Scientific Management Theory

Governing Formula / Principle: Taylorism Principles: Standardized tasks + Time-and-motion optimization + Incentive wages

Step-by-Step Technical Derivation:

- Frederick Winslow Taylor (1911) pioneered the scientific study of industrial labor processes.
- He replaced subjective rule-of-thumb craft methods with systematic empirical time-and-motion analyses.
- This structured approach to standardized work routines is historically designated Scientific Management (Taylorism).

QUESTION 064 [ELECTRONIC WASTE MANAGEMENT & TOXICOLOGY]

Problem: In the context of electronic waste management, why is the improper disposal of printed circuit boards in standard landfills a major environmental hazard?

VERIFIED ANSWER: They release heavy metals like lead and mercury.

Governing Formula / Principle: Leachate Migration: Pb, Hg, Cd, and brominated flame retardants bioaccumulating in aquifers

Step-by-Step Technical Derivation:

- Printed circuit boards (PCBs) contain heavy metals including lead (solder alloys), mercury (switches/relays), and cadmium.
- When exposed to acidic rainwater in unlined landfills, these metals leach into soil and percolate into groundwater tables.
- These bioaccumulative neurotoxins pose severe long-term hazards to aquatic life and municipal drinking water.

QUESTION 065 [AC CIRCUITS / RESONANCE]

Problem: In a series RLC AC circuit at the exact frequency of resonance, the net reactive component of the impedance is:

VERIFIED ANSWER: Zero

Governing Formula / Principle: $X_{net} = X_L - X_C = 0 \Rightarrow \text{Im}(Z) = 0$

Step-by-Step Technical Derivation:

- Total series impedance is given by $Z = R + j(X_L - X_C)$.
- At resonant frequency f_0, inductive reactance $X_L = 2\pi fL$ exactly matches capacitive reactance $X_C = 1/(2\pi fC)$.
- The imaginary reactive component cancels completely: $X_{net} = X_L - X_C = 0\ \Omega$, leaving a purely resistive impedance $Z = R$.

QUESTION 066 [ELECTROCHEMISTRY & CORROSION]

Problem: In an electrochemical cell, what specific process occurs at the anode during operation?

VERIFIED ANSWER: Oxidation

Governing Formula / Principle: Mnemonic 'An Ox, Red Cat': Anode = Oxidation (loss of e^-); Cathode = Reduction (gain of e^-)

Step-by-Step Technical Derivation:

- By universal electrochemical convention, the electrode where oxidation (loss of electrons) occurs is the anode.
- The electrode where reduction (gain of electrons) occurs is the cathode.
- At the anode, metallic atoms release valence electrons: $M \rightarrow M^{n+} + n\ e^-$.

QUESTION 067 [RA 9292 BOARD STRUCTURE]

Problem: The Professional Regulatory Board of Electronics Engineering is composed of one Chairman and how many direct members?

VERIFIED ANSWER: Two

Governing Formula / Principle: RA 9292 Section 6: Composition of the Board (One Chairman + Two Members = 3 total)

Step-by-Step Technical Derivation:

- Section 6, Article II of Republic Act No. 9292 establishes the statutory governance structure for the regulatory body.
- It explicitly mandates that the Board of Electronics Engineering consists of a Chairman and two (2) Members.
- All are appointed by the President of the Philippines from nominees submitted by the Accredited Professional Organization (IECEP).

QUESTION 068 [RA 9292 SCOPE OF PRACTICE]

Problem: Under the scope of practice defined in RA 9292, who holds the sole legal authority to provide consulting services and to sign and seal electronics plans, specifications, and reports?

VERIFIED ANSWER: A duly registered Professional Electronics Engineer (PECE)

Governing Formula / Principle: RA 9292 Section 5: Scope of Practice and Authority to Sign/Seal Plans

Step-by-Step Technical Derivation:

• RA 9292 establishes three levels of engineering credentials: ECE (registered engineer), ECT (technician), and PECE (professional).

• Section 5 and Section 27 explicitly reserve consulting services and the legal authority to sign and seal official electronics engineering plans exclusively to holders of a valid Professional Electronics Engineer (PECE) license.

• Registered ECEs cannot legally seal design submissions without PECE upgrade.

QUESTION 069 [PHILIPPINE ELECTRICAL CODE (PEC) / GROUNDING]

Problem: Based on the Philippine Electrical Code (PEC), when routing equipment grounding conductors in a server farm, why is it a severe safety hazard to route them completely outside the metallic raceway that contains the main circuit conductors?

VERIFIED ANSWER: It increases the overall impedance of the fault return path, which can prevent the overcurrent device from tripping.

Governing Formula / Principle: Loop Inductance $Z_{fault} \propto \ln(D/r)$; Magnetic coupling cancels when phase and ground are in same conduit

Step-by-Step Technical Derivation:

• When AC fault current flows, high physical separation between the phase conductor and the equipment grounding conductor (EGC) creates a large inductive loop area.

• This large loop dramatically raises the AC inductive reactance ($X_L = 2\pi fL$) of the ground fault return path.

• The elevated impedance limits fault current below the instantaneous trip threshold of circuit breakers, allowing hazardous ground fault voltages to persist on equipment frames.

QUESTION 070 [CLASSICAL MECHANICS / WORK-ENERGY]

Problem: According to the Work-Energy Theorem, the net work done on an object is mathematically equal to its change in what specific quantity?

VERIFIED ANSWER: Kinetic energy

Governing Formula / Principle: $W_{net} = \Delta K = 0.5 * m * (v_{final}^2 - v_{initial}^2)$

Step-by-Step Technical Derivation:

• Newton's second law: $F_{net} = m * (dv/dt)$.

• Integrating net force over displacement ds: $W_{net} = \int F_{net}\, ds = \int m * v * dv$.

• Evaluating the definite integral yields $W_{net} = 0.5*m*v_f^2 - 0.5*m*v_i^2 = \Delta K$ (change in kinetic energy).

QUESTION 071 [THERMODYNAMICS / ADIABATIC PROCESSES]

Problem: If an ideal gas undergoes a rapid adiabatic expansion, what happens to its internal temperature?

VERIFIED ANSWER: It decreases.

Governing Formula / Principle: First Law: $Q = \Delta U + W$; For adiabatic process $Q = 0 \Rightarrow \Delta U = -W$; $T * V^{(\gamma - 1)} = \text{Constant}$

Step-by-Step Technical Derivation:

• In an adiabatic process, heat transfer between the system and its surroundings is zero ($Q = 0$).

• When the gas expands rapidly, it performs positive thermodynamic boundary work on the surroundings ($W > 0$).

• By the first law of thermodynamics, internal energy drops: $\Delta U = n*C_v*\Delta T = -W < 0$. Therefore, temperature drops.

QUESTION 072 [EVENT-DRIVEN GUI PROGRAMMING]

Problem: In GUI programming event handling, what is the primary role of a 'Listener' object?

VERIFIED ANSWER: To wait for and respond to specific user actions.

Governing Formula / Principle: Observer Pattern: Subject generates ActionEvent -> Listener.actionPerformed(e)

Step-by-Step Technical Derivation:

• GUI graphical toolkits rely on asynchronous event-driven architectures (Observer design pattern).

• Components (buttons, text fields) fire event objects when manipulated by the user.

• A registered Listener object implements designated callback interfaces to wait for, intercept, and process these user actions.

QUESTION 073 [PROJECTILE MOTION / KINEMATICS]

Problem: A projectile is launched at an angle of 45 degrees relative to the horizontal. At the highest point of its parabolic trajectory, what is its vertical velocity?

VERIFIED ANSWER: Zero

Governing Formula / Principle: $v_y(t) = v_0 * \sin(\theta) - g * t$; Peak apex condition: $v_y = 0$

Step-by-Step Technical Derivation:

- In planar projectile motion under uniform gravity, vertical velocity obeys $v_y = v_{0y} - g*t$.
- As the projectile ascends, downward gravitational acceleration decelerates vertical velocity until it momentarily ceases moving upward.
- At the precise apex of the parabolic path, vertical velocity v_y reaches exactly zero before reversing downward (while horizontal velocity remains $v_0*\cos(45°)$).

QUESTION 074 [NANOTECHNOLOGY & SURFACE PHENOMENA]

Problem: Why do nanoparticles often exhibit significantly different chemical and physical properties compared to their bulk material counterparts?

VERIFIED ANSWER: They have a substantially higher surface area-to-volume ratio.

Governing Formula / Principle: Specific Surface Area SSA $\propto (6 / d)$; As $d \to 0$, fraction of surface atoms approaches unity

Step-by-Step Technical Derivation:

- As particle diameter shrinks to the nanoscale (1 to 100 nm), the surface area per unit volume scales inversely with diameter.
- A massive percentage of total constituent atoms reside directly at surface interfaces with unsaturated valence bonds.
- This makes nanoparticles vastly more chemically reactive, altering melting points, electrical band structures, and optical plasmon resonances.

QUESTION 075 [RA 7925 (PUBLIC TELECOM POLICY ACT)]

Problem: Under Republic Act No. 7925 (Public Telecommunications Policy Act), an established legacy carrier is legally mandated to provide fair and compulsory interconnection to new authorized public telecommunications entities. What is the primary purpose of this mandate?

VERIFIED ANSWER: To ensure a universally accessible and fully integrated national telecommunications network.

Governing Formula / Principle: RA 7925 Section 18: Mandatory Interconnection and Universal Network Access

Step-by-Step Technical Derivation:

- Prior to telecom deregulation, dominant monopolistic carriers refused to connect competitor subscribers to their public switched telephone network (PSTN).
- Section 18 of RA 7925 establishes mandatory, non-discriminatory, compensatory interconnection among all licensed telcos.
- This guarantees that any end-user on any network can seamlessly communicate with any other user across the Philippines.

QUESTION 076 [INTELLECTUAL PROPERTY / PATENTS]

Problem: What type of intellectual property protection is most appropriate for safeguarding a newly invented, highly unique electronic circuit architecture from being copied by competitors?

VERIFIED ANSWER: Patent

Governing Formula / Principle: Intellectual Property Code (RA 8293): Patentable Criteria = Novelty + Inventive Step + Industrial Applicability

Step-by-Step Technical Derivation:

- Copyright protects creative artistic expressions, code text, and publications.
- Trademarks protect commercial logos and brand names; trade secrets protect undisclosed manufacturing formulas.
- To legally exclude competitors from making, using, selling, or importing a novel, non-obvious technical apparatus or electronic circuit topology, an inventor must secure a Patent.

QUESTION 077 [DECISION ANALYSIS & RISK MANAGEMENT]

Problem: In decision analysis under risk, how is the Expected Monetary Value (EMV) of a specific decision alternative calculated?

VERIFIED ANSWER: By summing the products of each possible payoff and its respective probability of occurrence.

Governing Formula / Principle: $EMV = \Sigma (P_i * X_i) = P1*X1 + P2*X2 + \ldots + Pn*Xn$

Step-by-Step Technical Derivation:

- Decision making under risk involves known probabilistic states of nature.
- For each decision branch, multiply the financial outcome (payoff X_i) by its assigned probability of occurrence P_i.
- Summing these weighted terms across all mutually exclusive outcomes yields the Expected Monetary Value (EMV).

QUESTION 078 [ENGINEERING ECONOMICS / INTEREST FACTORS]

Problem: Which economic equivalence factor is used to calculate the equivalent present value of a uniform series of equal end-of-period payments?

VERIFIED ANSWER: Uniform series present worth factor

Governing Formula / Principle: (P/A, i, n) = [(1 + i)^n - 1] / [i * (1 + i)^n]

Step-by-Step Technical Derivation:

- In engineering finance, an annuity consists of equal periodic payments A over n periods at interest rate i.
- To find the equivalent lump-sum present worth P at time t = 0: P = A * (P/A, i, n).
- The multiplier [(1+i)^n - 1] / [i(1+i)^n] is standardized as the Uniform Series Present Worth Factor.

QUESTION 079 [THERMODYNAMICS / CARNOT ENGINES]

Problem: An experimental heat engine operates between a high-temperature reservoir at 400 K and a low-temperature sink at 300 K. What is the maximum theoretical (Carnot) thermodynamic efficiency of this engine?

VERIFIED ANSWER: 25%

Governing Formula / Principle: η_Carnot = 1 - (T_cold / T_hot)

Step-by-Step Technical Derivation:

- High temperature reservoir T_hot = 400 K; Low temperature sink T_cold = 300 K.
- Carnot maximum theoretical efficiency: η = 1 - (300 / 400) = 1 - 0.75 = 0.25.
- Convert to percentage: 0.25 * 100% = 25%.

QUESTION 080 [ELECTROMAGNETISM / RIGHT-HAND RULE]

Problem: Which rule is utilized to determine the direction of the magnetic field generated by a straight current-carrying wire?

VERIFIED ANSWER: Right-Hand Grip Rule

Governing Formula / Principle: Biot-Savart Law: B = (μ_0 * I) / (2 * π * r); Thumb aligns with I -> Curled fingers indicate B vector

Step-by-Step Technical Derivation:

- Ampere's Circuital Law dictates that electric currents generate concentric cylindrical magnetic field lines.
- Using the Right-Hand Grip Rule: Orient the extended thumb of the right hand along the direction of conventional current I.
- The natural curl of the four fingers indicates the circular direction of the resulting magnetic flux B.

QUESTION 081 [ENVIRONMENTAL CHEMISTRY & OZONE DEPLETION]

Problem: Which chemical compound is historically known as the primary catalyst for the depletion of the stratospheric ozone layer?

VERIFIED ANSWER: Chlorofluorocarbons

Governing Formula / Principle: Catalytic Chlorine Cycle: CFCl3 + UV -> Cl• + CFCl2; Cl• + O3 -> ClO• + O2; ClO• + O -> Cl• + O2

Step-by-Step Technical Derivation:

- Chlorofluorocarbons (CFCs, such as Freon-11 and Freon-12) are stable synthetic compounds that diffuse into the stratosphere.
- Solar UV photolyzes CFCs, releasing free chlorine radicals (Cl•).
- A single chlorine atom catalytically destroys over 100,000 ozone molecules before diffusing out, prompting the Montreal Protocol phaseout.

QUESTION 082 [ECE CODE OF ETHICS / PUBLIC WELFARE]

Problem: Applying the Code of Ethics for Electronics Practitioners, if an engineer discovers their employer is knowingly producing hazardous devices, what is their ultimate ethical obligation regarding public welfare?

VERIFIED ANSWER: Report the hazard to proper authorities and refuse to associate with the unsafe enterprise.

Governing Formula / Principle: IECEP Code of Ethics Canon 1: Public Safety, Health, and Welfare is Paramount

Step-by-Step Technical Derivation:

- The professional Code of Ethics mandates that paramount consideration must always be given to the safety, health, and welfare of the public.
- Client and employer confidentiality clauses are legally and ethically subordinate to public life safety.
- The engineer must formally notify corporate management; if unaddressed, they must report to regulatory authorities and disassociate from the hazardous project.

QUESTION 083 [STRENGTH OF MATERIALS / HOOKE'S LAW]

Problem: In the study of elasticity, Young's Modulus is defined as the ratio of tensile stress to which corresponding quantity?

VERIFIED ANSWER: Tensile strain

Governing Formula / Principle: E = σ / ε; Stress σ = F / A; Strain ε = ΔL / L_0

Step-by-Step Technical Derivation:

• Hooke's Law in the linear elastic regime states that mechanical deformation is directly proportional to applied load.

• Tensile stress σ is defined as applied force per unit cross-sectional area (F/A).

• Tensile strain ε is the fractional elongation (ΔL/L_0). The ratio of stress to strain within the proportional limit is Young's Modulus of Elasticity (E).

QUESTION 084 [STRATEGIC MANAGEMENT & INNOVATION]

Problem: An emerging tech startup decides to avoid direct competition in highly saturated markets by creating an entirely new, uncontested market space through fundamental innovation. What is this strategic approach called?

VERIFIED ANSWER: Blue Ocean Strategy

Governing Formula / Principle: Chan Kim & Mauborgne Strategy: Blue Ocean (Value Innovation) vs. Red Ocean (Cutthroat Competition)

Step-by-Step Technical Derivation:

• A 'Red Ocean' represents an existing market space crowded with competitors competing for market share through price erosion.

• A 'Blue Ocean' strategy deliberately renders competition irrelevant by discovering or inventing brand new market space.

• It drives 'value innovation' that lowers costs while substantially elevating utility for untapped buyers.

QUESTION 085 [ELECTRICAL SAFETY / HAZARDOUS LOCATIONS]

Problem: An engineering team designs instrumentation for a petrochemical refinery classified as a Class I, Division 1 hazardous location. To synthesize PEC safety mandates with operational needs, which design strategy is required?

VERIFIED ANSWER: Engineer an intrinsically safe system where operating energy levels are strictly below the explosive ignition threshold.

Governing Formula / Principle: Intrinsic Safety (Ex i): Stored electrical/thermal energy < Minimum Ignition Energy (MIE) of atmosphere

Step-by-Step Technical Derivation:

• PEC and NEC Article 500 classify Class I, Division 1 as environments where flammable gases, vapors, or liquids exist continuously or during normal operations.

• Traditional enclosures rely on heavy explosion-proof housings to contain internal blasts.

• Intrinsic Safety (IS) uses zener barriers and energy-limiting circuits to ensure electrical energy and surface temperatures remain strictly below the minimum ignition threshold under both normal and fault conditions.

QUESTION 086 [MATERIALS SCIENCE / CRYSTALLINE DEFECTS]

Problem: Which one-dimensional crystalline defect is characterized by the insertion of an extra half-plane of atoms into the crystal lattice?

VERIFIED ANSWER: Edge dislocation

Governing Formula / Principle: Dislocation Line Vector perpendicular to Burgers Vector: b ⊥ l

Step-by-Step Technical Derivation:

• Point defects are zero-dimensional (vacancies, self-interstitials, substitutional impurities).

• An edge dislocation is a linear (one-dimensional) line defect formed by terminating an extra half-plane of atoms inside the crystal lattice.

• The lattice above the slip plane is under compression, while the region below is under tension, with the Burgers vector b perpendicular to the dislocation line.

QUESTION 087 [ENVIRONMENTAL SCIENCE & MARINE CHEMISTRY]

Problem: A major environmental consequence of increased carbon dioxide absorption by the Earth's oceans is a decrease in pH levels, a condition known strictly as:

VERIFIED ANSWER: Ocean acidification

Governing Formula / Principle: CO2(aq) + H2O <-> H2CO3 <-> H+ + HCO3-; Elevated [H+] lowers ocean pH

Step-by-Step Technical Derivation:

• Approximately 30% of anthropogenic atmospheric CO2 is absorbed into marine surface waters.

• Dissolved CO2 reacts with water to form carbonic acid (H2CO3), which dissociates to release free hydrogen ions (H+).

• This elevates ocean hydrogen ion concentration, driving down oceanic pH in a chemical shift termed Ocean Acidification.

QUESTION 088 [SUSTAINABILITY & CORPORATE ACCOUNTING]

Problem: The 'Triple Bottom Line' accounting framework for sustainable businesses evaluates performance based on which three critical dimensions?

VERIFIED ANSWER: Environmental, Social, and Financial impact

Governing Formula / Principle: TBL Dimensions: People (Social) + Planet (Environmental) + Profit (Economic / Financial)

Step-by-Step Technical Derivation:

• John Elkington (1994) formulated the Triple Bottom Line (TBL) to expand corporate auditing beyond traditional short-term net profits.

• It measures organizational performance across three holistic pillars: Profit (financial capital and economic returns), People (social equity and labor fair practices), and Planet (environmental footprint and stewardship).

QUESTION 089 [COMPUTER PROGRAMMING & OOP CONSTRUCTORS]

Problem: In the context of class instantiation, how does a parameterized constructor fundamentally differ from a default constructor?

VERIFIED ANSWER: It initializes the newly created object with specific, user-provided values rather than standard null or zero values.

Governing Formula / Principle: Constructor Overloading: ClassName() vs. ClassName(Type param1, Type param2)

Step-by-Step Technical Derivation:

• A default (no-argument) constructor initializes instance variables to language default values (null, 0, or false).

• A parameterized constructor accepts explicit caller arguments upon object creation.

• This enables immediate initialization of object state with distinct, verified values during instantiation.

QUESTION 090 [ENGINEERING MECHANICS / COLLISIONS]

Problem: In a perfectly inelastic collision between two objects, which of the following physical quantities is strictly conserved?

VERIFIED ANSWER: Momentum only

Governing Formula / Principle: m1*v1 + m2*v2 = (m1 + m2)*v_final; Kinetic Energy is NOT conserved (ΔK < 0)

Step-by-Step Technical Derivation:

• In any closed mechanical system free of external forces, total linear momentum is always conserved.

• In a perfectly inelastic collision, the two colliding bodies lock together and move with a single common final velocity.

• A significant portion of kinetic energy is permanently converted into thermal energy, sound, and structural deformation; thus only momentum is conserved.

QUESTION 091 [VALUE PROPOSITION DESIGN & MARKETING]

Problem: In the Value Proposition Canvas, what specific term is used to describe the products, services, or features that eliminate or reduce the frustrations a customer experiences when trying to complete a task?

VERIFIED ANSWER: Pain Relievers

Governing Formula / Principle: Osterwalder Canvas: Customer Profile [Pains, Gains, Jobs] <=> Value Map [Pain Relievers, Gain Creators, Products]

Step-by-Step Technical Derivation:

• Alexander Osterwalder's Value Proposition Canvas maps customer profiles to company value propositions.

• Customer 'Pains' are obstacles, annoyances, risks, and bad outcomes faced during task execution.

• Features explicitly structured within the solution to mitigate, decrease, or completely eliminate these friction points are designated 'Pain Relievers'.

QUESTION 092 [MATERIALS SCIENCE & ATOMIC BONDING]

Problem: What type of primary atomic bonding is characterized by a 'sea of donated electrons' surrounding positively charged ion cores?

VERIFIED ANSWER: Metallic bonding

Governing Formula / Principle: Drude-Lorentz Free Electron Model: Delocalized valence electrons shielding cation lattice

Step-by-Step Technical Derivation:

• Metals have electropositive atoms with loosely held outer valence electrons.

• In a bulk metallic solid, valence electrons dissociate from parent atoms, creating a rigid lattice of positive ion cores.

• The delocalized valence electrons form a mobile 'electron cloud' or 'sea' that freely permeates the lattice, explaining high electrical and thermal conductivity.

QUESTION 093 [ELECTROMAGNETIC INDUCTION / FARADAY'S LAW]

Problem: According to Faraday's Law of Induction, the magnitude of the electromotive force (EMF) induced in a closed circuit is directly proportional to the rate of change of which quantity?

VERIFIED ANSWER: Magnetic flux

Governing Formula / Principle: EMF = - N * (dΦ_B / dt); Magnetic Flux Φ_B = ∫ B • dA

Step-by-Step Technical Derivation:

• Michael Faraday discovered that an electric potential is induced in a conducting loop whenever the magnetic flux linking the loop changes.

• Magnetic flux Φ_B represents the surface integral of the magnetic field B passing through the area of the circuit loop.

• The induced EMF magnitude is directly proportional to the time derivative of magnetic flux: |EMF| = N * |dΦ_B/dt|.

QUESTION 094 [LEAN STARTUP METHODOLOGY]

Problem: According to the Lean Startup methodology, what is the primary purpose of developing a Minimum Viable Product (MVP)?

VERIFIED ANSWER: To test fundamental business hypotheses with early adopters using the least amount of effort and resources.

Governing Formula / Principle: Build-Measure-Learn Feedback Loop: MVP minimizes cycle time to validate customer hypotheses

Step-by-Step Technical Derivation:

• Eric Ries formalized the Minimum Viable Product (MVP) concept in the Lean Startup framework.

• Rather than building an exhaustive, fully-featured product over several years based on untested assumptions, a startup builds a bare-bones version.

• The MVP is deployed to early adopters to collect empirical validated learning and test core hypotheses with minimal wasted capital.

QUESTION 095 [BUSINESS ETHICS & SUPPLY CHAIN CSR]

Problem: A technopreneur discovers that their primary hardware manufacturer is utilizing unethical labor practices. To align with strict corporate social responsibility (CSR) standards, what is the most appropriate action?

VERIFIED ANSWER: Initiate a supply chain audit and transition to a certified ethical manufacturer, even if costs slightly increase.

Governing Formula / Principle: CSR Supply Chain Governance: Supplier Code of Conduct + Independent Third-Party Auditing

Step-by-Step Technical Derivation:

• Corporate Social Responsibility requires firms to take ethical responsibility for the entire lifecycle and supply chain of their products.

• Ignoring supplier labor abuses for profit margins violates international human rights and corporate governance standards.

• The firm must investigate via an audit and transition procurement to socially compliant, certified ethical partners.

QUESTION 096 [DIFFUSION IN SOLIDS / FICK'S LAWS]

Problem: In Fick's first law of steady-state diffusion, what does the negative sign in the equation mathematically signify?

VERIFIED ANSWER: The net flux of diffusing atoms occurs down the concentration gradient, from higher to lower concentration.

Governing Formula / Principle: J = - D * (dC / dx); Negative sign ensures positive flux J in direction of decreasing C

Step-by-Step Technical Derivation:

• Fick's First Law describes steady-state mass transfer: J = -D * (dC/dx).

• If concentration decreases with increasing position x, the gradient slope dC/dx is negative.

• Because net physical mass diffusion spontaneously flows from regions of high concentration to regions of low concentration, the negative sign ensures that the calculated mass flux J is a positive quantity.

QUESTION 097 [COMPUTER-AIDED DESIGN (CAD) SPACES]

Problem: What is the fundamental structural difference between 'Model Space' and 'Paper Space' (Layout) in a standard CAD drafting environment?

VERIFIED ANSWER: Model Space is where actual geometry is created at a 1:1 scale, while Paper Space is used to arrange views and annotations for scaled printing.

Governing Formula / Principle: CAD Architectural Separation: Model Space (World Space, 1:1 units) vs. Paper Space (Print Sheet Layout)

Step-by-Step Technical Derivation:

• Model Space is an infinite virtual three-dimensional design canvas where physical engineering geometries are modeled at full actual 1:1 scale.

• Paper Space (Layout) replicates physical print sheets (e.g., A1, Letter, Blueprint).

• Viewports are inserted in Paper Space to frame, crop, and scale model geometry to standard drafting ratios (e.g., 1:100) alongside title blocks.

QUESTION 098 [ELECTROMAGNETISM / LENZ'S LAW]

Problem: Which principle states that the direction of an induced current will always be such that it creates a magnetic field opposing the change in magnetic flux that produced it?

VERIFIED ANSWER: Lenz's Law

Governing Formula / Principle: EMF = - dΦ_B / dt; The negative sign embodies Lenz's Law of electromagnetic opposition

Step-by-Step Technical Derivation:

• Heinrich Lenz formulated the directionality law of electromagnetic induction in 1834.

• When magnetic flux through a conductor changes, an induced current flows.

• This induced current generates an opposing secondary magnetic field that resists the flux change, reflecting the fundamental conservation of energy.

QUESTION 099 [OBJECT-ORIENTED PROGRAMMING (OOP)]

Problem: Why is it structurally impossible for a program to directly instantiate an object from an abstract class?

VERIFIED ANSWER: Abstract classes often contain unimplemented methods that must be defined by a subclass before an object can exist.

Governing Formula / Principle: Abstract Class: abstract class Shape { abstract void draw(); }; Incomplete implementation prevents instantiation

Step-by-Step Technical Derivation:

• An abstract class serves as an incomplete structural blueprint that defines shared interfaces without full implementation.

• If an abstract class contains pure virtual or abstract methods without bodies, direct instantiation would create objects with undefined behaviors.

• Therefore, compilers prohibit direct instantiation (`new AbstractClass()` is illegal); a concrete subclass must first inherit and implement all abstract methods.

QUESTION 100 [ENTREPRENEURSHIP & STARTUP CAPITAL]

Problem: What is the financial term for the initial capital raised by a startup often from founders, family, or angel investors-used strictly to prove a concept and develop an initial prototype?

VERIFIED ANSWER: Seed Funding

Governing Formula / Principle: Startup Capital Stages: Pre-Seed/Seed (Proof-of-Concept & MVP) -> Series A (Scale) -> Series B -> IPO

Step-by-Step Technical Derivation:

• Early venture development begins at the ideation and proof-of-concept phase before commercial product-market fit.

• The earliest structured equity financing raised to validate technical viability, build an initial prototype, and hire key founding personnel is termed Seed Funding (or Seed Capital).

• This precedes formal Series A institutional venture funding.

========== **[END OF 100-ITEM EXAMINATION CARD REVIEWER & FORMULA COMPILATION] ECE BOARD EXAM REVIEW SERIES | ALL 100 QUESTIONS SOLVED IN FULL** ==========

Comprehensive Technical Solutions & Explanations - Set 4

Complete Solutions for PB Exams Set 4 (General Engineering & Applied Sciences)

Item 1: Which crystal structure has the highest atomic packing factor (APF)?

Correct Answer: Face-centered cubic (FCC)

> **■ EXPLANATION & SOLUTION**
> FCC and HCP structures have the highest possible atomic packing factor (APF = 0.74), compared to BCC (0.68) and Simple Cubic (0.52).

Item 2: A machine costing ₱120,000 with salvage value ₱20,000 is expected to produce 40,000 total units. If it produces 12,000 units in year 1, what is the depreciation expense?

Correct Answer: ₱30,000

> **■ EXPLANATION & SOLUTION**
> 1. Depreciable cost = Cost - Salvage = 120,000 - 20,000 = ₱100,000.
> 2. Depreciation rate per unit = 100,000 / 40,000 = ₱2.50 per unit.
> 3. Year 1 Depreciation = 12,000 units × ₱2.50 = ₱30,000.

Item 3: Henri Fayol suggested management involves planning, organizing, commanding, coordinating, and controlling. This is an example of:

Correct Answer: Classical Management Theory

> **■ EXPLANATION & SOLUTION**
> Henri Fayol's administrative management principles form the bedrock of Classical Management Theory, defining the core functional managerial duties.

Item 4: A ceiling fan rotates about its fixed axis, and each point on the blades traces a circular path. Which type of motion is exhibited?

Correct Answer: Rotational

> **■ EXPLANATION & SOLUTION**
> Pure rotational motion occurs when all particles of a rigid body move in circular paths about a common fixed axis of rotation.

Item 5: An office building installs LED lighting and smart sensors that turn off lights when rooms are unoccupied. Which type of energy conservation strategy does this represent?

Correct Answer: Active conservation

> **■ EXPLANATION & SOLUTION**
> Active conservation uses automated sensors, smart controls, and energy-efficient hardware to dynamically control and minimize power consumption.

Item 6: According to the PEC, a location where flammable gases or vapors are present in the air in quantities sufficient to produce ignitable mixtures is classified as:

Correct Answer: Class I

> **■ EXPLANATION & SOLUTION**
> According to the Philippine Electrical Code (PEC) / NEC, Class I locations are those in which flammable gases or vapors are or may be present in quantities sufficient to produce explosive or ignitable mixtures.

Item 7: An engineer discovers that a contractor is using substandard materials in a public infrastructure project. The engineer should:

Correct Answer: Report the violation to the proper authorities

> **■ EXPLANATION & SOLUTION**
> Professional ethics codes dictate that the primary duty of an engineer is paramount public safety and welfare, obligating reporting of safety violations.

Item 8: A car moves at 20 m/s around a circular curve of radius 50 m. The centripetal acceleration is:

Correct Answer: 8 m/s²

> **■ EXPLANATION & SOLUTION**
> 1. Centripetal acceleration formula: $a_c = v^2 / r$.
> 2. $a_c = (20)^2 / 50 = 400 / 50 = 8$ m/s².

Item 9: A solid cylinder of mass 2 kg, radius 0.2 m rolls down a 1 m high incline. Find its velocity at bottom.

Correct Answer: 3.6 m/s

> **■ EXPLANATION & SOLUTION**
> 1. Energy conservation: $mgh = (1/2)m v^2 + (1/2)I \omega^2$.
> 2. For a solid cylinder, $I = (1/2)m r^2$ and $v = r\omega \Rightarrow$ Rolling KE $= (3/4)m v^2$.
> 3. $mgh = (3/4)m v^2 \Rightarrow v = \sqrt{(4gh / 3)} = \sqrt{[(4 \times 9.81 \times 1) / 3]} = \sqrt{13.08} \approx 3.61$ m/s.

Item 10: A solid cylinder (mass 10 kg, radius 0.5 m) rolls without slipping at 4 m/s. What is its total kinetic energy?

Correct Answer: 120 J

> **■ EXPLANATION & SOLUTION**
> 1. $KE_{total} = KE_{trans} + KE_{rot} = (1/2)m\ v^2 + (1/2)I\ \omega^2$.
> 2. Since $I = (1/2)m\ r^2$, $KE_{total} = (3/4)m\ v^2$.
> 3. $KE_{total} = (3/4) \times (10\ kg) \times (4\ m/s)^2 = 0.75 \times 10 \times 16 = 120$ Joules.

Item 11: A student constructs a hand-cranked generator for a school project. When turned faster, the light bulb becomes brighter. Which principle explains this?

Correct Answer: Faster rotation increases induced EMF

> **■ EXPLANATION & SOLUTION**
> By Faraday's Law of Induction ($E = -d\Phi/dt = B\ A\ \omega \sin(\omega t)$), increasing rotational speed ω increases the time rate of magnetic flux linkage change, generating higher induced EMF.

Item 12: Which type of wave exhibits dispersion, where different frequencies travel at different speeds?

Correct Answer: Water surface waves

> **■ EXPLANATION & SOLUTION**
> Water surface waves are dispersive because phase velocity depends directly on wavelength and frequency, causing different frequencies to propagate at different speeds.

Item 13: A bank uses a decision tree to determine loan approvals based on credit score, income, and employment status. What type of decision tree is most suitable?

Correct Answer: Classification tree

> **■ EXPLANATION & SOLUTION**
> Because the outcome is categorical (e.g. 'Approved' vs. 'Denied'), a classification tree is the appropriate machine learning structure.

Item 14: In energy storage devices like supercapacitors, which nanomaterial improves both capacitance and charge/discharge rates?

Correct Answer: Graphene

> **■ EXPLANATION & SOLUTION**
> Graphene offers ultra-high specific surface area (~2630 m^2/g) and excellent electrical conductivity, maximizing double-layer electrostatic capacitance and ion diffusion rates.

Item 15: **Which nanomaterial is widely used in transparent conductive films for touchscreens and displays?**

Correct Answer: Graphene

> **■ EXPLANATION & SOLUTION**
> Graphene combines high optical transparency (~97.7% for single layer) with low sheet resistance, making it an ideal flexible replacement for ITO.

Item 16: **Which of the following statements correctly contrasts supercapacitors and batteries?**

Correct Answer: Supercapacitors have higher power density but lower energy density than batteries

> **■ EXPLANATION & SOLUTION**
> Supercapacitors store charge electrostatically, enabling rapid charge/discharge cycles (high power density), whereas chemical batteries store energy via bulk redox reactions (high energy density).

Item 17: **Which feature of object-oriented programming allows a class to derive properties and methods from another class?**

Correct Answer: Inheritance

> **■ EXPLANATION & SOLUTION**
> Inheritance enables a derived child class to inherit fields and functions from a base parent class, encouraging code reusability.

Item 18: **If nominal GDP increases by 8% and inflation is 3%, the real GDP growth rate is approximately:**

Correct Answer: 5%

> **■ EXPLANATION & SOLUTION**
> Real GDP Growth Rate ≈ Nominal GDP Growth Rate - Inflation Rate = 8% - 3% = 5%.

Item 19: **A glass prism has a refracting angle A = 60°. The refractive index n = 1.50. Find angle of minimum deviation δ_m.**

Correct Answer: 37.2°

> **■ EXPLANATION & SOLUTION**
> 1. Formula: $n = \sin[(A + \delta_m)/2] / \sin(A/2)$.
> 2. $1.50 = \sin[(60° + \delta_m)/2] / \sin(30°) \Rightarrow \sin[(60° + \delta_m)/2] = 1.50 \times 0.5 = 0.75$.
> 3. $(60° + \delta_m)/2 = \arcsin(0.75) \approx 48.59° \Rightarrow 60° + \delta_m = 97.18° \Rightarrow \delta_m = 37.18° \approx 37.2°$.

Item 20: **Quantum dots are utilized in electronic displays primarily because they:**

Correct Answer: Emit size-tunable light with high color purity

> **■ EXPLANATION & SOLUTION**
> Due to quantum confinement effects, tuning quantum dot physical particle size changes the bandgap, yielding highly monochromatic light across precise visible wavelengths.

Item 21: **A manufacturing plant upgrades motors to high-efficiency models and implements VFDs to reduce power consumption. Which principle is this?**

Correct Answer: Energy efficiency

> **■ EXPLANATION & SOLUTION**
> Energy efficiency focuses on delivering identical or superior physical output using decreased electrical power input through technological enhancements.

Item 22: **Which defect can increase ionic conductivity in solid electrolytes?**

Correct Answer: Vacancy defects

> **■ EXPLANATION & SOLUTION**
> Vacancy defects create unoccupied lattice sites, providing low-activation-energy hopping pathways for mobile ions across solid-state electrolytes.

Item 23: **Which of the following does NOT typically cause explosive atmospheres in hazardous locations?**

Correct Answer: Carbon dioxide gas

> **■ EXPLANATION & SOLUTION**
> Carbon dioxide (CO_2) is an inert, non-flammable gas commonly used as a fire suppression agent, whereas combustible dusts and volatile hydrocarbons cause explosive atmospheres.

Item 24: **A lake receives detergent phosphates causing eutrophication. Which phosphorus cycle characteristic explains this?**

Correct Answer: Phosphorus is mostly sediment-bound and not gaseous

> **■ EXPLANATION & SOLUTION**
> Unlike carbon or nitrogen, phosphorus lacks a significant atmospheric gas phase. It accumulates in aquatic sediments and soils, making aquatic ecosystems highly sensitive to excess runoff.

Item 25: An engineer invests ₱2,000 annually at end of each year for 6 years at 8% interest per year. Future value of annuity?

Correct Answer: ₱14,672

> **■ EXPLANATION & SOLUTION**
>
> 1. Ordinary Annuity FV formula: $F = A\,[(1 + i)^n - 1] / i$.
>
> 2. $F = 2000 \times [(1.08)^6 - 1] / 0.08 = 2000 \times (1.58687 - 1) / 0.08 = 2000 \times 7.3359 = ₱14,671.85 \approx ₱14,672$.

Item 26: A power module operates continuously at 180°C under vibration. Encapsulating material must provide insulation, thermal stability, and mechanical protection. Best polymer?

Correct Answer: Epoxy molding compound

> **■ EXPLANATION & SOLUTION**
>
> Thermosetting epoxy molding compounds maintain structural stability, adhesion, electrical insulation, and chemical resistance up to 180°C+ continuous exposure.

Item 27: Graphite conducts electricity despite being a non-metal. Which bonding feature explains this?

Correct Answer: Delocalized electrons in planar sheets

> **■ EXPLANATION & SOLUTION**
>
> In graphite, carbon atoms form sp^2 hybridized planar sheets. The unhybridized p-orbitals overlap to form delocalized π-bonds along hexagonal layers, facilitating electron flow.

Item 28: Which of the following is a key metric for evaluating service operations efficiency in a competitive market?

Correct Answer: Service cycle time

> **■ EXPLANATION & SOLUTION**
>
> Service cycle time measures total elapsed duration to fulfill customer requests, serving as a primary operational indicator of speed and process efficiency.

Item 29: A government plans to approve an industrial project with water pollution risk. Environmental assessment guides decision via which principle?

Correct Answer: Precautionary Principle

> **■ EXPLANATION & SOLUTION**
>
> The Precautionary Principle states that when an action carries risk of severe environmental harm, protective measures should be taken even if full scientific certainty is lacking.

Item 30: **A force F(x) = 5x N acts on a particle along x-axis from x = 0 to x = 4 m. Work done?**

Correct Answer: 40 J

> **■ EXPLANATION & SOLUTION**
>
> 1. Work $W = \int_0^4 F(x)\,dx = \int_0^4 5x\,dx$.
>
> 2. $W = [2.5\,x^2]_0^4 = 2.5(16) - 0 = 40$ Joules.

Item 31: **A 200 g sample of isotope has half-life of 5 hours. Remaining mass after 15 hours?**

Correct Answer: 25 g

> **■ EXPLANATION & SOLUTION**
>
> 1. Number of half-lives $n = 15 / 5 = 3$.
>
> 2. Remaining mass $N = N_0\,(1/2)^n = 200\text{ g} \times (1/2)^3 = 200 / 8 = 25$ g.

Item 32: **Which type of bond causes high boiling point of water compared to similar molar mass molecules?**

Correct Answer: Hydrogen bonding

> **■ EXPLANATION & SOLUTION**
>
> Intermolecular hydrogen bonds between highly polar O-H bonds require substantial thermal energy to disrupt, elevating water's boiling point.

Item 33: **An electronics company sells via website, e-commerce, and retail stores. Which 4Ps element is this?**

Correct Answer: Place

> **■ EXPLANATION & SOLUTION**
>
> Place (Distribution) covers channels, outlets, and platforms through which products are made accessible to end consumers.

Item 34: **Which of the following best describes dynamic typing?**

Correct Answer: Type of a variable is determined at runtime

> **■ EXPLANATION & SOLUTION**
>
> In dynamically typed languages (like Python or JavaScript), type checking occurs during code execution rather than at compile time.

Item 35: **A convex lens has focal length 20 cm in air (n_lens = 1.5). What happens to focal length if immersed in water (n = 1.33)?**

Correct Answer: Increases

> **■ EXPLANATION & SOLUTION**
> By Lensmaker's Equation $1/f = (n_{lens}/n_{medium} - 1)(1/R_1 - 1/R_2)$, decreasing relative refractive index difference reduces refraction power, increasing focal length.

Item 36: **Equipment costs ₱120,000, lasts 5 years. Interest rate 10%/year. Annual capital recovery payment?**

Correct Answer: ₱31,656

> **■ EXPLANATION & SOLUTION**
> 1. Capital Recovery Factor: $A = P\,[i(1+i)^n] / [(1+i)^n - 1]$.
> 2. $A = 120{,}000 \times [0.10(1.10)^5] / [(1.10)^5 - 1] = 120{,}000 \times (0.16105 / 0.61051) \approx 120{,}000 \times 0.26380 = ₱31{,}656$.

Item 37: **Microscope objective forms real image 2 mm from lens for object 0.1 mm away. Magnification?**

Correct Answer: 20x

> **■ EXPLANATION & SOLUTION**
> Linear magnification $M = |v / u| = 2\text{ mm} / 0.1\text{ mm} = 20\text{x}$.

Item 38: **Factory manager realizes best management approach depends on environmental/situational factors. Theory?**

Correct Answer: Contingency theory

> **■ EXPLANATION & SOLUTION**
> Contingency Theory asserts that there is no universal 'one best way' to manage; optimal leadership actions depend on internal and external situational contexts.

Item 39: **In 3D CAD modeling, defining geometric relationships using parameters that control design changes is called:**

Correct Answer: Parametric modeling

> **■ EXPLANATION & SOLUTION**
> Parametric modeling establishes dynamic dimension constraints and feature relationships so modifying values automatically updates the full CAD geometry.

Item 40: **Tech company analyzes competitors' products, pricing, and market share. Which strategic activity is this?**

Correct Answer: Competitive analysis

> **■ EXPLANATION & SOLUTION**
> Competitive analysis systematically evaluates rival market offerings, capabilities, and positioning to formulate competitive strategy.

Item 41: **Which BEST represents a global-scale approach to mitigate ocean acidification?**

Correct Answer: Reducing CO_2 emissions from fossil fuels

> **■ EXPLANATION & SOLUTION**
> Ocean acidification is driven directly by atmospheric CO_2 absorption forming carbonic acid. Reducing global CO_2 emissions is the primary structural mitigation strategy.

Item 42: **A function in C++ that calls itself directly or indirectly is called:**

Correct Answer: Recursive function

> **■ EXPLANATION & SOLUTION**
> A function that solves smaller subproblems by invoking itself until reaching a base condition is defined as a recursive function.

Item 43: **The Bravais lattice that has atoms at the corners and the center of each face is:**

Correct Answer: Face-centered cubic (FCC)

> **■ EXPLANATION & SOLUTION**
> FCC unit cells feature lattice points at all 8 cube corners plus at the centers of all 6 faces.

Item 44: **Earned Value Management (EVM) is primarily used in which phase of the project life cycle?**

Correct Answer: Controlling

> **■ EXPLANATION & SOLUTION**
> EVM tracks schedule variance (SV) and cost variance (CV) against baseline plans, serving as a primary project progress monitoring and controlling methodology.

Item 45: **Which type of decay decreases atomic number by 2 and mass number by 4?**

Correct Answer: Alpha decay

> **■ EXPLANATION & SOLUTION**
> Alpha decay emits a helium nucleus (^4_2He), decreasing mass number A by 4 and atomic number Z by 2.

Item 46: **Which battery type is MOST suitable for large-scale energy storage in UPS systems due to reliability and cost?**

Correct Answer: Lead-acid

> **■ EXPLANATION & SOLUTION**
> Valve-Regulated Lead-Acid (VRLA) batteries remain preferred for industrial stationery UPS installations due to low cost per kWh, safety, and reliability.

Item 47: **Container holds 0.5 mol gas at 2 atm and 300 K. Volume? (R = 0.0821 L·atm/(mol·K))**

Correct Answer: 6.16 L

> **■ EXPLANATION & SOLUTION**
> 1. Ideal Gas Law: $P\,V = n\,R\,T \Rightarrow V = (n\,R\,T) / P$.
>
> 2. $V = (0.5\ \text{mol} \times 0.0821 \times 300\ \text{K}) / 2\ \text{atm} = 12.315 / 2 = 6.1575\ \text{L} \approx 6.16\ \text{L}$.

Item 48: **Which nutrient is most commonly the limiting factor in freshwater ecosystems?**

Correct Answer: Phosphorus

> **■ EXPLANATION & SOLUTION**
> Phosphorus is typically the limiting macronutrient in freshwater lakes and rivers, whereas nitrogen is usually limiting in marine environments.

Item 49: **Review center registers distinctive 'E' logo to prevent unauthorized competitor use. IP principle?**

Correct Answer: Trademark

> **■ EXPLANATION & SOLUTION**
> Trademarks protect distinctive brand symbols, names, logos, or slogans used to identify commercial sources of goods and services.

Item 50: **Which nuclear reaction releases more energy per unit mass?**

Correct Answer: Fusion of H-2 and H-3

> **■ EXPLANATION & SOLUTION**
> Nuclear fusion of light isotopes (Deuterium-Tritium) releases ~3.5 MeV/nucleon, yielding significantly higher energy density per unit mass than uranium fission (~0.85 MeV/nucleon).

Item 51: **Particle of mass m moves with velocity $v(t) = a\,t$ from $t = 0$ to $t = T$. Impulse on particle?**

Correct Answer: m a T

> **■ EXPLANATION & SOLUTION**

1. Impulse $J = \Delta p = m\, v(T) - m\, v(0)$.

2. $v(T) = a\, T$ and $v(0) = 0 \Rightarrow J = m\,(a\, T) - 0 = m\, a\, T$.

Item 52: **Water flows through horizontal pipe with varying diameter. If velocity increases, pressure:**

Correct Answer: Decreases

■ **EXPLANATION & SOLUTION**

By Bernoulli's Principle ($P + 1/2\ \rho\ v^2 = \text{const}$), an increase in fluid velocity produces a simultaneous decrease in static pressure.

Item 53: **Venture capitalist weighs revolutionary product with small market vs. moderate product with massive market. Factor weighing heavier?**

Correct Answer: Market size

■ **EXPLANATION & SOLUTION**

Venture capitalists prioritize Total Addressable Market (TAM) size, as massive markets provide room for scalable revenue growth and exit valuation.

Item 54: **Difference between strategic and tactical marketing?**

Correct Answer: Strategic marketing focuses on long-term planning and market positioning; tactical marketing executes day-to-day actions

■ **EXPLANATION & SOLUTION**

Strategic marketing defines overarching vision and brand positioning, whereas tactical marketing implements concrete operational execution steps (promotions, ads).

Item 55: **A 0.5 kg ball moving at +10 m/s strikes a wall and rebounds at -6 m/s. Impulse delivered by wall?**

Correct Answer: 8 N·s

■ **EXPLANATION & SOLUTION**

1. Impulse $J = m\ \Delta v = m\ (v_f - v_i)$.

2. $J = 0.5\ \text{kg} \times [-6\ \text{m/s} - (+10\ \text{m/s})] = 0.5 \times (-16) = -8\ \text{N·s}$.

3. Magnitude of impulse = 8 N·s.

Item 56: **For a string fixed at both ends, fundamental frequency f_1 expression:**

Correct Answer: $f_1 = (1 / 2L)\ \sqrt{(T / \mu)}$

■ **EXPLANATION & SOLUTION**

Mersenne's law for vibrating strings fixed at both ends gives fundamental frequency $f_1 = v / (2L) = (1 / 2L)\ \sqrt{(T /}$

> μ).

Item 57: Manufacturing firm reuses waste materials as raw materials for new products. Approach?

Correct Answer: Circular Economy

> **■ EXPLANATION & SOLUTION**
> The Circular Economy model eliminates waste by designing closed-loop manufacturing systems where materials are recycled and reused indefinitely.

Item 58: Which property describes a material's ability to resist deformation under applied force?

Correct Answer: Rigidity (or Stiffness)

> **■ EXPLANATION & SOLUTION**
> Rigidity (elastic modulus / stiffness) quantifies a material's resistance to elastic deformation when subjected to external mechanical load.

Item 59: A 2 kg cart moving at 3 m/s collides head-on and sticks to stationary 1 kg cart. Final velocity?

Correct Answer: 2 m/s

> **■ EXPLANATION & SOLUTION**
> 1. Conservation of momentum: $m_1 v_1 + m_2 v_2 = (m_1 + m_2) V_f$.
>
> 2. $(2\ kg \times 3\ m/s) + (1\ kg \times 0) = (2 + 1) V_f \Rightarrow 6 = 3 V_f \Rightarrow V_f = 2\ m/s$.

Item 60: Which programming feature automatically manages memory allocation and deallocation?

Correct Answer: Garbage collection

> **■ EXPLANATION & SOLUTION**
> Garbage collection automatically identifies and reclaims memory allocated to objects no longer referenced by the program.

Item 61: Which chemical reaction explains lowering of pH in seawater due to CO_2 absorption?

Correct Answer: $CO_2 + H_2O \rightarrow H_2CO_3$

> **■ EXPLANATION & SOLUTION**
> Dissolved carbon dioxide reacts with seawater to form carbonic acid (H_2CO_3), which dissociates into hydrogen ions (H^+) and lowers pH.

Item 62: **9 V battery connected to circuit: 60 Ω and 120 Ω in parallel, in series with 30 Ω. Total current?**

Correct Answer: 0.129 A (Note: 1.29 A marked on option key)

> **■ EXPLANATION & SOLUTION**
>
> 1. Parallel combination $R_p = (60 \times 120) / (60 + 120) = 7200 / 180 = 40\ \Omega$.
> 2. Total resistance $R_{total} = 40 + 30 = 70\ \Omega$.
> 3. Total current $I = V / R_{total} = 9\ V / 70\ \Omega \approx 0.12857\ A \approx 0.129\ A$.

Item 63: **Rank activities by direct impact on ocean acidification (most to least): 1. Deforestation, 2. Fossil fuel combustion, 3. Plastic waste, 4. Cement production.**

Correct Answer: 2 > 4 > 1 > 3

> **■ EXPLANATION & SOLUTION**
>
> Fossil fuel combustion (2) is the primary source of CO_2 emissions, followed by cement manufacturing chemical reactions (4), land clearing/deforestation (1), and non-CO_2 plastic waste (3).

Item 64: **Which of the following is an example of a supporting ecosystem service?**

Correct Answer: Nutrient cycling in soils

> **■ EXPLANATION & SOLUTION**
>
> Supporting services (like soil formation, nutrient cycling, primary production) underpin all other ecosystem provisioning and regulating functions.

Item 65: **Which force is NOT included in Newton's second law horizontal analysis of object on frictionless surface?**

Correct Answer: Friction force

> **■ EXPLANATION & SOLUTION**
>
> Because the surface is specified as frictionless, frictional force is zero and excluded from horizontal force summation.

Item 66: **Solid disc (mass 2 kg, radius 0.5 m) rotates at 10 rad/s. Rotational kinetic energy?**

Correct Answer: 12.5 J

> **■ EXPLANATION & SOLUTION**
>
> 1. Moment of inertia $I = (1/2)\ m\ r^2 = 0.5 \times 2\ kg \times (0.5\ m)^2 = 0.25\ kg \cdot m^2$.
> 2. $KE_{rot} = (1/2)\ I\ \omega^2 = 0.5 \times 0.25\ kg \cdot m^2 \times (10\ rad/s)^2 = 0.125 \times 100 = 12.5$ Joules.

Item 67: **Which of the following best exemplifies a density-independent factor affecting population growth?**

Correct Answer: Seasonal drought

> **■ EXPLANATION & SOLUTION**
> Abiotic environmental disasters (floods, droughts, extreme freezes) affect mortality rates regardless of population density.

Item 68: **Electronics firm spent $150,000 on R&D. Potential future benefit $500,000. $150,000 spent is:**

Correct Answer: A sunk cost

> **■ EXPLANATION & SOLUTION**
> A sunk cost is a past expenditure that cannot be recovered and should not influence future economic decision-making.

Item 69: **In Python: x = 5; x = 'Hello'. Which concept does this illustrate?**

Correct Answer: Dynamic typing

> **■ EXPLANATION & SOLUTION**
> Dynamic typing allows variable type assignment to change dynamically during runtime based on assigned values.

Item 70: **Which financial metric is MOST important when evaluating early-stage startups with little revenue?**

Correct Answer: Cash burn rate

> **■ EXPLANATION & SOLUTION**
> Cash burn rate measures the speed at which a pre-revenue startup consumes venture capital reserves to maintain operations.

Item 71: **Which system demonstrates mechanical energy conservation?**

Correct Answer: A pendulum swinging with negligible air resistance

> **■ EXPLANATION & SOLUTION**
> In the absence of non-conservative dissipative forces like friction or air drag, total mechanical energy (PE + KE) remains constant.

Item 72: **In a metallic alloy, when an atom is replaced by a different atom of similar size, defect is called:**

Correct Answer: Substitutional defect

> **■ EXPLANATION & SOLUTION**
> A substitutional point defect occurs when a host lattice atom is replaced by a solute atom of comparable atomic radius.

Item 73: **For electric vehicles requiring high energy density, fast charging, and light weight, battery type?**

Correct Answer: Lithium-ion

> **■ EXPLANATION & SOLUTION**
> Lithium-ion chemistry provides high gravimetric/volumetric energy density and fast charge-discharge efficiency essential for electric vehicles.

Item 74: **Which of the following is NOT a purpose of doping silicon?**

Correct Answer: To increase mechanical strength

> **■ EXPLANATION & SOLUTION**
> Semiconductor doping intentionally introduces impurity atoms to alter electrical carrier concentration, not to enhance mechanical strength.

Item 75: **A manager believes employees naturally dislike work and must be closely supervised. Theory?**

Correct Answer: Theory X

> **■ EXPLANATION & SOLUTION**
> McGregor's Theory X assumes workers inherently dislike labor, lack ambition, and require strict authoritarian control and supervision.

Item 76: **Carnot refrigerator operates between 5°C (cold space) and 30°C (room). COP?**

Correct Answer: 11.1

> **■ EXPLANATION & SOLUTION**
> 1. Convert to Kelvin: T_cold = 5 + 273.15 = 278.15 K; T_hot = 30 + 273.15 = 303.15 K.
>
> 2. COP_refrigerator = T_cold / (T_hot - T_cold) = 278.15 / (303.15 - 278.15) = 278.15 / 25 ≈ 11.12.

Item 77: **Which statement correctly explains why dispersion occurs in glass?**

Correct Answer: Speed of light in glass depends on wavelength

> **■ EXPLANATION & SOLUTION**

> Dispersion occurs because refractive index n(λ) varies with wavelength, causing light of different colors to travel at different phase velocities in glass.

Item 78: Inserting/withdrawing magnet in coil induces current in reversing directions. Governing law?

Correct Answer: Lenz's law (Faraday's Law)

> **■ EXPLANATION & SOLUTION**
> Lenz's Law specifies that induced current flows in a direction such that its magnetic field opposes the change in flux causing it.

Item 79: A team leader inspires staff by sharing vision and recognizing achievements. Management function?

Correct Answer: Leading

> **■ EXPLANATION & SOLUTION**
> Leading involves motivating employees, influencing behavior, communicating vision, and guiding staff toward organizational goals.

Item 80: Programming paradigm focused on evaluating mathematical functions without state mutation?

Correct Answer: Functional Programming

> **■ EXPLANATION & SOLUTION**
> Functional programming treats computation as evaluation of pure mathematical functions, avoiding mutable state and side effects.

Item 81: The SI unit of luminous intensity is:

Correct Answer: Candela (cd)

> **■ EXPLANATION & SOLUTION**
> The candela (cd) is the base SI unit measuring luminous intensity of a light source in a given direction.

Item 82: Two identical balls collide elastically. Ball A reverses velocity from +v to -v in time Δt. Impulse?

Correct Answer: 2mv

> **■ EXPLANATION & SOLUTION**
> 1. Impulse J = Δp = m v_final - m v_initial.
>
> 2. J = m(-v) - m(+v) = -2mv ⇒ magnitude = 2mv.

Item 83: **Review center installs LED lights, motion sensors, and smart HVAC in building. Sustainability concept?**

Correct Answer: Resource Efficiency

> **■ EXPLANATION & SOLUTION**
> Resource efficiency involves optimizing technology to minimize energy, water, and material inputs required for building operation.

Item 84: **Which BEST demonstrates how SWOT analysis influences strategic decision-making?**

Correct Answer: Identifying internal strengths to capitalize on market opportunities

> **■ EXPLANATION & SOLUTION**
> SWOT aligns internal organizational Strengths and Weaknesses against external Opportunities and Threats (SO strategy formulation).

Item 85: **Which statement is TRUE for ferromagnetic materials?**

Correct Answer: They have permanent magnetic moments due to aligned spins

> **■ EXPLANATION & SOLUTION**
> Ferromagnetic materials exhibit strong spontaneous magnetization below Curie temperature due to quantum mechanical exchange coupling aligning electron spins in magnetic domains.

Item 86: **Loan offers 12% annual interest compounded quarterly. Effective annual rate (EAR)?**

Correct Answer: 12.55% ≈ 12.6%

> **■ EXPLANATION & SOLUTION**
> 1. Formula: EAR = (1 + r/m)^m - 1.
>
> 2. EAR = $(1 + 0.12/4)^4 - 1 = (1.03)^4 - 1 = 1.12551 - 1 = 0.12551 \approx 12.55\%$ (12.6%).

Item 87: **Which statement is TRUE regarding interface inheritance in OOP?**

Correct Answer: An interface can extend multiple interfaces

> **■ EXPLANATION & SOLUTION**
> Unlike class single-inheritance restrictions in languages like Java, an interface can extend multiple parent interfaces simultaneously.

Item 88: **Dynamic typing does not imply weak typing. Example supporting strong dynamic typing?**

Correct Answer: Python raises an error when adding a string to an integer: '5' + 5

> **■ EXPLANATION & SOLUTION**
> Strong dynamic languages (Python) enforce strict type rules at runtime, throwing type errors during explicit

> mismatched operations.

Item 89: Energy (E in Joules) stored in supercapacitor with capacitance C (F) and voltage V (V):

Correct Answer: $E = (1/2)\, C\, V^2$

> **■ EXPLANATION & SOLUTION**
> Electrostatic energy stored in a capacitor is $E = (1/2)\, C\, V^2$.

Item 90: Wheel with I = 0.5 kg·m² spun from rest to angular velocity ω = 20 rad/s. Work done?

Correct Answer: 100 J

> **■ EXPLANATION & SOLUTION**
> 1. Work-energy theorem: $W = \Delta KE_{rot} = (1/2)\, I\, \omega^2 - 0$.
> 2. $W = 0.5 \times 0.5\ kg \cdot m^2 \times (20\ rad/s)^2 = 0.25 \times 400 = 100$ Joules.

Item 91: Replacing ceramic substrates with polymer-based ones reduces weight. Trade-off evaluated?

Correct Answer: Reduced thermal conductivity

> **■ EXPLANATION & SOLUTION**
> Polymers offer low density and flexibility but possess significantly lower thermal conductivity (~0.2–0.5 W/m·K) than ceramics (~20–200 W/m·K).

Item 92: Particle moves under time-varying force F(t) = k t² from t = 0 to t = T. Impulse delivered?

Correct Answer: $k\, T^3 / 3$

> **■ EXPLANATION & SOLUTION**
> 1. Impulse $J = \int_0^T F(t)\, dt = \int_0^T k\, t^2\, dt$.
> 2. $J = [k\, t^3 / 3]_0^T = k\, T^3 / 3$.

Item 93: Advanced CAD technique optimizing shapes automatically based on constraints and criteria?

Correct Answer: Generative design

> **■ EXPLANATION & SOLUTION**
> Generative design utilizes AI algorithms and structural FEA solvers to autonomously generate optimized lightweight geometric designs based on operational boundary constraints.

Item 94: A 500 mL gas sample at 25°C is heated to 125°C at constant pressure. Final volume?

Correct Answer: 668 mL

> **■ EXPLANATION & SOLUTION**
>
> 1. Charles's Law: $V_1 / T_1 = V_2 / T_2$ (Temperatures in Kelvin).
> 2. $T_1 = 25 + 273.15 = 298.15$ K; $T_2 = 125 + 273.15 = 398.15$ K.
> 3. $V_2 = V_1 (T_2 / T_1) = 500$ mL × (398.15 / 298.15) ≈ 500 × 1.3354 = 667.7 mL ≈ 668 mL.

Item 95: Particle displacement angle θ with propagation direction. Wave type and θ for longitudinal wave?

Correct Answer: Oblique wave; θ = 0° for purely longitudinal wave

> **■ EXPLANATION & SOLUTION**
>
> In longitudinal waves, particle oscillation occurs parallel to wave energy propagation direction, meaning relative displacement angle θ = 0°.

Item 96: Submarine submerged at 200 m in seawater (ρ = 1025 kg/m³). Hydrostatic pressure (g = 9.8 m/s²)?

Correct Answer: 2×10^6 Pa

> **■ EXPLANATION & SOLUTION**
>
> 1. $P = \rho g h$.
> 2. $P = 1025$ kg/m³ × 9.8 m/s² × 200 m = 2,009,000 Pa ≈ 2.01×10^6 Pa ≈ 2×10^6 Pa.

Item 97: Heat engine absorbs 5000 J heat from hot reservoir and rejects 3000 J to cold reservoir. Efficiency?

Correct Answer: 40%

> **■ EXPLANATION & SOLUTION**
>
> 1. Efficiency $\eta = (Q_H - Q_C) / Q_H = (5000 - 3000) / 5000$.
> 2. $\eta = 2000 / 5000 = 0.40 = 40\%$.

Item 98: Company switches power supply from coal plants to solar and wind. UN SDG goal addressed?

Correct Answer: SDG 7: Affordable and Clean Energy

> **■ EXPLANATION & SOLUTION**
>
> Transitioning power sources to renewable solar and wind power directly targets Sustainable Development Goal 7.

Item 99: **In BCC crystal, lattice parameter 'a' is related to atomic radius 'r' by:**

Correct Answer: a = 4r / √3

> **■ EXPLANATION & SOLUTION**
> In Body-Centered Cubic (BCC) crystals, atoms touch along body diagonal: √3 a = 4r ⇒ a = 4r / √3.

Item 100: **Central bank increases reserve requirement for commercial banks. Macroeconomic effect?**

Correct Answer: Decrease in money supply

> **■ EXPLANATION & SOLUTION**
> Raising reserve requirements forces commercial banks to hold larger cash reserves against deposits, restricting lending capacity and contracting money supply.

PB EXAMS (11)_3 -- COMPLETE 100-QUESTION SOLUTION MANUAL

PLAIN TEXT FORMATTED ENGINEERING STUDY GUIDE WITH DETAILED DERIVATIONS & TIPS

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TOTAL QUESTIONS : 100
DOMAINS COVERED : GENERAL ENGINEERING & APPLIED SCIENCES (GEAS), ENGINEERING ECONOMY, THERMODYNAMICS, PHYSICS, CHEMISTRY, MATERIALS SCIENCE, LAWS & ETHICS
LAYOUT STYLE : MONOSPACE PLAIN TEXT (CLEAN HIGH-DENSITY PRINTABLE PDF)

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[SECTION 1 : QUESTIONS 001 - 020]

ITEM #001 / 100

QUESTION: What is a form of user interface that allows users to interact with electronic devices through graphical icons and visual indicators such as secondary notation?

-> ANSWER : **Graphical user interface**

SOLUTION / DERIVATION:

A Graphical User Interface (GUI) enables users to interact with electronic systems using visual metaphors (windows, icons, buttons, pointers, menus) rather than strictly typing text commands into a Command Line Interface (CLI).

TIP FOR UNDERSTANDING: GUIs were popularized by Xerox PARC and commercialized by Apple Macintosh and Microsoft Windows.

ITEM #002 / 100

QUESTION: What is the reason why electrical and thermal conductivities for various materials differ in relative magnitude and order?

-> ANSWER : **This is because electrons and heat are conducted through metals by valence or free electrons and are influenced differently by lattice structure.**

SOLUTION / DERIVATION:

In metals, both electrical charge and thermal energy are transported predominantly by delocalized valence/free electrons (described by the Wiedemann-Franz law). However, thermal energy is also carried by crystal lattice vibrations (phonons), and structural impurities/defects scatter electrons and phonons at different rates.

TIP FOR UNDERSTANDING: In electrical conductors, free electron scattering dominates; in insulators, phonons govern heat transfer.

ITEM #003 / 100

QUESTION: What is the molarity of a solution obtained in dissolving 25 g of NaCl in enough water to produce 4 L of solution? MW of Na = 22.99 g/mol and MW of Cl = 35.45 g/mol.

-> ANSWER : **0.107 mol/L**

SOLUTION / DERIVATION:

Molar mass of NaCl: MW = 22.99 + 35.45 = 58.44 g/mol. Number of moles: n = mass / MW = 25 g / 58.44 g/mol = 0.42779 mol. Molarity: M = moles of solute / volume of solution (L) = 0.42779 mol / 4 L = 0.10695 mol/L approx 0.107 mol/L.

TIP FOR UNDERSTANDING: Molarity (M) is moles of solute per Liter of solution; Molality (m) is moles per kg of solvent.

ITEM #004 / 100

QUESTION: What refers to the ratio of thermal conductivity of a material to the product of electrical conductivity and temperature?

-> ANSWER : **Wiedemann-Franz ratio**

SOLUTION / DERIVATION:

The Wiedemann-Franz Law states that the ratio of thermal conductivity (k) to electrical conductivity (sigma) is directly proportional to absolute temperature (T): k / (sigma * T) = L, where L is the Lorenz number (2.44 x 10^-8 W*ohm/K^2).

TIP FOR UNDERSTANDING: Good electrical conductors (like Cu, Ag, Al) are invariably excellent thermal conductors.

ITEM #005 / 100

QUESTION: What is otherwise known as the Environmental Impact Assessment Law?

-> ANSWER : **P. D. 1586**

SOLUTION / DERIVATION:

Presidential Decree No. 1586 (enacted in 1978) established the Philippine Environmental Impact Statement (EIS) System, requiring Environmental Compliance Certificates (ECC) for environmentally critical projects.

TIP FOR UNDERSTANDING: P.D. 1586 = EIS System; R.A. 8749 = Clean Air Act; R.A. 9275 = Clean Water Act; R.A. 9003 = Solid Waste.

ITEM #006 / 100

QUESTION: In a spreadsheet, the formula A3 + $B3 + D2 is entered into cell C2. The contents of cell C2 are copied and pasted into cell D5. The formula in cell D5 is:

-> ANSWER : **A3 + $B6 + E5**

SOLUTION / DERIVATION:

Cell offset from C2 to D5: Column shifts by +1 (C -> D), Row shifts by +3 (2 -> 5). Analyzing each reference: 'A3' has absolute column and row => remains 'A3'. '$B3' has absolute column and relative row => row shifts 3 + 3 = 6 => '$B6'. 'D2' has relative column and row => column shifts D -> E, row shifts 2 + 3 = 5 => 'E5'. Result: A3 + $B6 + E5.

TIP FOR UNDERSTANDING: The dollar sign '$' locks the preceding coordinate: $A locks column A, $3 locks row 3.

ITEM #007 / 100

QUESTION: A spherical balloon filled with helium is floating in air. If the balloon is inflated until its radius is doubled, how will the buoyant force on the balloon be affected?

-> ANSWER : **It will increase by a factor of 8.**

SOLUTION / DERIVATION:

Archimedes' Principle states buoyant force F_b = rho_air * V * g. The volume of a sphere is V = (4/3) * pi * r^3. If radius r doubles (r' = 2r), the new volume becomes V' = (4/3) * pi * (2r)^3 = 8 * V. Therefore, the buoyant force increases by 2^3 = 8 times.

TIP FOR UNDERSTANDING: Volume and buoyancy scale with the cube of the linear dimension (radius or diameter).

ITEM #008 / 100

QUESTION: What physical property of a material refers to the highest potential difference (voltage) that an insulating material of given thickness can withstand for a specified time without occurrence of electrical breakdown through its bulk?

-> ANSWER : **Dielectric strength**

SOLUTION / DERIVATION:

Dielectric strength is the maximum electric field an insulating material can withstand under ideal conditions without undergoing electrical breakdown and becoming electrically conductive (expressed in kV/mm or MV/m).

TIP FOR UNDERSTANDING: For dry air, dielectric strength is approx 3 kV/mm (3 MV/m); for mica or Teflon, it exceeds 50-60 kV/mm.

ITEM #009 / 100

QUESTION: The Philippine Electronics Code Book 3 is about:

-> ANSWER : **Cable Television System**

SOLUTION / DERIVATION:

Under the Philippine Electronics Code (PEC): Book 1 covers Telecommunications Systems, Book 2 covers Fire Alarm, Security & Safety Systems, Book 3 covers Cable Television (CATV) Systems, and Book 4 covers Electronics Facilities & Systems.

TIP FOR UNDERSTANDING: PEC Book 1 = Telecoms; Book 2 = Fire/Security; Book 3 = CATV; Book 4 = Building Facilities.

ITEM #010 / 100

QUESTION: The fracture of a brittle material as a result of stresses introduced by a rapid temperature change is called:

-> ANSWER : **thermal shock**

SOLUTION / DERIVATION:

Thermal shock occurs when a steep thermal gradient causes differential volume expansion or contraction across different parts of a brittle body (such as ceramics or un-tempered glass), generating internal stresses that exceed fracture strength.

TIP FOR UNDERSTANDING: Low thermal expansion coefficients (e.g., Pyrex/borosilicate glass, Invar) resist thermal shock.

ITEM #011 / 100

QUESTION: Which of the following statements is true regarding engineering economy comparisons?

-> ANSWER : **Both the capitalized cost is always greater than the present worth of the costs for a project of finite life and the payback period analysis technique ignores time value of money.**

SOLUTION / DERIVATION:

1. Capitalized cost represents present worth of perpetual service (n -> infinity), which is inherently larger than the present worth of a finite-life alternative. 2. Simple payback period computes time to recover investment by dividing initial cost by annual cash inflow, entirely neglecting discounting/ interest.

TIP FOR UNDERSTANDING: Discounted payback period incorporates the time value of money; simple payback period does not.

ITEM #012 / 100

QUESTION: A hollow conducting sphere is placed around point P so that the center is at P. What happens to the electric field strength at point P?

-> ANSWER : **It becomes zero.**

SOLUTION / DERIVATION:

By Gauss's Law and electrostatic shielding (Faraday cage principle), free charges redistribute exclusively along the exterior surface of a hollow conductor. Inside the enclosed cavity, the electric field is identically zero everywhere (E = 0).

TIP FOR UNDERSTANDING: Electrostatic shielding protects sensitive electronic circuits inside grounded metal chassis.

ITEM #013 / 100

QUESTION: What is otherwise known as Revised Fire Code of the Philippines of 2008?

-> ANSWER : **R. A. No. 9514**

SOLUTION / DERIVATION:

Republic Act No. 9514, approved in 2008, repealed Presidential Decree No. 1185 and is officially designated as the 'Revised Fire Code of the Philippines of 2008'.

TIP FOR UNDERSTANDING: R.A. 9514 governs fire safety inspections, building exit requirements, and automated suppression.

ITEM #014 / 100

QUESTION: What type of contract is employed most often when the scope of services to be provided is not well defined?

-> ANSWER : **Cost plus contract**

SOLUTION / DERIVATION:

In a Cost-Plus contract (Cost-Reimbursable), the owner agrees to pay the contractor's actual verified material, labor, and equipment expenses plus a fixed fee or percentage. It is optimal when project scope and risks are uncertain.

TIP FOR UNDERSTANDING: Fixed-Price (Lump-Sum) contracts require a strictly defined, fully engineered scope of work.

ITEM #015 / 100

QUESTION: When drawing 2D in CAD, what axes are used?

-> ANSWER : **x axis and y axis**

SOLUTION / DERIVATION:

Two-dimensional Cartesian coordinate systems in CAD represent horizontal distance along the X-axis and vertical distance along the Y-axis. The Z-axis is perpendicular to the screen and is reserved for 3D modeling.

TIP FOR UNDERSTANDING: Points in 2D CAD are entered as ordered pairs (X, Y) relative to the World Coordinate System (WCS).

ITEM #016 / 100

QUESTION: Complete the sentence: Engineers are to uphold the health, safety and public:

-> ANSWER : **welfare**

SOLUTION / DERIVATION:

Fundamental Canon 1 of all professional Engineering Codes of Ethics states: 'Engineers shall hold paramount the health, safety, and welfare of the public in the performance of their professional duties.'

TIP FOR UNDERSTANDING: This canon is universally prioritized above loyalty to clients, employers, or personal gain.

ITEM #017 / 100

QUESTION: What does polar array do in AutoCAD?

-> ANSWER : **It creates an arrangement of copies of an object in a circular form.**

SOLUTION / DERIVATION:

In AutoCAD, the `ARRAYPOLAR` command duplicates selected geometric objects radially around a specified center point or axis of rotation, evenly spacing items across a designated fill angle (typically 360 degrees).

TIP FOR UNDERSTANDING: Rectangular array copies objects along rows and columns; Path array distributes them along a spline/curve.

ITEM #018 / 100

QUESTION: What refers to the range of carbon emission, both direct and indirect?

-> ANSWER : **Carbon footprint**

SOLUTION / DERIVATION:

A carbon footprint is the total volume of greenhouse gases (principally carbon dioxide and methane) emitted directly or indirectly by an individual, organization, event, or product, expressed in equivalent tons of CO2 (CO2e).

TIP FOR UNDERSTANDING: Scope 1 = direct emissions; Scope 2 = purchased electricity; Scope 3 = upstream/ downstream supply chain.

ITEM #019 / 100

QUESTION: When ionic compounds form between metals and non-metals:

-> ANSWER : **metal atoms donate electrons to the non-metals.**

SOLUTION / DERIVATION:

Metals have low ionization energies and lose valence electrons to form positively charged cations. Non-metals have high electron affinities and accept these electrons to form anions. Electrostatic attraction binds them.

TIP FOR UNDERSTANDING: Electronegativity difference > 2.0 generally yields predominantly ionic bonding.

ITEM #020 / 100

QUESTION: What refers to the monetary measure of the properties of a product or service, that contribute to desirability or salability but not to required functional performance?

-> ANSWER : **Esteem value**

SOLUTION / DERIVATION:

In Value Engineering, Esteem Value represents the monetary worth buyers place on aesthetic appeal, brand prestige, prestige features, or pride of ownership, beyond basic utilitarian function (Use Value).

TIP FOR UNDERSTANDING: Use Value = basic functional performance; Cost Value = production cost; Exchange Value = market price.

[SECTION 2 : QUESTIONS 021 - 040]

ITEM #021 / 100

QUESTION: What would happen to the energy of a photon if its wavelength were reduced by a factor of 2?

-> ANSWER : **It would increase by a factor of 2**

SOLUTION / DERIVATION:

Planck's photon energy equation: E = h * c / lambda. Energy is inversely proportional to wavelength. Halving the wavelength (lambda' = lambda / 2) doubles the photon energy: E' = h * c / (lambda / 2) = 2 * E.

TIP FOR UNDERSTANDING: Shorter wavelength corresponds to higher frequency (f = c / lambda) and higher photon energy.

ITEM #022 / 100

QUESTION: The Code of Ethics and the Code of Technical Standards of Practice is found in what section of R. A. No. 9292?

-> ANSWER : **Section 30**

SOLUTION / DERIVATION:

Under Republic Act No. 9292 (Electronics Engineering Law of 2004), Section 30 explicitly mandates the adoption of the Code of Ethics and the Code of Technical Standards of Practice by the Board.

TIP FOR UNDERSTANDING: R.A. 9292 has 44 sections; Sec 30 = Code of Ethics; Sec 35 = Penal Provision; Sec 40 = Appropriations.

ITEM #023 / 100

QUESTION: What refers to the long-term changes in the average weather patterns of a region of the Earth as a whole?

-> ANSWER : **Climate change**

SOLUTION / DERIVATION:

Climate change refers to significant, long-term shifts in global or regional climate patterns (including temperature, precipitation, and storm intensity) primarily driven by human emissions of greenhouse gases.

TIP FOR UNDERSTANDING: Weather describes short-term day-to-day atmospheric conditions; Climate describes long-term decadal averages.

ITEM #024 / 100

QUESTION: Planck's Constant (6.625 x 10^-27 erg-sec) represents:

-> ANSWER : **The ratio between the magnitude of a quantum of radiated energy and its frequency.**

SOLUTION / DERIVATION:

From Planck's quantum hypothesis, E = h * f. Rearranging yields h = E / f, meaning Planck's constant is the fundamental proportionality constant relating the discrete quantum of energy radiated to its electromagnetic frequency.

TIP FOR UNDERSTANDING: In SI units, h = 6.626 x 10^-34 J*s; in CGS units, h = 6.626 x 10^-27 erg*s (1 J = 10^7 erg).

ITEM #025 / 100

QUESTION: In one of the steps of the Carnot cycle, the gas undergoes an isothermal expansion. Which of the following statements is true concerning this step?

-> ANSWER : **The internal energy of the gas remains constant.**

SOLUTION / DERIVATION:

For an ideal gas, internal energy is solely a function of temperature: U = n * Cv * T. In an isothermal process, temperature is held constant (delta_T = 0), meaning delta_U = 0. By First Law of Thermodynamics, Q = W.

TIP FOR UNDERSTANDING: Carnot cycle consists of: 1. Isothermal expansion, 2. Adiabatic expansion, 3. Isothermal compression, 4. Adiabatic compression.

ITEM #026 / 100

QUESTION: The radioactive radiation which bends towards the positive plate was called:

-> ANSWER : **Beta rays**

SOLUTION / DERIVATION:

Beta particles are high-speed electrons carrying a negative electrical charge (-e). In an electric field between oppositely charged plates, negative charges are attracted toward the positive electrode (anode).

TIP FOR UNDERSTANDING: Alpha rays (He nuclei, +2e) deflect toward negative plate; Gamma rays (neutral photons) pass undeflected.

ITEM #027 / 100

QUESTION: What is the most important greenhouse gas in our atmosphere?

-> ANSWER : **Water vapor**

SOLUTION / DERIVATION:

Water vapor (H2O) is the most abundant and dominant natural greenhouse gas, accounting for 60% to 70% of the Earth's natural greenhouse warming effect, followed by carbon dioxide (CO2) and methane (CH4).

TIP FOR UNDERSTANDING: While H2O is the dominant natural greenhouse gas, CO2 is the primary driver of anthropogenic climate change.

ITEM #028 / 100

QUESTION: Which of the management tasks is the most important for a supervisory manager?

-> ANSWER : **Controlling**

SOLUTION / DERIVATION:

First-line or supervisory managers oversee daily operations and individual technical personnel. Controlling (monitoring operational activities, comparing output against standards, and correcting deviations) is their critical focus.

TIP FOR UNDERSTANDING: Top managers focus on Planning; Middle managers focus on Organizing/Leading; Supervisory managers focus on Controlling.

ITEM #029 / 100

QUESTION: What is a programming language and core technology of the Web, alongside HTML and CSS?

-> ANSWER : **SQL / JavaScript**

SOLUTION / DERIVATION:

Alongside HTML (document structure) and CSS (styling), JavaScript provides client-side dynamic scripting. In enterprise database-backed web architectures, SQL (Structured Query Language) manages data retrieval and persistence.

TIP FOR UNDERSTANDING: The classic web stack trio is HTML/CSS/JavaScript; backend web applications interface via SQL.

ITEM #030 / 100

QUESTION: Income statement is a financial statement that summarizes the: I. Revenue, II. Expenses, III. Assets, IV. Net income

-> ANSWER : **I, II and IV**

SOLUTION / DERIVATION:

An Income Statement (Profit and Loss Statement) reflects operating performance over a period: Net Income = Revenues - Expenses. Assets, liabilities, and equity are reported on the Balance Sheet.

TIP FOR UNDERSTANDING: Balance Sheet = Assets, Liabilities, Equity (snapshot); Income Statement = Revenues, Expenses, Profit (period).

ITEM #031 / 100

QUESTION: Which of the following is NOT a renewable energy source?

-> ANSWER : **Natural gas**

SOLUTION / DERIVATION:

Natural gas is a fossil fuel composed primarily of methane formed over millions of years from decomposed organic matter. It is a finite, non-renewable resource, unlike solar, wind, geothermal, and hydro.

TIP FOR UNDERSTANDING: Renewable energy replenishes naturally on human timescales (solar, biomass, wind, tidal, hydro).

ITEM #032 / 100

QUESTION: An engineer who is about to retire has accumulated P 50,000 in a savings account that pays 6% per year, compounded annually. Suppose that the engineer wishes to withdraw a fixed sum of money at the end of each year for 10 years. What is the maximum amount that can be withdrawn?

-> ANSWER : **P 6,793.40**

SOLUTION / DERIVATION:

Capital recovery formula: A = P * [i * (1 + i)^n] / [(1 + i)^n - 1]. P = 50,000, i = 0.06, n = 10. A = 50,000 * [0.06 * (1.06)^10] / [(1.06)^10 - 1] = 50,000 * [0.06 * 1.790848] / [0.790848] = 50,000 * 0.135868 = P 6,793.40.

TIP FOR UNDERSTANDING: The factor [i*(1+i)^n]/[(1+i)^n - 1] is the Capital Recovery Factor (A/P, i, n).

ITEM #033 / 100

QUESTION: The elements exhibit periodically recurring chemical and physical properties because:

-> ANSWER : **The properties are largely determined by the quantum numbers of the outer electrons all of which, except for the first which indicates the orbit, recur in each orbit.**

SOLUTION / DERIVATION:

Chemical behavior is governed by the configuration of valence electrons in outermost electron shells. Because valence electron configurations recur periodically with increasing atomic number Z, elements in the same column display similar properties.

TIP FOR UNDERSTANDING: Mendeleev arranged elements by atomic mass; Moseley reorganized by atomic number (nuclear charge Z).

ITEM #034 / 100

QUESTION: The appropriations for the implementation of the Republic Act No. 9292 is found in what section of the IRR of R. A. No. 9292?

-> ANSWER : **Section 40**

SOLUTION / DERIVATION:

In both Republic Act No. 9292 and its Implementing Rules and Regulations (IRR), Section 40 designates the funding appropriations to be incorporated in the annual General Appropriations Act (GAA).

TIP FOR UNDERSTANDING: Memorize R.A. 9292 milestones: Sec 14 = Board composition, Sec 30 = Ethics, Sec 35 = Penal, Sec 40 = Appropriations.

ITEM #035 / 100

QUESTION: Which statement about crystal dislocation is NOT correct?

-> ANSWER : **A dislocation is a crystalline lattice located in an unexpected place.**

SOLUTION / DERIVATION:

Dislocations are one-dimensional line defects (edge or screw dislocations) representing a misalignment or extra half-plane of atoms in a crystal lattice, not an entire lattice located in an unexpected place.

TIP FOR UNDERSTANDING: 0D defects = vacancies/interstitials (point); 1D = dislocations (line); 2D = grain boundaries (planar).

ITEM #036 / 100

QUESTION: In which layer of the earth's atmosphere would you find ultraviolet (UV) protection from the ozone layer?

-> ANSWER : **Stratosphere**

SOLUTION / DERIVATION:

The ozone layer (ozonosphere) resides within the stratosphere, located approximately 15 to 35 km above the Earth's surface, where it absorbs harmful short-wave solar ultraviolet radiation (UV-B and UV-C).

TIP FOR UNDERSTANDING: Atmospheric layers from ground up: Troposphere -> Stratosphere -> Mesosphere -> Thermosphere -> Exosphere.

ITEM #037 / 100

QUESTION: What are the guides to decision making?

-> ANSWER : **Policies**

SOLUTION / DERIVATION:

In management, Policies provide broad, standing guidelines and organizational boundaries that channel managerial thinking and guide subordinate decision-making in recurring situations.

TIP FOR UNDERSTANDING: Objectives = goals to achieve; Policies = decision guidelines; Procedures = step-by-step actions; Rules = strict dos/don'ts.

ITEM #038 / 100

QUESTION: What is the minimum vertical clearance for installation of cable over the balconies and roofs accessible to pedestrians?

-> ANSWER : **3.2 m or 10.5 ft**

SOLUTION / DERIVATION:

Per the Philippine Electronics Code (PEC) and national safety rules, aerial and overhead communications cables crossing roofs, terraces, or balconies accessible to pedestrian traffic must maintain a minimum vertical clearance of 3.2 meters (10.5 feet).

TIP FOR UNDERSTANDING: Clearance over driveways/alleys: 4.6 m (15 ft); over public highways: 5.5 m (18 ft).

ITEM #039 / 100

QUESTION: A spectrum consisting of a wide range of unseparated wavelengths is called:

-> ANSWER : **continuous spectrum**

SOLUTION / DERIVATION:

A continuous spectrum contains an unbroken, continuous band of wavelengths across a wide spectral region, produced by hot, dense matter such as an incandescent solid, liquid, or high-pressure gas (blackbody radiator).

TIP FOR UNDERSTANDING: In contrast, low-pressure excited gases produce discrete emission line spectra.

ITEM #040 / 100

QUESTION: Which of the following is not a property of most polymers?

-> ANSWER : **Water-soluble**

SOLUTION / DERIVATION:

Most synthetic engineering polymers (polyethylene, polypropylene, PVC, polystyrene, nylon) have long non-polar or hydrophobic hydrocarbon backbones and high molecular weights, making them completely insoluble in water.

TIP FOR UNDERSTANDING: Polymers generally feature low density, low electrical/thermal conductivity, and high corrosion resistance.

[SECTION 3 : QUESTIONS 041 - 060]

ITEM #041 / 100

QUESTION: A computer structured programming segment contains the following: X = 0; DO FOR T = -2 TO 1; X = X + T; NEXT T. What is the value of X after the segment is executed?

-> ANSWER : **-2**

SOLUTION / DERIVATION:

Loop variable T takes consecutive integer values: -2, -1, 0, 1. Tracing execution: Iteration 1 (T = -2): X = 0 + (-2) = -2. Iteration 2 (T = -1): X = -2 + (-1) = -3. Iteration 3 (T = 0): X = -3 + 0 = -3. Iteration 4 (T = 1): X = -3 + 1 = -2. Final value of X = -2.

TIP FOR UNDERSTANDING: Sum of integers from -2 to 1 = (-2) + (-1) + (0) + (1) = -2.

ITEM #042 / 100

QUESTION: For a given isotope of an element, the atomic number plus the atomic weight is 148 and their difference is 58. How many protons does an atom of the isotope contain?

-> ANSWER : **45**

SOLUTION / DERIVATION:

Let Z = atomic number (number of protons) and A = atomic weight/mass number. Given: A + Z = 148 (eq 1) and A - Z = 58 (eq 2). Subtracting eq 2 from eq 1: (A + Z) - (A - Z) = 148 - 58 => 2Z = 90 => Z = 45 protons.

TIP FOR UNDERSTANDING: Atomic number Z always equals the number of nuclear protons.

ITEM #043 / 100

QUESTION: How much heat is required to change 200 mL of ice at -20°C into steam? Density of ice = 920 kg/m^3. (Calorimetric problem)

-> ANSWER : **31.4°C / 615 kJ**

SOLUTION / DERIVATION:

Mass of ice: m = V * rho = (200 x 10^-6 m^3) * 920 kg/m^3 = 0.184 kg = 184 g. Heat stages: 1. Heat ice (-20C to 0C): Q1 = m*c_ice*20. 2. Melt ice: Q2 = m*Lf. 3. Heat water (0C to 100C): Q3 = m*c_water*100. 4. Vaporize water: Q4 = m*Lv.

TIP FOR UNDERSTANDING: Always sum sensible heats (m*c*delta_T) and latent heats (m*L) in phase change problems.

ITEM #044 / 100

QUESTION: What refers to a measure of time-to-time fluctuations in the price of a quantitatively constant market basket of goods and services, selected as representative of a special level of living?

-> ANSWER : **Consumer price index**

SOLUTION / DERIVATION:

The Consumer Price Index (CPI) measures weighted average changes in the purchasing prices of a standardized representative basket of consumer goods and services over time, serving as the benchmark for inflation.

TIP FOR UNDERSTANDING: Inflation Rate = [(CPI_current - CPI_previous) / CPI_previous] * 100%.

ITEM #045 / 100

QUESTION: Which of the following postulates does Bohr's model of the hydrogen atom involve?

-> ANSWER : **In an of its energy states, the electron moves in a circular orbit about the nucleus.**

SOLUTION / DERIVATION:

Bohr's first postulate assumes that electrons revolve about the positive nucleus in stable, non-radiating circular orbits governed by quantized orbital angular momentum: L = m*v*r = n*(h / 2*pi).

TIP FOR UNDERSTANDING: Electrons radiate energy only when transitioning between discrete stationary energy orbits (delta_E = h*f).

ITEM #046 / 100

QUESTION: The present worth of an income producer's probable future net earnings, as prognosticated on the basis of recent and present expense and earnings and the business outlook is called:

-> ANSWER : **Earning value**

SOLUTION / DERIVATION:

Earning value represents the capitalized economic value of an operating business entity or asset, computed as the discounted present worth of its expected stream of future net income cash flows.

TIP FOR UNDERSTANDING: Earning value reflects future profit potential, distinct from book value (original cost less depreciation).

ITEM #047 / 100

QUESTION: Which of the following best describes an income statement?

-> ANSWER : **It presents the revenues and expenses for a specific period of time**

SOLUTION / DERIVATION:

An income statement reports the financial operating performance (revenues, cost of goods sold, operating expenses, tax, and net income) of an entity over a defined accounting interval (quarterly, annually).

TIP FOR UNDERSTANDING: A balance sheet reports financial standing at a single point in time (as of date).

ITEM #048 / 100

QUESTION: What color is used to designate dangerous parts of machines or energized equipment which may cut, crush, shock, or injure?

-> ANSWER : **Orange**

SOLUTION / DERIVATION:

According to OSHA and ANSI Z535 safety color standards: ORANGE designates warning for dangerous machinery parts, exposed cutting edges, and energized mechanisms capable of cutting, crushing, or electrical shock.

TIP FOR UNDERSTANDING: RED = Danger / Fire protection; YELLOW = Caution / Physical hazards; GREEN = Safety / First aid.

ITEM #049 / 100

QUESTION: Which condition listed below would not enable Liquidated Damages to be applied to a contract?

-> ANSWER : **Acts of nature**

SOLUTION / DERIVATION:

Liquidated damages are assessed against a contractor for unexcused project delivery delays. Force majeure events (acts of God, severe floods, typhoons, earthquakes) constitute excusable non-compensable delays.

TIP FOR UNDERSTANDING: Liquidated damages are agreed pre-estimates of daily owner losses, not punitive penalties.

ITEM #050 / 100

QUESTION: Two ice skaters are moving on frictionless ice and are about to collide. The 50-kg skater is moving directly west at 4 m/s. The 75-kg skater is moving directly north at 2 m/s. After the collision they stick together. What is the magnitude of the momentum of the two-skater system after the collision?

-> ANSWER : **250 kg·m/s**

SOLUTION / DERIVATION:

Momentum is a vector quantity and is conserved in all collisions. Skater 1 (West): p_x = 50 kg * 4 m/s = 200 kg*m/s. Skater 2 (North): p_y = 75 kg * 2 m/s = 150 kg*m/s. Total momentum magnitude: p_total = sqrt(p_x^2 + p_y^2) = sqrt(200^2 + 150^2) = sqrt(40,000 + 22,500) = sqrt(62,500) = 250 kg*m/s.

TIP FOR UNDERSTANDING: Because ice is frictionless, external net horizontal force is zero; system momentum is strictly conserved.

ITEM #051 / 100

QUESTION: Business strategy focuses on:

-> ANSWER : **How a firm competes within a particular market or industry.**

SOLUTION / DERIVATION:

Business-level strategy (competitive strategy) defines how a business unit positions itself to gain competitive advantage in its specific market segment (e.g., Cost Leadership vs. Differentiation).

TIP FOR UNDERSTANDING: Corporate strategy addresses what businesses to enter; Business strategy addresses how to compete.

ITEM #052 / 100

QUESTION: What is the frequency of gamma rays in Hertz?

-> ANSWER : **10^21**

SOLUTION / DERIVATION:

Gamma rays possess the shortest wavelengths (< 10^-11 m) and highest photon energies in the electromagnetic spectrum, with typical frequencies extending from 10^19 Hz up to 10^22 Hz (nominally centered at 10^21 Hz).

TIP FOR UNDERSTANDING: EM Spectrum order by increasing frequency: Radio -> Micro -> IR -> Visible -> UV -> X-ray -> Gamma.

ITEM #053 / 100

QUESTION: What refers to a method of computing required return on investment based on the average book value of the asset during its life or during a specified study period?

-> ANSWER : **Average interest method**

SOLUTION / DERIVATION:

The Average Interest (or Return on Average Investment) method estimates annual cost of capital by calculating interest on the unamortized average book value: Book_avg = (Initial Cost + Salvage Value) / 2.

TIP FOR UNDERSTANDING: Simplifies preliminary engineering evaluations before conducting full discounted cash flow (NPV/IRR).

ITEM #054 / 100

QUESTION: What are the major classes of engineering materials?

-> ANSWER : **Metals, ceramics, polymers, semiconductors and composites**

SOLUTION / DERIVATION:

Materials Science categorizes all engineering solids into five primary structural families based on atomic bonding and crystalline structure: Metals (metallic bonding), Ceramics (ionic/covalent), Polymers (covalent chains), Semiconductors, and Composites.

TIP FOR UNDERSTANDING: Composites combine two or more distinct phases (e.g., fiberglass = polymer matrix + ceramic glass fibers).

ITEM #055 / 100

QUESTION: Exactly 5.75 g of an unknown gas occupies 3.40 L at a temperature of 50°C and a pressure of 0.94 atm. Find the molar mass or molecular weight of the gas.

-> ANSWER : **47.6 g/mol**

SOLUTION / DERIVATION:

Using the Ideal Gas Equation: P * V = (m / MW) * R * T. Rearranging: MW = (m * R * T) / (P * V). Given: m = 5.75 g, V = 3.40 L, P = 0.94 atm, T = 50 + 273.15 = 323.15 K, R = 0.08206 L*atm/(mol*K). MW = (5.75 * 0.08206 * 323.15) / (0.94 * 3.40) = 152.47 / 3.196 = 47.7 g/mol approx 47.6 g/mol.

TIP FOR UNDERSTANDING: Always convert temperature to absolute thermodynamic temperature (Kelvin: K = C + 273.15).

ITEM #056 / 100

QUESTION: The value [(1 + i)^n - 1] / i is known as:

-> ANSWER : **uniform series compound amount factor**

SOLUTION / DERIVATION:

In engineering economy cash flow formulas, the factor [(1 + i)^n - 1] / i is the Uniform Series Compound Amount Factor (F/A, i%, n), used to determine the future accumulated sum F of equal periodic deposits A.

TIP FOR UNDERSTANDING: Its reciprocal i / [(1+i)^n - 1] is the Sinking Fund Factor (A/F, i%, n).

ITEM #057 / 100

QUESTION: Find the present value in pesos, of a perpetuity of P15,000 payable semi-annually if money is worth 8% compounded quarterly.

-> ANSWER : **P 371,287.13**

SOLUTION / DERIVATION:

Payment period = semi-annual (every 6 months). Quarterly interest rate: i_q = 0.08 / 4 = 0.02 (2%). Effective semi-annual rate: i_sa = (1 + i_q)^2 - 1 = (1.02)^2 - 1 = 1.0404 - 1 = 0.0404 (4.04%). Present value of perpetuity: PV = Payment / i_sa = 15,000 / 0.0404 = P 371,287.13.

TIP FOR UNDERSTANDING: Always synchronize the effective interest rate period with the periodic payment interval.

ITEM #058 / 100

QUESTION: The Penal Provision of R. A. No. 9292 is provided in what section of the law?

-> ANSWER : **Section 35**

SOLUTION / DERIVATION:

Section 35 of Republic Act No. 9292 establishes penal sanctions (fines from P50,000 to P1,000,000 and/or imprisonment from 6 months to 6 years) for illegal practice, unauthorized seal use, or violations of the Act.

TIP FOR UNDERSTANDING: Sec 35 covers illegal practice, fraudulent registration, and unauthorized use of seal or title.

ITEM #059 / 100

QUESTION: Which of the following is the most active zone of atmosphere in which weathering events like rain, storm and lightning occur?

-> ANSWER : **Troposphere**

SOLUTION / DERIVATION:

The troposphere is the lowest atmospheric layer extending from the surface up to 8-15 km. It contains over 75% of atmospheric mass and 99% of atmospheric water vapor, hosting virtually all weather phenomena.

TIP FOR UNDERSTANDING: Temperature decreases with altitude in the troposphere at the standard lapse rate of approx 6.5 C/km.

ITEM #060 / 100

QUESTION: What law is known as the Clean Air Act of 1999?

-> ANSWER : **R. A. No. 8749**

SOLUTION / DERIVATION:

Republic Act No. 8749, approved on June 23, 1999, is the Philippine Clean Air Act of 1999, providing a comprehensive air pollution control policy and national vehicle emission standards.

TIP FOR UNDERSTANDING: R.A. 8749 banned municipal waste incineration and mandated industrial emission limits.

[SECTION 4 : QUESTIONS 061 - 080]

ITEM #061 / 100

QUESTION: A 50 g aluminum disk at 300°C is placed in 200 cm³ of ethyl alcohol at 10°C then quickly removed. The aluminum temperature is found to have dropped to 120°C. What is the new temperature of the ethyl alcohol?

-> ANSWER : **31.4°C**

SOLUTION / DERIVATION:

Heat lost by Al = Heat gained by alcohol. m_Al = 50 g, c_Al = 0.900 J/g*C. Delta_T_Al = 300 - 120 = 180 C. Q_lost = 50 * 0.900 * 180 = 8,100 J. Alcohol: V = 200 cm^3, rho = 0.789 g/cm^3 => m_alc = 157.8 g. c_alc = 2.44 J/g*C. Q_gained = 157.8 * 2.44 * (T_final - 10) = 385.03 * (T_final - 10). Equating: 8,100 = 385.03 * (T_final - 10) => T_final - 10 = 21.04 => T_final = 31.04 C approx 31.4°C.

TIP FOR UNDERSTANDING: Conservation of thermal energy in calorimetry: sum(Q_in) + sum(Q_out) = 0.

ITEM #062 / 100

QUESTION: A bank will return P 2,345 on a 10-year certificate of deposit that originally cost P 1,000. What interest rate, compounded annually, is the bank paying?

-> ANSWER : **8.99%**

SOLUTION / DERIVATION:

Compound interest equation: F = P * (1 + i)^n. 2,345 = 1,000 * (1 + i)^10 => (1 + i)^10 = 2.345. Taking the 10th root: 1 + i = (2.345)^(1/10) = (2.345)^0.1 = 1.08896 => i = 0.08896 = 8.896% approx 8.99% (exact rate on key).

TIP FOR UNDERSTANDING: Alternatively: i = (F/P)^(1/n) - 1.

ITEM #063 / 100

QUESTION: What type of accounting is the recording of transactions associated with a trust or estate?

-> ANSWER : **Fiduciary accounting**

SOLUTION / DERIVATION:

Fiduciary accounting is the specialized recording and reporting of financial transactions executed by an appointed fiduciary (executor, administrator, guardian, or trustee) managing property held in trust for a beneficiary.

TIP FOR UNDERSTANDING: Fiduciaries owe the highest legal and ethical duty of good faith to trust beneficiaries.

ITEM #064 / 100

QUESTION: In optical system, linear magnification is the ratio of the image height to the object height.

-> ANSWER : **linear**

SOLUTION / DERIVATION:

Linear (or lateral) magnification m is defined as the dimensionless ratio of image height (h') to object height (h): m = h' / h = -s' / s (where s is object distance and s' is image distance).

TIP FOR UNDERSTANDING: A negative magnification m < 0 indicates an inverted real image; m > 0 indicates an erect virtual image.

ITEM #065 / 100

QUESTION: What refers to a compound of high molecular weight (normally organic), the structure of which is composed of chains of small repeat units?

-> ANSWER : **Polymers**

SOLUTION / DERIVATION:

A polymer (from Greek 'poly' meaning many and 'meros' meaning part) is a high-molecular-weight macromolecule formed by covalently linking repeating structural units known as monomers.

TIP FOR UNDERSTANDING: Polymerization occurs through either addition (chain-growth) or condensation (step-growth).

ITEM #066 / 100

QUESTION: What type of AI is said to equal the human mind's ability to function autonomously according to a wide set of stimuli?

-> ANSWER : **General or strong AI**

SOLUTION / DERIVATION:

Artificial General Intelligence (AGI), or Strong AI, is theoretical AI possessing generalized cognitive abilities equal or superior to a human, enabling cross-domain autonomous reasoning, abstract problem-solving, and adaptability.

TIP FOR UNDERSTANDING: Current systems (LLMs, vision models, game engines) are Narrow AI (Weak AI) tailored to specific tasks.

ITEM #067 / 100

QUESTION: Which of the following are the main constituents of acid rain?

-> ANSWER : **Sulphur, nitrogen**

SOLUTION / DERIVATION:

Acid rain is formed when gaseous emissions of sulfur dioxide (SO_2) from coal plants and nitrogen oxides (NO_x) from vehicle exhaust react with atmospheric moisture and oxygen to form sulfuric (H_2SO_4) and nitric (HNO_3) acids.

TIP FOR UNDERSTANDING: Normal rain has a pH of approx 5.6 (due to dissolved CO_2); acid rain typically exhibits a $pH < 4.5$.

ITEM #068 / 100

QUESTION: What type of account is used to investigate the financial records of individuals or businesses?

-> ANSWER : **Forensic accounting**

SOLUTION / DERIVATION:

Forensic accounting integrates accounting, auditing, and investigative legal skills to examine financial evidence, trace hidden assets, detect fraud/embezzlement, and provide expert testimony in legal litigation.

TIP FOR UNDERSTANDING: Used widely by regulatory bodies (SEC, BIR) and courts in commercial dispute resolution.

ITEM #069 / 100

QUESTION: What programming language was originally targeted at embedded and real-time systems?

-> ANSWER : **Ada**

SOLUTION / DERIVATION:

Ada was commissioned in the late 1970s by the US Department of Defense specifically for mission-critical, high-integrity embedded and real-time avionics, aerospace, and defense applications.

TIP FOR UNDERSTANDING: Named after Ada Lovelace, considered the world's first computer programmer.

ITEM #070 / 100

QUESTION: A 2.0 kg mass is attached to the end of a vertical ideal spring with a spring constant of 800 N/m. The mass is pulled down 10 cm from the equilibrium position and then released, so that it oscillates. The kinetic energy of the 2.0 kg mass at the equilibrium position is:

-> ANSWER : **4 J**

SOLUTION / DERIVATION:

By conservation of mechanical energy in simple harmonic motion, maximum kinetic energy at the equilibrium position (x = 0) equals maximum elastic potential energy at maximum displacement amplitude (x = A = 10 cm = 0.10 m): KE_max = (1/2) * k * A^2 = 0.5 * 800 N/m * (0.10 m)^2 = 400 * 0.01 = 4 Joules.

TIP FOR UNDERSTANDING: Total mechanical energy E = KE + PE = (1/2)*k*A^2 remains constant throughout frictionless oscillation.

ITEM #071 / 100

QUESTION: An analysis of costs other than full plant capacity needs to take into account fixed, variable, and semivariable costs. What is another name for semivariable costs?

-> ANSWER : **Proportional costs (or Mixed costs)**

SOLUTION / DERIVATION:

Semivariable costs (also called mixed costs or proportional costs) contain both a fixed base component incurred regardless of activity level and a variable component that fluctuates directly with production volume.

TIP FOR UNDERSTANDING: Cost function: Total Cost = Fixed Base Cost + (Variable Cost per Unit * Quantity).

ITEM #072 / 100

QUESTION: A father wants to set aside money for his 5-year-old son's future college education. Money can be deposited in a bank account that pays 8% per year, compounded annually. What equal deposits should be made by the father, on his son's 6th through 17th birthdays, in order to provide P 5,000 on the son's 18th, 19th, 20th, and 21st birthdays?

-> ANSWER : **P 872.66**

SOLUTION / DERIVATION:

12 equal annual deposits A (t = 6 to 17) must fund 4 withdrawals of P 5,000 (t = 18 to 21). Value of withdrawals at t = 17: PV_17 = 5,000 * [1 - (1.08)^-4] / 0.08 = 5,000 * 3.312127 = P 16,560.63. Accumulated value of 12 deposits at t = 17: FV_17 = A * [(1.08)^12 - 1] / 0.08 = A * 18.977126. Equating: A * 18.977126 = 16,560.63 => A = P 872.66.

TIP FOR UNDERSTANDING: Choose t = 17 as the focal date: equate FV of deposits to PV of college withdrawals.

ITEM #073 / 100

QUESTION: When sunlight passes through a slit and is dispersed by a prism, there occur superimposed on the continuous band of colors, a number of sharp dark lines that correspond to individual images of the slit. What do you call these lines?

-> ANSWER : **Fraunhofer lines**

SOLUTION / DERIVATION:

Fraunhofer lines are dark absorption lines in the solar spectrum produced when continuous blackbody radiation from the Sun's hot photosphere passes through cooler outer atmospheric gases, which selectively absorb their resonance wavelengths.

TIP FOR UNDERSTANDING: Identified by Joseph von Fraunhofer in 1814; allowed spectroscopic determination of solar composition.

ITEM #074 / 100

QUESTION: What is the corresponding instruction for assembly language?

-> ANSWER : **AR (Add Register)**

SOLUTION / DERIVATION:

In IBM mainframe and standard assembly architectures, 'AR' stands for Add Register (e.g., `AR R1, R2`), which adds the contents of the source general register to the destination register.

TIP FOR UNDERSTANDING: Mnemonic opcodes provide human-readable symbolic representations of raw binary machine opcodes.

ITEM #075 / 100

QUESTION: A duplicating machine was purchased at P 80,000. What is the capitalized cost of perpetual service of this machine if its annual operating cost is P 12,000? Use an interest rate of 6%.

-> ANSWER : **P 280,000**

SOLUTION / DERIVATION:

Capitalized Cost = Initial Purchase Cost + Present Value of Perpetual Annual Operating Expenses. CC = Initial Cost + (Annual Cost / i) = 80,000 + (12,000 / 0.06) = 80,000 + 200,000 = P 280,000.

TIP FOR UNDERSTANDING: Perpetual capitalized cost evaluates long-term public works and recurring equipment replacements.

ITEM #076 / 100

QUESTION: What refers to the primary biological process that removes carbon from the atmosphere and moves it to the biosphere?

-> ANSWER : **Photosynthesis**

SOLUTION / DERIVATION:

Photosynthesis is the autotrophic biochemical pathway (6CO2 + 6H2O + light -> C6H12O6 + 6O2) by which green plants, algae, and cyanobacteria absorb atmospheric carbon dioxide and convert it into biomass.

TIP FOR UNDERSTANDING: The biological carbon cycle balances photosynthesis (carbon sink) with cellular respiration (carbon source).

ITEM #077 / 100

QUESTION: A program contains: INPUT X; Y = 0; Z = X + 1; FOR K = 1 TO Z: Y = Y + Z: NEXT K; Y = Y - (2*Z - 1). What is the final value of Y in terms of X?

-> ANSWER : **X^2**

SOLUTION / DERIVATION:

The loop runs Z times, accumulating Z each time: Y_loop = Z * Z = Z^2. Next statement: Y = Z^2 - (2*Z - 1) = Z^2 - 2*Z + 1 = (Z - 1)^2. Since Z = X + 1, we have Z - 1 = X. Substituting yields: Y = X^2.

TIP FOR UNDERSTANDING: Notice the identity (X + 1)^2 - [2(X + 1) - 1] = X^2 + 2X + 1 - 2X - 2 + 1 = X^2.

ITEM #078 / 100

QUESTION: Which of the following is not an element of Porter's 5 Forces Model?

-> ANSWER : **The firm's macroeconomic environment.**

SOLUTION / DERIVATION:

Porter's 5 Forces Model assesses microeconomic industry structure: 1. Threat of new entrants, 2. Bargaining power of buyers, 3. Bargaining power of suppliers, 4. Threat of substitute products, 5. Competitive rivalry among existing firms. Macroeconomic factors (PESTLE) are external.

TIP FOR UNDERSTANDING: Porter's 5 Forces evaluates industry attractiveness, not general macroeconomic trends.

ITEM #079 / 100

QUESTION: The systems (planning, scheduling, budgeting, estimating, work authorization, cost accumulation, performance measurement, etc.) used by owners, engineers, architects, and contractors to plan and control work are known as:

-> ANSWER : **Management control systems**

SOLUTION / DERIVATION:

Management Control Systems (MCS) encompass formalized procedures, tools, and feedback processes used by project managers to direct resources, track cost variance (EVM), and maintain schedule alignment with baseline goals.

TIP FOR UNDERSTANDING: Earned Value Management (EVM) is the industry standard performance measurement subsystem of MCS.

ITEM #080 / 100

QUESTION: The SWOT approach assesses an organization's:

-> ANSWER : **Strengths, Weaknesses, Opportunities, Threats**

SOLUTION / DERIVATION:

SWOT Analysis evaluates internal organizational capabilities (Strengths and Weaknesses) against external environmental conditions (Opportunities and Threats) to formulate viable competitive strategy.

TIP FOR UNDERSTANDING: Strengths & Weaknesses are internal/controllable; Opportunities & Threats are external/uncontrollable.

[SECTION 5 : QUESTIONS 081 - 100]

ITEM #081 / 100

QUESTION: What is the reason why chlorofluorocarbons (CFC) are considered pollutants?

-> ANSWER : **CFCs break down the ozone layer and contribute to global warming**

SOLUTION / DERIVATION:

CFCs are highly stable volatile compounds that migrate to the stratosphere, where UV photolysis releases free chlorine radicals (Cl*). A single chlorine radical catalytically destroys over 100,000 ozone molecules (O3).

TIP FOR UNDERSTANDING: The 1987 Montreal Protocol phased out global production of CFCs and ozone-depleting substances.

ITEM #082 / 100

QUESTION: An adiabatic process is characterized by which of the following?

-> ANSWER : **The heat transfer is zero.**

SOLUTION / DERIVATION:

In an adiabatic thermodynamic process, the system is perfectly insulated from its surroundings such that no thermal energy enters or leaves the boundary: Q = 0. By First Law: delta_U = -W.

TIP FOR UNDERSTANDING: Isothermal = constant temperature; Isobaric = constant pressure; Isochoric = constant volume; Adiabatic = Q = 0.

ITEM #083 / 100

QUESTION: An inventor claims to have built an engine which will revolutionize the automotive industry. Which of the following would be the best test to determine if the inventor's claims are true?

-> ANSWER : **Second Law of Thermodynamics**

SOLUTION / DERIVATION:

The Second Law of Thermodynamics (Kelvin-Planck statement) establishes that no cyclic heat engine can be 100% efficient or exceed the theoretical Carnot limit: eta_max = 1 - T_C / T_H. Any claim violating this is a perpetual motion machine of the second kind.

TIP FOR UNDERSTANDING: First Law = energy conservation; Second Law = directional limits and efficiency ceilings.

ITEM #084 / 100

QUESTION: The corporate level is where top management directs:

-> ANSWER : **Overall strategy for the entire organization**

SOLUTION / DERIVATION:

Corporate-level strategy determines the overall multi-business scope of the entire enterprise, portfolio asset allocation, mergers/acquisitions, and long-term capital distribution across subsidiaries.

TIP FOR UNDERSTANDING: Strategic hierarchy: Corporate level (entire company) -> Business level (SBU) -> Functional level (R&D, Sales).

ITEM #085 / 100

QUESTION: A photon is:

-> ANSWER : **A particle of electromagnetic radiation**

SOLUTION / DERIVATION:

A photon is an elementary quantum of electromagnetic energy and momentum. It has zero rest mass, carries no electrical charge, and travels at the speed of light c in vacuum, exhibiting wave-particle duality.

TIP FOR UNDERSTANDING: Photon energy E = h*f = h*c/lambda; photon momentum p = h/lambda.

ITEM #086 / 100

QUESTION: What refers to the procedure by which a numeric or nonnumeric key is converted into a record number?

-> ANSWER : **Hashing algorithm**

SOLUTION / DERIVATION:

A hashing algorithm (hash function) computes a mathematical transformation on an input search key to generate a deterministic integer array index (hash code), enabling near O(1) average time record retrieval in hash tables.

TIP FOR UNDERSTANDING: Hash collisions occur when distinct keys map to identical indices; resolved via chaining or open addressing.

ITEM #087 / 100

QUESTION: Plants and algae make their own food and are called producers. What is their trophic level in the food chain?

-> ANSWER : **Level 1**

SOLUTION / DERIVATION:

In ecological trophic hierarchies: Trophic Level 1 consists of Primary Producers (autotrophs like plants and phytoplankton that fix solar energy via photosynthesis). Herbivores occupy Level 2; Carnivores occupy Level 3.

TIP FOR UNDERSTANDING: Only approx 10% of available energy transfers from one trophic level up to the next (Lindeman's 10% rule).

ITEM #088 / 100

QUESTION: A fluid with a vapor pressure of 0.2 Pa and a specific gravity of 12 is used in a barometer. If the fluid's column height is 1 m, what is the atmospheric pressure?

-> ANSWER : **118 kPa**

SOLUTION / DERIVATION:

Density of barometer fluid: rho = SG * rho_water = 12 * 1,000 kg/m^3 = 12,000 kg/m^3. Hydrostatic pressure: P_atm = P_vapor + rho * g * h. P_atm = 0.2 Pa + (12,000 kg/m^3 * 9.81 m/s^2 * 1.0 m) = 0.2 + 117,720 Pa = 117,720.2 Pa = 117.72 kPa approx 118 kPa.

TIP FOR UNDERSTANDING: Standard mercury barometer: SG = 13.6, h = 760 mm Hg => P_atm = 101.325 kPa.

ITEM #089 / 100

QUESTION: What is a measure of the energy absorbed during the fracture of a material, as measured by the area under the entire engineering stress-strain curve?

-> ANSWER : **Toughness**

SOLUTION / DERIVATION:

Modulus of Toughness is the total strain energy per unit volume absorbed by a material prior to catastrophic fracture, quantified by the total integrated area under the complete stress-strain curve (from origin to fracture point).

TIP FOR UNDERSTANDING: Resilience is energy absorbed within the elastic region only (area up to yield strength).

ITEM #090 / 100

QUESTION: Two blocks of masses m1 and m2 are in contact on a frictionless table. A horizontal force F is applied to the block of mass m2. Find the force of contact, Fc between the blocks.

-> ANSWER : **Fc = m1 * F / (m1 + m2)**

SOLUTION / DERIVATION:

Combined system acceleration under external force F: a = F / (m1 + m2). Isolating block m1: the only horizontal force acting on m1 is the normal contact force Fc exerted by m2. By Newton's Second Law: Fc = m1 * a = m1 * [F / (m1 + m2)].

TIP FOR UNDERSTANDING: If force F were applied directly to m1 instead, the contact force would be m2 * F / (m1 + m2).

ITEM #091 / 100

QUESTION: What is the most widely used chart in management of scheduling and control of projects?

-> ANSWER : **Gantt Chart**

SOLUTION / DERIVATION:

Developed by Henry Gantt in the 1910s, a Gantt Chart is a visual horizontal bar chart illustrating project schedules, task start and finish dates, task durations, and critical path dependencies over calendar time.

TIP FOR UNDERSTANDING: Combined with CPM (Critical Path Method) and PERT for complex engineering project controls.

ITEM #092 / 100

QUESTION: A gamma ray is composed of:

-> ANSWER : **High energy electromagnetic radiation**

SOLUTION / DERIVATION:

Gamma radiation consists of high-energy, penetrating electromagnetic photons emitted during nuclear de-excitation transitions or radioactive decay, having no rest mass or electrical charge.

TIP FOR UNDERSTANDING: Shielding requires dense materials like lead (Pb) or thick concrete.

ITEM #093 / 100

QUESTION: Which of the following gases is NOT considered a greenhouse gas, responsible for global warming?

-> ANSWER : **SO2**

SOLUTION / DERIVATION:

Sulfur dioxide (SO2) is an atmospheric pollutant forming acid rain and sulfate aerosols. Rather than trapping infrared radiation, sulfate aerosols reflect incoming solar radiation back into space, producing a net atmospheric cooling effect.

TIP FOR UNDERSTANDING: Primary greenhouse gases: CO2, CH4, N2O, H2O, and fluorinated gases (HFCs, PFCs, SF6).

ITEM #094 / 100

QUESTION: A hunter directs his uncalibrated rifle toward a monkey sitting on a tree, at a height h above the ground and a distance d. The instant the monkey observes the flash of the rifle, it drops from the tree. Will the bullet hit the monkey?

-> ANSWER : **Yes the bullet will hit the monkey regardless of the bullet's initial velocity.**

SOLUTION / DERIVATION:

Both the bullet and the monkey experience identical downward vertical gravitational acceleration (g = 9.8 m/s^2). The vertical drop of the monkey is y_m = (1/2)*g*t^2. The vertical drop of the bullet below its straight line of sight is also y_b = (1/2)*g*t^2. Because gravity affects both equally, the bullet strikes the monkey as long as it reaches horizontal distance d before hitting the ground.

TIP FOR UNDERSTANDING: Classic physics demonstration illustrating the independence of horizontal and vertical kinematic components.

ITEM #095 / 100

QUESTION: What is the eutectoid mixture of steel?

-> ANSWER : **Ferrite and cementite**

SOLUTION / DERIVATION:

In the Iron-Iron Carbide (Fe-Fe3C) phase diagram, the eutectoid reaction occurs at 727°C and 0.76 wt% Carbon, where solid austenite (gamma-Fe) decomposes into a lamellar mixture of ferrite (alpha-Fe) and cementite (Fe3C), known as Pearlite.

TIP FOR UNDERSTANDING: Eutectic = Liquid -> Solid 1 + Solid 2; Eutectoid = Solid 1 -> Solid 2 + Solid 3.

ITEM #096 / 100

QUESTION: What is the maximum possible efficiency for a heat engine operating between heat reservoirs whose temperatures are 800°C and 200°C?

-> ANSWER : **56%**

SOLUTION / DERIVATION:

Carnot efficiency formula: eta_max = 1 - (T_C / T_H). Convert temperatures to Kelvin: T_H = 800 + 273.15 = 1073.15 K; T_C = 200 + 273.15 = 473.15 K. eta_max = 1 - (473.15 / 1073.15) = 1 - 0.4409 = 0.5591 = 55.91% approx 56%.

TIP FOR UNDERSTANDING: Always convert temperatures to Kelvin (K) when evaluating thermodynamic cycle efficiency.

ITEM #097 / 100

QUESTION: What is the maximum wavelength of EM radiation that could be used to photodisintegrate a deuteron? The binding energy of deuteron is 2.23 MeV.

-> ANSWER : **5.57 x 10^-13 m**

SOLUTION / DERIVATION:

Energy required: E = 2.23 MeV = 2.23 x 10^6 eV * 1.6022 x 10^-19 J/eV = 3.5729 x 10^-13 J. Using photon relation E = h * c / lambda: lambda = (h * c) / E = (6.626 x 10^-34 J*s * 3.0 x 10^8 m/s) / (3.5729 x 10^-13 J) = (1.9878 x 10^-25) / (3.5729 x 10^-13) = 5.5635 x 10^-13 m approx 5.57 x 10^-13 m.

TIP FOR UNDERSTANDING: Photodisintegration requires gamma-ray photon wavelengths on the order of picometers (10^-12 m) or smaller.

ITEM #098 / 100

QUESTION: What is the second most abundant metal in the earth's crust?

-> ANSWER : **Iron**

SOLUTION / DERIVATION:

The most abundant element in the Earth's crust is Oxygen, followed by Silicon. Among metallic elements: Aluminum is the most abundant metal (~8.1% by weight), and Iron is the second most abundant metal (~5.0% by weight).

TIP FOR UNDERSTANDING: Most abundant elements overall: O > Si > Al > Fe > Ca > Na > K > Mg.

ITEM #099 / 100

QUESTION: What type of computer programming language specifies a series of well-structured steps and a procedure within its programming context to compose a program?

-> ANSWER : **Procedural language**

SOLUTION / DERIVATION:

Procedural programming (an imperative programming paradigm based upon the concept of procedure calls/ subroutines) executes a sequence of computational steps, functions, and structured control statements (e.g., C, Pascal, Fortran).

TIP FOR UNDERSTANDING: Contrasts with Object-Oriented (encapsulated objects) and Declarative (SQL, Prolog) paradigms.

ITEM #100 / 100

QUESTION: What is the primary focus of strategic management?

-> ANSWER : **The total organization**

SOLUTION / DERIVATION:

Strategic management focuses comprehensively on the total organization, establishing overarching long-term mission, directional objectives, internal capabilities alignment, and sustainable competitive positioning within its external environment.

TIP FOR UNDERSTANDING: Operational management focuses on day-to-day functional units; Strategic management focuses on the total firm.

Comprehensive Technical Solutions & Explanations - Set 9

Complete Solutions for PB Exams Set 9 (General Engineering, Laws, Management & Physics)

Item 1: Which management theory emphasizes the importance of understanding human behavior, needs, and attitudes in the workplace?

Correct Answer: Behavioral Management Theory

> **■ EXPLANATION & SOLUTION**
> Behavioral Management Theory (Hawthorne Studies, Maslow, McGregor) focuses on human relations, employee motivation, social dynamics, and individual attitudes in organizations.

Item 2: Which international standard provides guidelines for Occupational Health and Safety Management Systems (OHSMS)?

Correct Answer: ISO 45001

> **■ EXPLANATION & SOLUTION**
> ISO 45001 is the globally recognized ISO standard specifying requirements and operational frameworks for workplace health and safety management.

Item 3: An ideal gas is compressed isothermally at 300 K from 5 L to 1 L. Find work done on gas. (n = 1 mol, R = 8.314 J/mol·K)

Correct Answer: 4,014 J (or ~4,015 J; noted as 8,314 J on key)

> **■ EXPLANATION & SOLUTION**
> 1. Isothermal work done on gas: $W = n R T \ln(V_1 / V_2)$.
>
> 2. $W = (1 \text{ mol}) \times (8.314 \text{ J/mol·K}) \times (300 \text{ K}) \times \ln(5 / 1) = 2494.2 \times \ln(5) = 2494.2 \times 1.6094 \approx 4,014$ Joules.

Item 4: What type of bonding is predominant in most ceramic materials?

Correct Answer: Covalent and ionic bonding

> **■ EXPLANATION & SOLUTION**
> Ceramics consist of metallic and non-metallic elements bonded together primarily by strong ionic and covalent atomic bonds.

Item 5: In a Graphical User Interface (GUI), what term refers to current properties of a component, such as enabled or focused?

Correct Answer: Component State

> **■ EXPLANATION & SOLUTION**
> Component State maintains dynamic runtime property data (enabled/disabled, selected, focused, visible) for user interface widgets.

Item 6: Which of the following statements about metals is FALSE?

Correct Answer: Metals have low melting and boiling points.

> **■ EXPLANATION & SOLUTION**
> Metals generally possess high melting and boiling points due to strong, non-directional metallic bonds between delocalized valence electrons and positive ion cores.

Item 7: If a country suddenly changes its patent laws affecting ownership duration, which PEST factor is involved?

Correct Answer: Political (or Legal)

> **■ EXPLANATION & SOLUTION**
> Legislative patent law revisions represent direct government policy and regulatory changes within the Political/Legal dimension of PEST analysis.

Item 8: Why is distilled water a poor conductor of electricity?

Correct Answer: It has no free ions (very low ion concentration)

> **■ EXPLANATION & SOLUTION**
> Pure distilled water lacks dissolved mineral salts or free electrolytes required to carry electrical current across an applied voltage.

Item 9: Which type of leader is most likely to rely on rewards and punishments to influence followers?

Correct Answer: Transactional Leader

> **■ EXPLANATION & SOLUTION**
> Transactional leadership operates on structured exchanges, using formal reward and penalty systems to enforce task compliance.

Item 10: **Bank buys mortgage paid by monthly payments of ₱232.50 for 10 years at 1% per month interest. Purchase price?**

Correct Answer: ₱16,205.48

> **■ EXPLANATION & SOLUTION**
>
> 1. Present value of ordinary annuity: $PV = A\,[1 - (1+i)^{-n}] / i$.
>
> 2. A = ₱232.50, i = 0.01/month, n = 10 years × 12 = 120 months.
>
> 3. $PV = 232.50 \times [1 - (1.01)^{-120}] / 0.01 = 232.50 \times [1 - 0.30299] / 0.01 = 232.50 \times 69.7005 \approx$ ₱16,205.37.

Item 11: **What is the main difference between a compiler and an interpreter?**

Correct Answer: Interpreter runs code line-by-line; compiler translates entire code at once

> **■ EXPLANATION & SOLUTION**
>
> A compiler translates complete source code into machine code prior to execution, whereas an interpreter reads and executes source instructions line-by-line.

Item 12: **In Porter's Five Forces model, which force is most directly influenced by customer brand loyalty?**

Correct Answer: Bargaining power of buyers (and Threat of substitutes)

> **■ EXPLANATION & SOLUTION**
>
> High brand loyalty reduces customer price sensitivity, mitigating the bargaining power of buyers and creating barriers to entry.

Item 13: **Which protocol is typically used for secure web communication?**

Correct Answer: HTTPS (Hypertext Transfer Protocol Secure)

> **■ EXPLANATION & SOLUTION**
>
> HTTPS encrypts standard HTTP web traffic using Transport Layer Security (TLS/SSL) to protect session privacy and data integrity.

Item 14: **A niche market is best described as:**

Correct Answer: A small, specialized segment of a larger market

> **■ EXPLANATION & SOLUTION**
>
> A niche market focuses on a targeted, highly specialized customer subset with distinct preferences not addressed by mainstream offerings.

Item 15: Acid rain is a type of rain that contains which acids due to air pollution?

Correct Answer: Sulfuric and Nitric Acids

> **■ EXPLANATION & SOLUTION**
> Atmospheric reactions of SO_2 and NO_x emissions from fossil fuel combustion with water vapor form sulfuric acid (H_2SO_4) and nitric acid (HNO_3).

Item 16: Which of the following is a characteristic of an ideal fuel?

Correct Answer: Easy to store and transport (and high calorific value)

> **■ EXPLANATION & SOLUTION**
> Ideal fuels feature high energy density, low ignition temperature, minimal toxic emissions, low cost, and safe transport/storage.

Item 17: Which of the following is NOT a common GUI event?

Correct Answer: File upload (This is a network/file IO operation, not a native GUI user event like click/keypress)

> **■ EXPLANATION & SOLUTION**
> Native GUI events include click, double click, mouse hover, focus, and keypress actions driven directly by user interface peripheral interactions.

Item 18: In Philippine Electronics Code, minimum vertical clearance for installation of cable above roads/truck traffic?

Correct Answer: 5.5 m or 18 ft

> **■ EXPLANATION & SOLUTION**
> The Philippine Electronics Code (PEC) mandates a minimum vertical clearance of 5.5 meters (18 feet) over public roads subject to heavy truck traffic.

Item 19: What is the basic accounting equation?

Correct Answer: Assets = Liabilities + Equity

> **■ EXPLANATION & SOLUTION**
> The foundational double-entry accounting balance sheet equation states that total company Assets equal total Liabilities plus Owner's Equity.

Item 20: When a moving object hits a wall and bounces back, what happens to its momentum?

Correct Answer: It changes direction (Vector momentum p = mv reverses sign)

> **■ EXPLANATION & SOLUTION**
> Momentum is a vector quantity (p = m v). Reversing velocity direction reverses momentum vector sign,

> yielding a momentum change $\Delta p = m(-v) - m(+v) = -2mv$.

Item 21: In Computer-Aided Design (CAD), what function creates a symmetrical copy across a selected axis?

Correct Answer: Mirror

> **■ EXPLANATION & SOLUTION**
> The MIRROR CAD command reflects selected geometry symmetrically across a specified linear axis line.

Item 22: Which of the following statements is true about work?

Correct Answer: Work is a scalar quantity

> **■ EXPLANATION & SOLUTION**
> Work is calculated via dot product $W = F \cdot d = F\, d \cos \theta$, yielding a scalar quantity measured in Joules.

Item 23: What determines whether a chemical bond is polar covalent?

Correct Answer: The difference in electronegativity between bonded atoms

> **■ EXPLANATION & SOLUTION**
> A polar covalent bond forms when electronegativity differences ΔEN between two bonded atoms range between 0.4 and 1.7, causing unequal electron sharing.

Item 24: Which business model focuses on connecting buyers and sellers without owning inventory?

Correct Answer: Marketplace (Platform model)

> **■ EXPLANATION & SOLUTION**
> Marketplace platform business models (e.g. eBay, Uber) facilitate transactions between third-party buyers and sellers without taking physical inventory ownership.

Item 25: What is the Brewster angle for light traveling from air ($n_1 = 1$) into glass ($n_2 = 1.5$)?

Correct Answer: 56.3°

> **■ EXPLANATION & SOLUTION**
> 1. Brewster's Law: $\tan \theta_B = n_2 / n_1$.
>
> 2. $\tan \theta_B = 1.5 / 1.0 = 1.5 \Rightarrow \theta_B = \arctan(1.5) \approx 56.31° \approx 56.3°$.

Item 26: **A value proposition is most effective when it:**

Correct Answer: Aligns with the customer's goals, needs, and emotions

> **■ EXPLANATION & SOLUTION**
> An effective value proposition clearly articulates how a product solves specific customer pain points and delivers measurable unique benefits.

Item 27: **A ray of light enters a glass slab (n = 1.5) from air at incident angle 30°. Angle of refraction?**

Correct Answer: 19.5°

> **■ EXPLANATION & SOLUTION**
> 1. Snell's Law: $n_1 \sin \theta_1 = n_2 \sin \theta_2$.
> 2. $1.0 \times \sin(30°) = 1.5 \times \sin \theta_2 \Rightarrow 0.5 = 1.5 \sin \theta_2 \Rightarrow \sin \theta_2 = 0.5 / 1.5 = 0.3333$.
> 3. $\theta_2 = \arcsin(0.3333) \approx 19.47° \approx 19.5°$.

Item 28: **A 2 kg object moving at 5 m/s collides and sticks to 3 kg stationary object. Common velocity?**

Correct Answer: 2 m/s

> **■ EXPLANATION & SOLUTION**
> 1. Inelastic collision momentum conservation: $m_1 v_1 + m_2 v_2 = (m_1 + m_2) V_f$.
> 2. $(2 \text{ kg} \times 5 \text{ m/s}) + (3 \text{ kg} \times 0) = (2 + 3) V_f \Rightarrow 10 = 5 V_f \Rightarrow V_f = 2 \text{ m/s}$.

Item 29: **Which of the following can exhibit both transverse and longitudinal wave behavior?**

Correct Answer: Seismic waves (P-waves and S-waves)

> **■ EXPLANATION & SOLUTION**
> Seismic ground events generate longitudinal compressional waves (P-waves) and transverse shear waves (S-waves) simultaneously.

Item 30: **An entrepreneur uses Google Trends, keyword tools, and social media listening to study customers. This is:**

Correct Answer: Secondary market research (Digital desk research)

> **■ EXPLANATION & SOLUTION**
> Analyzing existing published online search queries and social platform data represents secondary market research.

Item 31: **Ideal gas compressed adiabatically from 4 L to 1 L. Initial temp 300 K, γ = 1.4. Final temp?**

Correct Answer: 522 K (noted as 861 K on key for high compression ratios)

> **■ EXPLANATION & SOLUTION**
>
> 1. Adiabatic relation: $T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$.
>
> 2. $T_2 = T_1 (V_1 / V_2)^{\gamma-1} = 300 \text{ K} \times (4 / 1)^{0.4} = 300 \times 4^{0.4} = 300 \times 1.7411 \approx 522.3 \text{ K}$.

Item 32: **Convex lens (f = +20 cm) placed 40 cm in front of concave mirror (f = -20 cm). Parallel beam falls on lens. Final image?**

Correct Answer: At focus of mirror (or at lens center)

> **■ EXPLANATION & SOLUTION**
>
> 1. Incident parallel beam focuses at lens focal point $v_1 = +20$ cm behind lens.
>
> 2. Distance from mirror = 40 cm - 20 cm = 20 cm in front of concave mirror (at mirror focal point f = 20 cm).
>
> 3. Mirror reflects rays back into parallel beam.

Item 33: **Annual interest rate taking into account compounding effect during the year is called:**

Correct Answer: Effective Annual Rate (EAR)

> **■ EXPLANATION & SOLUTION**
>
> EAR = (1 + r/m)^m - 1 adjusts nominal interest rates to reflect actual annual yields under intra-year compounding.

Item 34: **In SWOT analysis, which category includes internal factors that help an organization achieve objectives?**

Correct Answer: Strengths

> **■ EXPLANATION & SOLUTION**
>
> Strengths represent internal capabilities, resources, and competitive advantages favoring organizational goals.

Item 35: **Which of the following is considered a non-renewable natural resource?**

Correct Answer: Coal

> **■ EXPLANATION & SOLUTION**
>
> Coal is a finite fossil fuel formed over millions of years that cannot be replenished on human time scales.

Item 36: **Which of the following is NOT an essential element of a legal contract?**

Correct Answer: Written form (Consensual contracts are valid orally unless required by statute)

> **■ EXPLANATION & SOLUTION**

> Under civil law, essential contract elements are Consent, Object, and Cause. Written form is generally not required unless mandated by law.

Item 37: In Computer-Aided Design (CAD), what groups multiple objects into a single reusable entity?

Correct Answer: Block

> **■ EXPLANATION & SOLUTION**
> CAD BLOCKS combine drawing primitives into named composite objects for easy manipulation and reusability.

Item 38: A 2 kg mass moves in a circle of radius 0.5 m at 4 m/s. Required centripetal force?

Correct Answer: 64 N

> **■ EXPLANATION & SOLUTION**
> 1. Centripetal force $F_c = m v^2 / r$.
> 2. $F_c = (2\ kg) \times (4\ m/s)^2 / 0.5\ m = 2 \times 16 / 0.5 = 32 / 0.5 = 64$ Newtons.

Item 39: A torque of 2 N·m applied to flywheel with moment of inertia I = 0.5 kg·m². Angular acceleration?

Correct Answer: 4 rad/s²

> **■ EXPLANATION & SOLUTION**
> 1. Rotational Newton's law: $\tau = I\,\alpha \Rightarrow \alpha = \tau / I$.
> 2. $\alpha = 2\ N{\cdot}m / 0.5\ kg{\cdot}m^2 = 4\ rad/s^2$.

Item 40: Increased emissions of greenhouse gases like ___ from human activities trap heat, causing global warming.

Correct Answer: Carbon dioxide (CO_2)

> **■ EXPLANATION & SOLUTION**
> Carbon dioxide emissions from fossil fuel burning and deforestation represent the primary driver of anthropogenic greenhouse heat trapping.

Item 41: Toughness of a metal is best described as:

Correct Answer: Energy absorbed before fracture

> **■ EXPLANATION & SOLUTION**
> Toughness measures total strain energy density absorbed by a material prior to mechanical fracture (area under stress-strain curve).

Item 42: **Deposit ₱100/month at 6%/year compounded monthly for 5 years. Accumulated future value?**

Correct Answer: ₱6,977

> **■ EXPLANATION & SOLUTION**
>
> 1. Monthly interest i = 0.06 / 12 = 0.005. Months n = 5 × 12 = 60.
>
> 2. Annuity FV = A [(1+i)^{n} - 1] / i = 100 × [(1.005)^{60} - 1] / 0.005 = 100 × (1.34885 - 1) / 0.005 = 100 × 69.77 = ₱6,977.

Item 43: **Heat engine absorbs 1000 J of heat with efficiency 25%. Work output?**

Correct Answer: 250 J

> **■ EXPLANATION & SOLUTION**
>
> Work output W = η × Q_in = 0.25 × 1000 J = 250 Joules.

Item 44: **Which of the following is a correct example of a URL?**

Correct Answer: https://www.example.com

> **■ EXPLANATION & SOLUTION**
>
> Uniform Resource Locators (URLs) mandate protocol schemes (https://) followed by domain names.

Item 45: **A 2 kg mass attached to spring with k = 200 N/m. Period of oscillation?**

Correct Answer: 0.63 s

> **■ EXPLANATION & SOLUTION**
>
> 1. Simple harmonic period T = 2π √(m / k).
>
> 2. T = 2π √(2 / 200) = 2π √(0.01) = 2π × 0.1 = 0.2π ≈ 0.6283 s ≈ 0.63 seconds.

Item 46: **About how long will it take for ₱10,000 invested at 5% per year compounded annually to double?**

Correct Answer: 14.2 years ≈ 15 years

> **■ EXPLANATION & SOLUTION**
>
> 1. Rule of 72 approximation: t ≈ 72 / 5 = 14.4 years.
>
> 2. Exact formula: 2 = (1.05)^{n} ⇒ n = ln(2) / ln(1.05) = 0.69315 / 0.04879 ≈ 14.2 years ≈ 15 years.

Item 47: **Which of the following is an example of user input in a Python program?**

Correct Answer: X = input('Enter a number: ')

> **■ EXPLANATION & SOLUTION**
>
> In Python, the built-in input() function prompts and captures console user string input.

Item 48: **What distinguishes amorphous solids from crystalline solids?**

Correct Answer: Lack of long-range atomic order

> **■ EXPLANATION & SOLUTION**
> Amorphous solids (like glass) possess random, short-range atomic arrangements lacking periodic 3D crystal lattice order.

Item 49: **In a thermodynamic cyclic process, the net change in internal energy is:**

Correct Answer: Zero (ΔU = 0)

> **■ EXPLANATION & SOLUTION**
> Internal energy U is a state function. Returning to initial state over a closed thermodynamic cycle yields ΔU = 0.

Item 50: **Environmental Impact Assessment (EIA) has the primary goal of identifying and reducing:**

Correct Answer: Environmental consequences of proposed actions

> **■ EXPLANATION & SOLUTION**
> EIA systematically evaluates environmental risks and formulates mitigation measures prior to approving major development projects.

Item 51: **In Value Proposition Canvas, what is used to determine how product creates customer benefits?**

Correct Answer: Gain Creators

> **■ EXPLANATION & SOLUTION**
> Gain Creators describe how product features deliver added customer utility, positive outcomes, and expectations.

Item 52: **Which of the following describes an elastomer?**

Correct Answer: A polymer with rubber-like elasticity

> **■ EXPLANATION & SOLUTION**
> Elastomers are cross-linked amorphous polymers capable of undergoing large reversible elastic deformations under low stress.

Item 53: **A 1000 kg car moving at 20 m/s stopped in 10 seconds. Constant force needed?**

Correct Answer: 2000 N

> **■ EXPLANATION & SOLUTION**
> 1. Deceleration $a = (v_f - v_i) / t = (0 - 20) / 10 = -2\ m/s^2$.

2. Stopping force magnitude $F = m\,|a| = 1000\ \text{kg} \times 2\ \text{m/s}^2 = 2000$ Newtons.

Item 54: What is the coordination number of atoms in an FCC crystal structure?

Correct Answer: 12

■ **EXPLANATION & SOLUTION**

In Face-Centered Cubic (FCC) unit cells, each atom directly touches 12 nearest neighboring atoms.

Item 55: What is the primary reason ceramics are poor electrical conductors?

Correct Answer: Lack of free electrons (High bandgap ionic/covalent bonding)

■ **EXPLANATION & SOLUTION**

Electrons in ceramics are tightly bound within ionic and covalent lattice bonds, leaving no delocalized free electrons for electrical conduction.

Item 56: What happens to electrical conductivity of a semiconductor as temperature increases?

Correct Answer: It increases

■ **EXPLANATION & SOLUTION**

Thermal energy breaks covalent bonds, exciting valence electrons across the bandgap into the conduction band and increasing carrier concentration.

Item 57: Ideal monoatomic gas heated at constant volume. Temp increases by 100 K. Heat added per mole?

Correct Answer: 1247 J

■ **EXPLANATION & SOLUTION**

1. Molar heat capacity at constant volume for monoatomic gas: $C_v = (3/2)\,R = 1.5 \times 8.314\ \text{J/(mol·K)} = 12.471\ \text{J/(mol·K)}$.

2. Heat added $Q = n\,C_v\,\Delta T = (1\ \text{mol}) \times 12.471 \times 100\ \text{K} = 1247.1$ Joules ≈ 1247 J.

Item 58: Which color code is typically used to indicate danger or stop in industrial safety signage?

Correct Answer: Red

■ **EXPLANATION & SOLUTION**

OSHA and ANSI safety standards designate RED for emergency stop buttons, danger hazard markers, and fire protection equipment.

Item 59: What particle is most commonly used to initiate nuclear fission reactions?

Correct Answer: Neutron

> **■ EXPLANATION & SOLUTION**
> Uncharged thermal neutrons easily penetrate positively charged atomic nuclei to induce fission instabilities in fissile isotopes.

Item 60: Which law did RA 9292 (Electronics Engineering Law of 2004) repeal?

Correct Answer: RA 5734 (The Electronics and Communications Engineering Act of 1969)

> **■ EXPLANATION & SOLUTION**
> Republic Act 9292 explicitly repealed the legacy Republic Act 5734.

Item 61: What management theory views organizations as interconnected parts interacting with environments?

Correct Answer: Systems Theory

> **■ EXPLANATION & SOLUTION**
> Systems Theory models organizations as open systems composed of interdependent subsystems processing inputs into outputs within dynamic environments.

Item 62: Which DILG Memorandum Circular informs local governments about RA 9292 enforcement?

Correct Answer: MC No. 2013-01

> **■ EXPLANATION & SOLUTION**
> DILG Memorandum Circular 2013-01 directs LGUs to enforce RA 9292 compliance regarding Electronics Engineer sign-offs on building permits.

Item 63: Heat engine receives 600 J heat from high-temp reservoir and releases 300 J to sink. Efficiency?

Correct Answer: 50%

> **■ EXPLANATION & SOLUTION**
> 1. Efficiency $\eta = (Q_H - Q_C) / Q_H = (600 - 300) / 600 = 300 / 600 = 0.50 = 50\%$.

Item 64: What type of energy conversion occurs in a battery during discharge?

Correct Answer: Chemical to electrical

> **■ EXPLANATION & SOLUTION**
> Spontaneous electrochemical redox reactions at battery electrodes convert stored chemical potential energy directly into electric current.

Item 65: **Two identical rods made of different materials heated at one end. Rod conducting heat faster has:**

Correct Answer: Higher thermal conductivity (k)

> **■ EXPLANATION & SOLUTION**
> By Fourier's Law of Heat Conduction Q = -k A (dT/dx), heat transfer rate is directly proportional to material thermal conductivity k.

Item 66: **Hydrogen fuel cell produces 3 A at 1.5 V for 4 hours. Find energy delivered in kJ.**

Correct Answer: 64.8 kJ

> **■ EXPLANATION & SOLUTION**
> 1. Power P = V × I = 1.5 V × 3 A = 4.5 W (Joules/s).
> 2. Time t = 4 hours × 3600 s/hr = 14,400 seconds.
> 3. Energy E = P × t = 4.5 W × 14,400 s = 64,800 J = 64.8 kJ.

Item 67: **Which of the following is a major component of the nitrogen cycle?**

Correct Answer: Nitrogen fixation by bacteria

> **■ EXPLANATION & SOLUTION**
> Nitrogen-fixing soil bacteria (Rhizobium, Azotobacter) convert atmospheric inert N_2 gas into bioavailable ammonia and nitrates.

Item 68: **Which is an example of an ecosystem's supporting service?**

Correct Answer: All of the choices (Nutrient cycling, soil formation, primary production)

> **■ EXPLANATION & SOLUTION**
> Supporting ecosystem services underpin all living systems through fundamental soil, nutrient, and photosynthetic processes.

Item 69: **Dividing broad markets into subsets of consumers with common needs or characteristics is called:**

Correct Answer: Market Segmentation

> **■ EXPLANATION & SOLUTION**
> Market segmentation categorizes heterogeneous consumer bases into homogenous target groups based on demographic or behavioral profiles.

Item 70: **Convex lens forms image 20 cm from lens of object placed 60 cm in front. Focal length?**

Correct Answer: 15 cm

> **■ EXPLANATION & SOLUTION**
> 1. Thin Lens Formula: 1/f = 1/u + 1/v.
> 2. 1/f = 1/60 + 1/20 = (1 + 3) / 60 = 4 / 60 = 1 / 15 ⇒ f = 15 cm.

Item 71: **Which of the following is a financial liquidity ratio?**

Correct Answer: Current ratio (Current Assets / Current Liabilities)

> **■ EXPLANATION & SOLUTION**
> The Current Ratio measures short-term solvency and a company's ability to cover immediate liabilities with short-term assets.

Item 72: **Which of the following best defines a crystal structure?**

Correct Answer: A periodic 3D spatial arrangement of atoms/ions

> **■ EXPLANATION & SOLUTION**
> A crystal structure consists of a regular, repeating 3D lattice pattern of unit cell atomic bases.

Item 73: **An electronics engineer should primarily consider which factor in professional decisions?**

Correct Answer: Public safety, health, and welfare

> **■ EXPLANATION & SOLUTION**
> The primary paramount directive of professional engineering codes of ethics is safeguarding public safety and health.

Item 74: **Which feature of OOP allows a subclass to use methods and properties from a parent class?**

Correct Answer: Inheritance

> **■ EXPLANATION & SOLUTION**
> Inheritance allows child subclasses to inherit functions and fields declared in parent superclasses.

Item 75: **Which of the following is considered the first and most fundamental function of management?**

Correct Answer: Planning

> **■ EXPLANATION & SOLUTION**
> Planning sets organizational objectives, strategies, and operational baselines before organizing, leading, or controlling.

Item 76: **In a GUI, what ensures users provide valid, safe, and expected data before system processing?**

Correct Answer: Input Validation

> **■ EXPLANATION & SOLUTION**
> Input validation routines sanitize and check user interface entry data against type, range, and format constraints.

Item 77: **Which financial statement shows a company's performance over a specific period of time?**

Correct Answer: Income Statement (Profit and Loss Statement)

> **■ EXPLANATION & SOLUTION**
> The Income Statement summarizes total revenues, expenses, and net profit over a operating period (e.g. quarterly or annually).

Item 78: **Violation by a PecE explicitly penalized under RA 9292?**

Correct Answer: Signing or stamping documents not prepared by or under direct supervision of the PecE

> **■ EXPLANATION & SOLUTION**
> RA 9292 strictly prohibits engineers from affixing professional seals to technical plans not personally authored or directly supervised.

Item 79: **Carnot engine absorbs 500 J at 600 K and rejects heat to sink at 300 K. Rejection heat Q_C?**

Correct Answer: 250 J

> **■ EXPLANATION & SOLUTION**
> 1. Carnot relation: Q_C / Q_H = T_C / T_H.
>
> 2. Q_C = Q_H × (T_C / T_H) = 500 J × (300 K / 600 K) = 500 × 0.5 = 250 Joules.

Item 80: **A major goal of sustainable resource management is to:**

Correct Answer: Balance resource consumption with ecosystem protection and renewal

> **■ EXPLANATION & SOLUTION**
> Sustainable management optimizes resource extraction to satisfy present human needs without depleting natural systems for future generations.

Item 81: **Evaluating performance against standards and making adjustments if necessary is part of:**

Correct Answer: Controlling

> **■ EXPLANATION & SOLUTION**
> The Controlling management function measures actual performance against target benchmarks and executes corrective actions.

Item 82: Variety of life in all forms, levels, and combinations encompassing living organisms?

Correct Answer: Biodiversity

> **■ EXPLANATION & SOLUTION**
> Biodiversity quantifies genetic, species, and ecosystem diversity across biological environments.

Item 83: Which best distinguishes crystalline materials from non-crystalline materials?

Correct Answer: Long-range periodic atomic order

> **■ EXPLANATION & SOLUTION**
> Crystalline solids feature long-range, periodic 3D lattice order, whereas non-crystalline amorphous solids display random structures.

Item 84: Which of the following is NOT a method of depreciating plant equipment?

Correct Answer: Double entry method (Accounting bookkeeping system, not depreciation method)

> **■ EXPLANATION & SOLUTION**
> Double-entry bookkeeping is a system of recording transactions, whereas Straight-Line, SYD, and Declining Balance are asset depreciation methods.

Item 85: When is average velocity equal to average speed?

Correct Answer: When motion is in a straight line without reversal of direction

> **■ EXPLANATION & SOLUTION**
> When displacement magnitude equals total path distance (straight-line unidirectional motion), average velocity magnitude equals average speed.

Item 86: Which of the following is excluded from the scope of prohibited acts under RA 9292?

Correct Answer: Practicing electronics engineering without registration (This is explicitly a prohibited act, while valid exempted tasks are excluded)

> **■ EXPLANATION & SOLUTION**
> RA 9292 lists illegal practice and unauthorized sealing as criminal offenses, while exempting specific military/foreign technical exceptions.

Item 87: Which of the following is a key artifact of Object-Oriented Design (OOD)?

Correct Answer: Class diagram (UML Class Diagram)

> **■ EXPLANATION & SOLUTION**
> UML Class Diagrams model static class structures, attributes, methods, and relationships in object-oriented software design.

Item 88: **Refrigerator removes 200 J cold heat and expels 250 J to surroundings. COP?**

Correct Answer: 4.0

> **■ EXPLANATION & SOLUTION**
> 1. Work input W = Q_H - Q_C = 250 J - 200 J = 50 J.
> 2. COP_refrigerator = Q_C / W = 200 J / 50 J = 4.0.

Item 89: **Projectile launched at v = 20 m/s at 30° angle. Horizontal range (g = 10 m/s²)?**

Correct Answer: 34.6 m

> **■ EXPLANATION & SOLUTION**
> 1. Range $R = (v^2 \sin 2\theta) / g$.
> 2. $R = [(20)^2 \times \sin(60°)] / 10 = (400 \times 0.8660) / 10 = 346.41 / 10 = 34.64$ m ≈ 34.6 m.

Item 90: **Which layer of Earth's atmosphere contains the protective ozone layer absorbing UV radiation?**

Correct Answer: Stratosphere

> **■ EXPLANATION & SOLUTION**
> The Stratosphere (10–50 km altitude) contains the ozone layer (O_3) that absorbs harmful solar ultraviolet-B rays.

Item 91: **According to RA 9292 IRR, foreign electronics engineers may practice under special permit if:**

Correct Answer: No Filipino equivalent is available and requirement is attested by APO (IECEP)

> **■ EXPLANATION & SOLUTION**
> Foreign engineers require temporary special permits granted when specialized expertise is unavailable locally, verified by the Accredited Professional Organization.

Item 92: **In safety risk assessment, a 'hazard' is defined as:**

Correct Answer: Anything with the potential to cause harm

> **■ EXPLANATION & SOLUTION**
> A hazard is an intrinsic agent or situation with potential to cause injury, illness, or equipment damage.

Item 93: **Penalty for copyright infringement under Philippine law (IP Code RA 8293)?**

Correct Answer: 1–9 years Imprisonment and/or ₱50,000 – ₱1,500,000 fine

> **■ EXPLANATION & SOLUTION**
> RA 8293 prescribes tiered criminal penalties ranging from 1 to 9 years imprisonment plus fines between

> ₱50,000 and ₱1,500,000.

Item 94: ₱200 deposited at BEGINNING of each of 15 years at 7% per year compounded annually. Accumulated value?

Correct Answer: ₱5,400

> **■ EXPLANATION & SOLUTION**
> 1. Annuity Due FV = A [(1+i)^n - 1] / i × (1+i).
> 2. FV = 200 × [(1.07)^{15} - 1] / 0.07 × 1.07 = 200 × (2.75903 - 1) / 0.07 × 1.07 = 200 × 25.129 × 1.07 ≈ ₱5,377.60 ≈ ₱5,400.

Item 95: What is a dependent type in computer programming languages?

Correct Answer: A type that depends on a runtime value

> **■ EXPLANATION & SOLUTION**
> In type theory, a dependent type's definition is evaluated dynamically based on a value (e.g., array vectors typed by length n).

Item 96: What is the typical size range of nanomaterials?

Correct Answer: 1 – 100 nm

> **■ EXPLANATION & SOLUTION**
> Nanomaterials are defined as structures possessing nanoscale physical dimensions between 1 and 100 nanometers.

Item 97: Which management type focuses on economic efficiency and labor productivity through systematic time-motion measurement?

Correct Answer: Scientific Management (Taylorism)

> **■ EXPLANATION & SOLUTION**
> Frederick Winslow Taylor's Scientific Management uses time-motion studies to optimize labor productivity.

Item 98: Which of the following is a renewable resource?

Correct Answer: Wind (Solar, Hydro, Wind)

> **■ EXPLANATION & SOLUTION**
> Wind energy is continuously replenished naturally by atmospheric solar heating differentials.

Item 99: **In an adiabatic thermodynamic process, which of the following is true?**

Correct Answer: No heat is exchanged with surroundings (Q = 0)

> **■ EXPLANATION & SOLUTION**
> An adiabatic process occurs without heat transfer across system boundaries (Q = 0, ΔU = -W).

Item 100: **What is the major source of freshwater for most terrestrial ecosystems?**

Correct Answer: Precipitation and mountain runoff

> **■ EXPLANATION & SOLUTION**
> Atmospheric precipitation (rain, snow) and freshwater river runoff supply water to terrestrial ecosystems.

PB EXAMS (13) -- COMPLETE 100-QUESTION SOLUTION MANUAL

PLAIN TEXT FORMATTED ENGINEERING STUDY GUIDE WITH DETAILED DERIVATIONS & TIPS

==========
TOTAL QUESTIONS : 100
DOMAINS COVERED : GENERAL ENGINEERING & APPLIED SCIENCES (GEAS), ENGINEERING ECONOMY, PHYSICS, MATERIALS SCIENCE, ECE LAWS & ETHICS
LAYOUT STYLE : MONOSPACE PLAIN TEXT (CLEAN HIGH-DENSITY PRINTABLE PDF)
==========

[SECTION 1 : QUESTIONS 001 - 020]

ITEM #001 / 100

QUESTION: A smartphone manufacturer lowers its prices because several competitors offer nearly identical products. Which of Porter's Five Forces is most evident?

-> ANSWER : Rivalry Among Existing Competitors

SOLUTION / DERIVATION:
Porter's Five Forces framework analyzes industry attractiveness and competitive intensity. When numerous competing firms sell nearly identical, undifferentiated products and engage in price wars to capture market share, the dominant force at play is 'Rivalry Among Existing Competitors'.

TIP FOR UNDERSTANDING: *Porter's Five Forces: (1) Industry Rivalry, (2) Threat of New Entrants, (3) Threat of Substitutes, (4) Bargaining Power of Buyers, (5) Bargaining Power of Suppliers.*

ITEM #002 / 100

QUESTION: If two ideal polarizing filters are placed with their transmission axes at right angles (90°), what happens to the intensity of the transmitted light?

-> ANSWER : It becomes zero

SOLUTION / DERIVATION:
According to Malus's Law: I = I_0 * cos^2(theta). When the transmission axes of two polarizing filters are perpendicular (crossed polarizers, theta = 90°): cos(90°) = 0 => I = I_0 * (0)^2 = 0. All light polarized by the first filter is completely absorbed/blocked by the second filter.

TIP FOR UNDERSTANDING: *Unpolarized light through 1st polarizer drops by 50% (I_1 = I_unpol / 2). Passing through 2nd polarizer follows I_2 = I_1 * cos^2(theta).*

ITEM #003 / 100

QUESTION: A project requires an initial investment of P500,000 and generates P100,000 per year. Ignoring interest, what is its payback period?

-> ANSWER : 5 years

SOLUTION / DERIVATION:
Simple Payback Period (ignoring interest): Payback Period = Initial Investment / Annual Cash Inflow = P500,000 / P100,000/year = 5.0 years.

TIP FOR UNDERSTANDING: *Simple payback ignores the time value of money and cash flows occurring after the payback threshold.*

ITEM #004 / 100

QUESTION: A radioactive sample has a half-life of 5 hours. After 15 hours, what fraction remains?

-> ANSWER : 1/8

SOLUTION / DERIVATION:
Radioactive decay formula: N(t) = N_0 * (1/2)^(t / t_half). Given half-life t_half = 5 hours and elapsed time t = 15 hours: Number of half-lives n = 15 / 5 = 3. Fraction remaining = (1/2)^3 = 1/8.

TIP FOR UNDERSTANDING: *After 1 half-life: 1/2 remains; after 2 half-lives: 1/4 remains; after 3 half-lives: 1/8 remains; after 4 half-lives: 1/16 remains.*

ITEM #005 / 100

QUESTION: In object-oriented analysis and design, what does the principle of low coupling signify?

-> ANSWER : A class has minimal dependencies and interacts loosely with others

SOLUTION / DERIVATION:
Coupling measures the degree of interdependence between software modules or classes. 'Low coupling' (loose coupling) means classes have minimal dependencies on each other's internal implementations, making the codebase modular, easy to test, maintain, and refactor without ripple effects.

TIP FOR UNDERSTANDING: *Good software engineering design strives for 'High Cohesion and Low Coupling'.*

ITEM #006 / 100

QUESTION: Which Philippine law is known as the Clean Air Act?

-> ANSWER : Republic Act 8749

SOLUTION / DERIVATION:
Republic Act No. 8749 is the 'Philippine Clean Air Act of 1999', enacted to achieve and maintain clean air that meets national ambient air quality standards and penalize vehicular and industrial polluters.

TIP FOR UNDERSTANDING: *Memorize ECE board environmental laws: RA 8749 = Clean Air Act; RA 9275 = Clean Water Act; RA 9003 = Ecological Solid Waste Management Act; RA 6969 = Toxic Substances and Hazardous Waste.*

ITEM #007 / 100

QUESTION: Which of the following best describes fossil fuels?

-> ANSWER : Formed from decomposed plant and animal matter over millions of years under heat and pressure

SOLUTION / DERIVATION:
Fossil fuels (coal, crude oil, petroleum, and natural gas) are hydrocarbon deposits formed from the ancient organic remains of prehistoric plants and aquatic microorganisms buried under layers of sediment and subjected to intense heat and pressure over millions of years.

TIP FOR UNDERSTANDING: *Fossil fuels are non-renewable because their natural formation requires geological epochs (millions of years).*

ITEM #008 / 100

QUESTION: A company states: "To be the leading renewable energy company in Southeast Asia by 2040." This statement is best classified as:

-> ANSWER : Vision

SOLUTION / DERIVATION:
A Vision Statement articulates the long-term, aspirational future state of an organization ('where we want to be'). A Mission Statement describes its current core purpose and primary business operations ('what we do and who we serve').

TIP FOR UNDERSTANDING: *Vision = Future destination; Mission = Present purpose; Objectives = Quantifiable interim milestones.*

ITEM #009 / 100

QUESTION: Why do ceramics typically have low thermal expansion compared to metals?

-> ANSWER : Their strong ionic/covalent bonds restrict atomic vibration

SOLUTION / DERIVATION:
Ceramics are bonded primarily by strong, directional ionic and covalent bonds with deep interatomic potential energy wells. These rigid bonds restrict the amplitude of thermal atomic vibrations, resulting in low coefficients of thermal expansion (CTE) and high melting points compared to metallic bonding.

TIP FOR UNDERSTANDING: *Metals have non-directional metallic bonds with free electron seas, allowing greater thermal lattice expansion than ceramics.*

ITEM #010 / 100

QUESTION: What is the main purpose of garbage collection in high-level programming languages?

-> ANSWER : To manage memory by automatically freeing unused objects and resources

SOLUTION / DERIVATION:
Garbage collection (GC) in managed runtimes (Java, Python, C#, Go) automatically scans heap memory, identifies objects that are no longer reachable by any active reference, and deallocates their memory back to the OS, preventing memory leaks and dangling pointers.

TIP FOR UNDERSTANDING: *Unmanaged languages like C/C++ require explicit manual memory management via malloc()/free() or new/delete.*

ITEM #011 / 100

QUESTION: Given a cache access time of 2 ns, a main memory access time of 80 ns, and a cache hit rate of 90%, what is the effective memory access time?

-> ANSWER : 10 ns

SOLUTION / DERIVATION:
Effective Memory Access Time (EMAT) formula: EMAT = (Hit Rate * T_cache) + (Miss Rate * T_main) = (0.90 * 2 ns) + (0.10 * 80 ns) = 1.8 ns + 8.0 ns = 9.8 ns ≈ 10 ns. (Or if cache is accessed first before miss penalty: EMAT = T_cache + Miss_Rate * T_main = 2 + 0.10(80) = 10 ns).

TIP FOR UNDERSTANDING: *A 90% cache hit rate dramatically cuts average memory latency from 80 ns down to 10 ns.*

ITEM #012 / 100

QUESTION: What is the main reason biofuels are regarded as sustainable energy sources?

-> ANSWER : They are derived from renewable biological materials

SOLUTION / DERIVATION:
Biofuels (such as bioethanol from sugarcane/corn and biodiesel from coconut/soybean) are derived from recently living biomass. Because crops absorb atmospheric CO2 through photosynthesis during growth, their combustion is roughly carbon-neutral and replenishable year after year.

TIP FOR UNDERSTANDING: *Biofuels recycle modern atmospheric carbon; fossil fuels release sequestered prehistoric carbon.*

ITEM #013 / 100

QUESTION: Which factor does not affect the properties of a polymer?

-> ANSWER : Isotopic mass of hydrogen

SOLUTION / DERIVATION:
Polymer properties are governed by macromolecular structure: degree of polymerization (chain length), branching, cross-linking, tacticity, crystallinity, and molecular weight distribution. The microscopic isotopic mass of hydrogen (H-1 vs deuterium) has no observable effect on bulk engineering mechanical properties.

TIP FOR UNDERSTANDING: *Key polymer determiners: Molecular weight, chain branching, cross-linking, crystallinity, and glass transition temperature (T_g).*

ITEM #014 / 100

QUESTION: A company discovers that customer acquisition costs are increasing faster than customer lifetime value. Which conclusion is most appropriate?

-> ANSWER : The business model may become financially unsustainable

SOLUTION / DERIVATION:
Customer Acquisition Cost (CAC) is the expense incurred to gain a new customer; Customer Lifetime Value (LTV) is the total net profit anticipated from that customer over the relationship. If CAC exceeds LTV (or CAC grows faster than LTV), the company loses money on every incremental user, rendering the business model financially unsustainable.

TIP FOR UNDERSTANDING: *A healthy unit economics benchmark requires an LTV:CAC ratio of at least 3:1.*

ITEM #015 / 100

QUESTION: An Electronics Engineer attends an IECEP national convention to earn CPD units and learn about emerging technologies. This activity primarily demonstrates IECEP's role in:

-> ANSWER : Professional development and lifelong learning

SOLUTION / DERIVATION:
The Institute of Electronics Engineers of the Philippines (IECEP) is the Accredited Integrated Professional Organization (AIPO) for ECEs. Conventions, technical seminars, and Continuing Professional Development (CPD) programs fulfill IECEP's statutory role in lifelong learning and professional advancement.

TIP FOR UNDERSTANDING: *RA 10912 mandates CPD credit units for the periodic renewal of PRC Professional Identification Cards.*

ITEM #016 / 100

QUESTION: What distinguishes renewable natural resources from non-renewable ones?

-> ANSWER : Renewable resources can regenerate naturally within a human lifespan and sustain long-term use

SOLUTION / DERIVATION:
Renewable resources (solar, wind, biomass, timber) naturally replenish through biological cycles or solar energy at a rate equal to or faster than their consumption rate within human generational lifespans.

TIP FOR UNDERSTANDING: *Non-renewable resources (coal, oil, uranium) exist in finite fixed reserves and deplete with extraction.*

ITEM #017 / 100

QUESTION: Which government agency is tasked with policy formulation under RA 12234 or the "Konektadong Pinoy Act"?

-> ANSWER : DICT

SOLUTION / DERIVATION:
Republic Act No. 12234, officially known as the 'Konektadong Pinoy Act' (Open Access in Data Transmission Act), designates the Department of Information and Communications Technology (DICT) as the primary policy-making, planning, coordinating, and implementing government agency.

TIP FOR UNDERSTANDING: *DICT sets policy and technical frameworks; NTC acts as the attached regulatory body overseeing spectrum and compliance.*

ITEM #018 / 100

QUESTION: A tourism development master plan for coastal resorts is being prepared. Which process ensures that marine biodiversity and cultural heritage are considered?

-> ANSWER : Strategic Environmental Assessment

SOLUTION / DERIVATION:
A Strategic Environmental Assessment (SEA) systematically evaluates the environmental, social, and cultural consequences of proposed government master plans, policies, or multi-sector regional programs before project-level implementation, ensuring biodiversity and coastal heritage protection.

TIP FOR UNDERSTANDING: *SEA applies at the high policy/plan level; EIA (Environmental Impact Assessment) applies at the individual project level.*

ITEM #019 / 100

QUESTION: According to Bernoulli's principle, as the velocity of an ideal fluid increases, its:

-> ANSWER : Pressure decreases

SOLUTION / DERIVATION:
Bernoulli's equation for steady, incompressible, frictionless fluid flow along a streamline: P + (1/2) rho v^2 + rho g h = constant. Assuming horizontal flow (constant elevation h), an increase in fluid velocity v must result in a simultaneous decrease in static pressure P.

TIP FOR UNDERSTANDING: *Bernoulli's principle explains aerodynamic lift: faster airflow over the curved upper surface of an airfoil creates a low-pressure zone above the wing.*

ITEM #020 / 100

QUESTION: A point charge is surrounded by a spherical Gaussian surface. If the radius of the sphere is doubled, the electric flux through the surface:

-> ANSWER : Remains unchanged

SOLUTION / DERIVATION:
By Gauss's Law for electrostatics: Total electric flux Phi_E through any closed surface is: Phi_E = Q_enclosed / epsilon_0. Because the net charge Q_enclosed inside the sphere remains unchanged, doubling the radius of the sphere has zero effect on the total flux.

TIP FOR UNDERSTANDING: *Total flux depends exclusively on the enclosed charge, not on the size or shape of the enclosing Gaussian surface.*

[SECTION 2 : QUESTIONS 021 - 040]

ITEM #021 / 100

QUESTION: At the equilibrium position of a mass-spring system, the energy is entirely:

-> ANSWER : Kinetic energy

SOLUTION / DERIVATION:
In simple harmonic motion (mass-spring system): Total Mechanical Energy E = (1/2) k x^2 + (1/2) m v^2. At the equilibrium position (x = 0), spring displacement is zero, so elastic potential energy is zero (U = 0). Velocity is at its maximum (v = v_max), so the total energy is entirely kinetic energy.

TIP FOR UNDERSTANDING: *At extremes (x = ±A): Energy is 100% potential; at equilibrium (x = 0): Energy is 100% kinetic.*

ITEM #022 / 100

QUESTION: A single slit is 0.20 mm wide and illuminated with 500 nm light. The screen is 2 m away. What is the distance from the central maximum to the first minimum?

-> ANSWER : 5.0 mm

SOLUTION / DERIVATION:
Single-slit diffraction minimum condition: a * sin(theta) = m * lambda. For the first minimum (m = 1) with small-angle approximation sin(theta) ≈ y / L: y = (L * lambda) / a Given slit width a = 0.20 mm = 2 × 10^-4 m, wavelength lambda = 500 nm = 5 × 10^-7 m, screen distance L = 2 m: y = [2 m * (5 × 10^-7 m)] / (2 × 10^-4 m) = (10^-6) / (2 × 10^-4) = 5 × 10^-3 m = 5.0 mm.

TIP FOR UNDERSTANDING: *Distance from central maximum to first minimum is y_1 = L*lambda/a; full width of central peak is 2*y_1 = 2*L*lambda/a = 10.0 mm.*

ITEM #023 / 100

QUESTION: Which CAD feature allows a designer to reuse frequently used components such as symbols or standard parts?

-> ANSWER : Blocks

SOLUTION / DERIVATION:
In Computer-Aided Design (AutoCAD), 'Blocks' are named collections of geometric objects combined to form a single reusable 2D or 3D component, allowing designers to insert repeated standardized symbols (e.g. resistors, logic gates, valves) efficiently.

TIP FOR UNDERSTANDING: *Using blocks reduces drawing file size and allows global updating: modifying a block definition automatically updates all inserted instances.*

ITEM #024 / 100

QUESTION: A 2-kg object moves in a circle of radius 4 m at a speed of 8 m/s. What is the required centripetal force?

-> ANSWER : 32 N

SOLUTION / DERIVATION:
Centripetal force formula: F_c = m * a_c = m * (v^2 / r). Given mass m = 2 kg, velocity v = 8 m/s, radius r = 4 m: F_c = 2 * (8)^2 / 4 = 2 * (64) / 4 = 128 / 4 = 32 N.

TIP FOR UNDERSTANDING: *Centripetal force always acts perpendicular to instantaneous velocity, directed inward toward the center of curvature.*

ITEM #025 / 100

QUESTION: An electronics company introduces standardized work procedures to ensure that every product meets the same specifications. Which operations objective is primarily addressed?

-> ANSWER : Quality consistency

SOLUTION / DERIVATION:
Standard Operating Procedures (SOPs) establish uniform, repeatable execution parameters across production cycles, minimizing operator variability and ensuring consistent product quality meeting predefined engineering specifications.

TIP FOR UNDERSTANDING: *Five core operations performance objectives: Quality, Speed, Dependability, Flexibility, and Cost.*

ITEM #026 / 100

QUESTION: Which of the following is a key reason why nanomaterials often have lower melting points compared to their bulk counterparts?

-> ANSWER : Higher surface-to-volume ratio

SOLUTION / DERIVATION:
In nanomaterials (particles < 100 nm), the surface-area-to-volume ratio increases dramatically. A large fraction of total atoms reside on the surface with uncoordinated, broken bonds and higher surface free energy, requiring less thermal kinetic energy to overcome cohesive forces and initiate melting.

TIP FOR UNDERSTANDING: *Melting point depression at the nanoscale is described by the Gibbs-Thomson relation.*

ITEM #027 / 100

QUESTION: Which of the following pollutants is primarily responsible for eutrophication in water bodies?

-> ANSWER : Nitrates and phosphates

SOLUTION / DERIVATION:
Eutrophication is caused by excessive nutrient runoff—specifically nitrates (NO3-) and phosphates (PO4 3-) from agricultural synthetic fertilizers and untreated municipal sewage, triggering algal blooms that deplete dissolved oxygen upon death and decay.

TIP FOR UNDERSTANDING: *Algal bloom -> Sunlight blocked -> Submerged plants die -> Bacterial decomposition consumes O2 -> Hypoxia/Fish kill.*

ITEM #028 / 100

QUESTION: What is otherwise known as "PRC modernization law"?

-> ANSWER : Republic Act No. 8981

SOLUTION / DERIVATION:
Republic Act No. 8981 is the 'PRC Modernization Act of 2000', which modernized the Professional Regulation Commission, granting it comprehensive regulatory, quasi-legislative, and quasi-judicial powers over 46 professional regulatory boards.

TIP FOR UNDERSTANDING: *RA 8981 repealed PD 223, establishing standardized examination automation and PRC administrative hierarchy.*

ITEM #029 / 100

QUESTION: Which of the following is a major consequence of habitat fragmentation?

-> ANSWER : Isolation of species leading to decreased genetic diversity and population decline

SOLUTION / DERIVATION:
Habitat fragmentation breaks large, contiguous ecological biomes into isolated patches separated by roads and urban development. This restricts animal migration, shrinks breeding gene pools, amplifies the 'edge effect', and leads to inbreeding depression and local extinction.

TIP FOR UNDERSTANDING: *Habitat corridors are engineering wildlife crossings designed to mitigate habitat fragmentation.*

ITEM #030 / 100

QUESTION: Which of the following is a common GUI framework for desktop applications in Python?

-> ANSWER : Tkinter

SOLUTION / DERIVATION:
Tkinter is Python's standard built-in de-facto Graphical User Interface (GUI) package, acting as an object-oriented Python wrapper around the Tcl/Tk cross-platform windowing toolkit.

TIP FOR UNDERSTANDING: *Python GUI toolkits: Tkinter (built-in standard), PyQt/PySide (Qt-based advanced), wxPython, Kivy (touch/mobile).*

ITEM #031 / 100

QUESTION: A machine claims to convert all absorbed heat into work without rejecting any heat to the surroundings. This machine violates:

-> ANSWER : Second Law of Thermodynamics (Kelvin-Planck statement)

SOLUTION / DERIVATION:
The Kelvin-Planck statement of the Second Law of Thermodynamics states: 'It is impossible for any device operating on a thermodynamic cycle to receive heat from a single thermal reservoir and produce a net amount of work.' A heat engine must reject waste heat to a low-temperature sink; 100% heat-to-work conversion is physically impossible.

TIP FOR UNDERSTANDING: *A perpetual motion machine of the second kind (PMM2) claims 100% thermal efficiency (Q_L = 0), violating the 2nd Law.*

ITEM #032 / 100

QUESTION: An electronics company introduces standardized work procedures to ensure that every product meets the same specifications. Which operations objective is primarily addressed?

-> ANSWER : Quality consistency

SOLUTION / DERIVATION:
Standardized work procedures eliminate variation in task execution, ensuring that every assembly or manufacturing step adheres strictly to tolerance limits and quality standards.

TIP FOR UNDERSTANDING: *Standardization is a cornerstone of Total Quality Management (TQM) and ISO 9001 certification.*

ITEM #033 / 100

QUESTION: How many atoms are effectively shared in a simple cubic unit cell corner atom?

-> ANSWER : 1/8

SOLUTION / DERIVATION:
In a simple cubic (SC) crystal lattice, 8 identical corner atoms surround the unit cell. Each corner atom is shared symmetrically among 8 adjacent unit cells sharing that vertex, contributing exactly (1/8) of an atom to the interior of each unit cell: 8 corners * (1/8) = 1 net atom per SC cell.

TIP FOR UNDERSTANDING: *SC: 8 * (1/8) = 1 atom/cell; BCC: 8*(1/8) + 1 center = 2 atoms/cell; FCC: 8*(1/8) + 6*(1/2) faces = 4 atoms/cell.*

ITEM #034 / 100

QUESTION: During nuclear fission, what happens to the atomic nucleus?

-> ANSWER : It splits into two smaller nuclei, releasing energy

SOLUTION / DERIVATION:
During nuclear fission, a heavy, unstable nucleus (e.g. Uranium-235) absorbs a thermal neutron and splits into two lighter daughter nuclei (fission fragments), releasing 2-3 prompt neutrons and massive kinetic energy according to Delta_E = Delta_m * c^2.

TIP FOR UNDERSTANDING: *Fission = Splitting of a heavy nucleus; Fusion = Combining of light nuclei (e.g. Deuterium + Tritium).*

ITEM #035 / 100

QUESTION: Which of the following best distinguishes crystalline materials from non-crystalline materials?

-> ANSWER : Long-range atomic order

SOLUTION / DERIVATION:
Crystalline materials (metals, quartz, silicon) possess long-range periodic translational order extending over thousands of unit cells. Non-crystalline (amorphous) materials (glass, polymers) exhibit only short-range chemical bonding order without periodic long-range lattice symmetry.

TIP FOR UNDERSTANDING: *Crystalline solids exhibit sharp melting points; amorphous solids soften gradually over a glass transition range.*

ITEM #036 / 100

QUESTION: What does the term "cumulative impact" refer to in Environmental Impact Assessment (EIA)?

-> ANSWER : The combined effects of past, present, and future projects within the same region

SOLUTION / DERIVATION:
In environmental planning, 'cumulative impact' represents the incremental environmental degradation resulting from the combined, aggregated effects of past existing infrastructure, the present proposed project, and reasonably foreseeable future developments within the same geographic watershed or airshed.

TIP FOR UNDERSTANDING: *Even if individual projects pass EIA thresholds, their cumulative impact can exceed the carrying capacity of an ecosystem.*

ITEM #037 / 100

QUESTION: A company compares two projects using sensitivity analysis. Project A remains profitable over a wide range of sales volumes, while Project B becomes unprofitable with a small decrease in sales. Which project has lower risk?

-> ANSWER : Project A

SOLUTION / DERIVATION:
Sensitivity analysis assesses how changes in input variables (e.g. sales volume, unit price) affect profitability. Project A remains profitable across wide sales fluctuations, demonstrating robust downside protection and lower financial risk, whereas Project B is highly sensitive and vulnerable to market downturns.

TIP FOR UNDERSTANDING: *Flatter NPV-sensitivity curves indicate lower risk; steep curves indicate high operational leverage and risk.*

ITEM #038 / 100

QUESTION: What is the effect of increasing temperature on diffusion in solids?

-> ANSWER : It increases diffusion rate

SOLUTION / DERIVATION:
Diffusion in solid crystalline materials follows the Arrhenius equation: D = D_0 * exp(-Q / [R * T]). Increasing temperature T increases atomic thermal kinetic energy, exponentially raising the probability that atoms overcome the activation energy barrier Q to jump into neighboring vacancies.

TIP FOR UNDERSTANDING: *Higher temperature = Exponentially higher diffusion rate (Fick's laws + Arrhenius temperature dependence).*

ITEM #039 / 100

QUESTION: Which Philippine law strengthened occupational safety and health standards in workplaces?

-> ANSWER : Republic Act No. 11058

SOLUTION / DERIVATION:
Republic Act No. 11058 is titled 'An Act Strengthening Compliance with Occupational Safety and Health Standards and Providing Penalties for Violations Thereof', mandating safe workplace practices, PPE provision, safety officers, and steep administrative fines for non-compliance.

TIP FOR UNDERSTANDING: *RA 11058 holds employers strictly liable for workplace health and safety compliance under DOLE inspection.*

ITEM #040 / 100

QUESTION: If an electron in hydrogen jumps from n = 5 to n = 2, the resulting radiation lies in:

-> ANSWER : Visible (Balmer series region)

SOLUTION / DERIVATION:
Electronic transitions terminating at the n = 2 energy level in a hydrogen atom belong to the Balmer Series. The transition from n = 5 to n = 2 emits a photon of wavelength lambda ≈ 434 nm (blue-violet light), which falls squarely within the visible electromagnetic spectrum.

TIP FOR UNDERSTANDING: *Lyman series (n -> 1): Ultraviolet; Balmer series (n -> 2): Visible; Paschen series (n -> 3): Infrared; Brackett series (n -> 4): Far Infrared.*

[SECTION 3 : QUESTIONS 041 - 060]

ITEM #041 / 100

QUESTION: Which Philippine law primarily promotes interconnection among public telecommunications entities to encourage competition and improve public service?

-> ANSWER : Republic Act No. 7925

SOLUTION / DERIVATION:
Republic Act No. 7925 is the 'Public Telecommunications Policy Act of the Philippines', which mandated compulsory interconnection among all public telecommunications entities (PTEs) to dismantle telecom monopolies and foster a competitive market.

TIP FOR UNDERSTANDING: *RA 7925 treats telecom facilities as common carriers required to interconnect without discrimination.*

ITEM #042 / 100

QUESTION: What does the Contingency Theory of Management state?

-> ANSWER : Managerial decisions should depend on the situation

SOLUTION / DERIVATION:
The Contingency (Situational) Theory of Management, pioneered by Fred Fiedler and others, asserts that there is no single 'best way' to manage or lead an organization. The optimal leadership style and managerial decision depend upon the internal and external contingencies of the specific situation.

TIP FOR UNDERSTANDING: *Classic Contingency rule: 'It depends.' Managerial actions must adapt to environmental volatility, team competence, and task structure.*

ITEM #043 / 100

QUESTION: Which of the following has the same unit as work?

-> ANSWER : Joule (or energy / torque unit N·m)

SOLUTION / DERIVATION:
Work is the dot product of force and displacement: W = F · d [N · m]. The SI unit of both work and energy is the Joule (1 J = 1 N · m = 1 kg · m^2 / s^2).

TIP FOR UNDERSTANDING: *Torque also has units of N·m, but work/energy is a scalar Joule, whereas torque is a pseudo-vector moment.*

ITEM #044 / 100

QUESTION: What is the main reason for using functions in programming?

-> ANSWER : To organize, reuse, and manage program code more effectively

SOLUTION / DERIVATION:
Functions (subroutines) encapsulate discrete computational tasks into reusable, modular blocks, preventing code duplication (DRY principle: Don't Repeat Yourself), enhancing readability, and simplifying debugging.

TIP FOR UNDERSTANDING: *Functions facilitate modular programming, abstraction, and scope isolation.*

ITEM #045 / 100

QUESTION: Which of the following explains the thermal conductivity of metals?

-> ANSWER : Motion of delocalized electrons

SOLUTION / DERIVATION:
In metals, valence electrons are not bound to individual atoms; they form a delocalized 'sea of conduction electrons'. Thermal energy is transferred rapidly throughout the lattice by the high-velocity kinetic motion and collisions of these free electrons (Wiedemann-Franz Law).

TIP FOR UNDERSTANDING: *Free electrons account for >90% of thermal and electrical conduction in metals; non-metals rely solely on slower lattice phonons.*

ITEM #046 / 100

QUESTION: What is a widget in graphical user interface (GUI) programming?

-> ANSWER : A graphical control element such as a button or text box

SOLUTION / DERIVATION:
In GUI toolkits (Tkinter, Qt, Windows Forms), a 'widget' (window gadget) is a modular graphical visual control component (such as a push button, label, checkbox, slider, or text entry field) that displays data and handles user events.

TIP FOR UNDERSTANDING: *Widgets fire callback events (e.g. onClick) to trigger application logic.*

ITEM #047 / 100

QUESTION: Which statement best describes the External Rate of Return (ERR) method?

-> ANSWER : It assumes interim cash flows are reinvested at an external rate, usually the MARR

SOLUTION / DERIVATION:
The External Rate of Return (ERR) method addresses the unrealistic reinvestment assumption of the standard IRR. ERR explicitly assumes that interim net positive cash flows are reinvested externally at the company's realistic Minimum Attractive Rate of Return (MARR) until project termination.

TIP FOR UNDERSTANDING: *IRR unrealistically assumes reinvestment at the IRR itself; ERR realistically assumes reinvestment at the MARR.*

ITEM #048 / 100

QUESTION: A company plans to introduce a premium electric bicycle designed specifically for urban professionals with high disposable incomes. Which marketing activity is being demonstrated?

-> ANSWER : Market segmentation

SOLUTION / DERIVATION:
Market segmentation is the process of dividing a broad heterogeneous consumer market into distinct, homogeneous subsets based on shared characteristics (e.g. demographics, income, lifestyle). Targeting high-income urban professionals specifically is an example of demographic and psychographic segmentation.

TIP FOR UNDERSTANDING: *STP framework: Segmentation -> Targeting -> Positioning.*

ITEM #049 / 100

QUESTION: A sound wave travels through air and hits a solid wall. Most of the wave energy is:

-> ANSWER : Reflected back into the air

SOLUTION / DERIVATION:
When an acoustic wave travelling in air encounters a rigid, dense solid wall, there is a massive acoustic impedance mismatch (Z_wall >> Z_air). By boundary transmission laws, the acoustic reflection coefficient is near 1.0, causing the vast majority of sound energy to be reflected back into the air as an echo.

TIP FOR UNDERSTANDING: *Impedance mismatch reflects wave energy; matching absorbs or transmits wave energy.*

ITEM #050 / 100

QUESTION: An engineer chooses the alternative with the highest Internal Rate of Return (IRR), but its IRR is still below the company's MARR. What should be done?

-> ANSWER : Reject the project

SOLUTION / DERIVATION:
In engineering economy, an investment alternative is economically acceptable if and only if its rate of return equals or exceeds the MARR: Accept if IRR >= MARR; Reject if IRR < MARR. Even if an alternative has the highest IRR among available projects, if it fails to clear the MARR hurdle, investing in it will destroy capital value.

TIP FOR UNDERSTANDING: *Never accept a project whose return is below the opportunity cost of capital (MARR).*

ITEM #051 / 100

QUESTION: Why is discounted decision tree analysis useful in engineering economy?

-> ANSWER : It evaluates risky decisions while accounting for the time value of money

SOLUTION / DERIVATION:
Decision trees graphically map sequential decision stages, uncertain future chance outcomes, and probabilistic payoffs. Discounted decision tree analysis discounts cash flows across branching paths back to present worth, evaluating risk while respecting the time value of money.

TIP FOR UNDERSTANDING: *Decision nodes (squares) represent choices; Chance nodes (circles) represent probabilistic outcomes.*

ITEM #052 / 100

QUESTION: If the radius of gyration of a body about an axis is doubled, its moment of inertia becomes:

-> ANSWER : Four times

SOLUTION / DERIVATION:
Moment of inertia relates to radius of gyration k by: I = m * k^2. If the radius of gyration is doubled (k' = 2k) while mass m is held constant: I' = m * (2k)^2 = 4 m k^2 = 4 I (it becomes four times as large).

TIP FOR UNDERSTANDING: *Moment of inertia scales with the square of the radius of gyration ($I \propto k^2$).*

ITEM #053 / 100

QUESTION: What is the use of a finally block in exception handling?

-> ANSWER : To execute code regardless of whether an exception occurred

SOLUTION / DERIVATION:
In try-catch-finally exception handling constructs, the 'finally' block contains critical cleanup code (such as closing database handles, releasing network sockets, or closing open file descriptors) that is guaranteed to execute unconditionally whether an exception is raised or not.

TIP FOR UNDERSTANDING: *Use 'finally' blocks to guarantee resource deallocation and prevent leaks.*

ITEM #054 / 100

QUESTION: A processor runs 5 billion instructions with an average CPI of 2 on a 2 GHz clock. What is the total execution time?

-> ANSWER : 5 seconds

SOLUTION / DERIVATION:
CPU execution time formula: Execution Time = (Instruction Count * CPI) / Clock Frequency Clock Cycles = 5 × 10^9 instructions * 2 cycles/instruction = 10 × 10^9 cycles. Time = (10 × 10^9 cycles) / (2 × 10^9 cycles/s) = 5.0 seconds.

TIP FOR UNDERSTANDING: *Execution Time = (IC * CPI) / f = (5e9 * 2) / 2e9 = 5 seconds.*

ITEM #055 / 100

QUESTION: Which Philippine law primarily protects intellectual property rights, including patents, trademarks, copyrights, and industrial designs?

-> ANSWER : Republic Act No. 8293

SOLUTION / DERIVATION:
Republic Act No. 8293 is the 'Intellectual Property Code of the Philippines', which consolidated and strengthened national laws governing patents, utility models, registered trademarks, copyrights, and industrial designs.

TIP FOR UNDERSTANDING: *Administered by the Intellectual Property Office of the Philippines (IPOPHL).*

ITEM #056 / 100

QUESTION: Which of the following quantities is dimensionless?

-> ANSWER : Poisson's ratio

SOLUTION / DERIVATION:
Poisson's ratio is the ratio of transverse strain to axial strain: nu = - (epsilon_transverse / epsilon_axial). Because strain is displacement per unit length (m/m, dimensionless), the ratio of two strains is purely dimensionless.

TIP FOR UNDERSTANDING: *Common dimensionless quantities: Strain, Poisson's ratio, Mach number, Reynolds number, friction factor.*

ITEM #057 / 100

QUESTION: Which color code on a chemical hazard diamond represents health hazard under NFPA (National Fire Protection Association)?

-> ANSWER : Blue

SOLUTION / DERIVATION:
The National Fire Protection Association (NFPA 704) standard uses a color-coded hazard diamond: - Blue (left quadrant): Health Hazard - Red (top quadrant): Flammability - Yellow (right quadrant): Instability / Reactivity - White (bottom quadrant): Special hazards (e.g. W with slash, OX).

TIP FOR UNDERSTANDING: *NFPA 704: Blue = Health; Red = Fire; Yellow = Reactivity; White = Special.*

ITEM #058 / 100

QUESTION: A company has a current ratio of 1.2 and a quick ratio of 0.8. What does this imply?

-> ANSWER : The company has excessive inventories

SOLUTION / DERIVATION:
Current Ratio = Current Assets / Current Liabilities = 1.2. Quick (Acid-Test) Ratio = (Current Assets - Inventory) / Current Liabilities = 0.8. The large drop from 1.2 to 0.8 reveals that illiquid inventory comprises a major fraction of short-term assets, implying potential overstocking or slow inventory turnover.

TIP FOR UNDERSTANDING: *Quick ratio isolates liquid assets (cash + receivables); a wide gap between current and quick ratios indicates heavy inventory dependency.*

ITEM #059 / 100

QUESTION: Which CAD model represents an object using only lines and curves without surfaces or volume?

-> ANSWER : Wireframe model

SOLUTION / DERIVATION:
A wireframe CAD model represents 3D physical objects strictly using boundary edges, vertices, and coordinate lines, devoid of surface skinning, face textures, mass properties, or internal volumetric definitions.

TIP FOR UNDERSTANDING: *CAD model progression: Wireframe (edges only) -> Surface (hollow skin) -> Solid (full volume, density, mass).*

ITEM #060 / 100

QUESTION: A heat engine receives 1200 kJ of heat from a high-temperature reservoir and rejects 750 kJ to the sink. Another engine receives the same heat input but rejects only 600 kJ. Which engine has the higher thermal efficiency?

-> ANSWER : The second engine

SOLUTION / DERIVATION:
Thermal efficiency of a heat engine: eta = Work Output / Heat Input = (Q_H - Q_L) / Q_H = 1 - (Q_L / Q_H). Both engines receive Q_H = 1200 kJ. Engine 1: eta_1 = 1 - 750 / 1200 = 1 - 0.625 = 0.375 (37.5%). Engine 2: eta_2 = 1 - 600 / 1200 = 1 - 0.500 = 0.500 (50.0%). Rejecting less waste heat means Engine 2 converts more heat into useful work, achieving higher thermal efficiency.

TIP FOR UNDERSTANDING: *Lower heat rejection Q_L for the same heat input Q_H directly translates to higher thermal efficiency.*

[SECTION 4 : QUESTIONS 061 - 080]

ITEM #061 / 100

QUESTION: A reversible heat engine operates between 600 K and 300 K. What is its maximum possible efficiency?

-> ANSWER : 50%

SOLUTION / DERIVATION:
Carnot efficiency sets the theoretical upper efficiency limit for any heat engine operating between two temperatures: eta_Carnot = 1 - (T_L / T_H) Given T_H = 600 K and T_L = 300 K: eta_max = 1 - (300 / 600) = 1 - 0.50 = 0.50 = 50%.

TIP FOR UNDERSTANDING: *Temperatures in thermodynamic efficiency formulas must strictly be expressed in absolute scale (Kelvin).*

ITEM #062 / 100

QUESTION: Which property best distinguishes a crystalline solid from an amorphous solid?

-> ANSWER : Crystalline solids have a regular, repeating atomic arrangement

SOLUTION / DERIVATION:
A crystalline solid is distinguished by a continuous, periodic 3D spatial arrangement of constituent atoms or molecules (crystal lattice), giving rise to sharp X-ray diffraction peaks, anisotropic physical properties, and a precise melting temperature.

TIP FOR UNDERSTANDING: *Crystalline: Long-range translational periodic order; Amorphous: Random, disordered network without long-range repetition.*

ITEM #063 / 100

QUESTION: Two mutually exclusive projects both satisfy the company's MARR. Which project should generally be selected?

-> ANSWER : The one with the highest economic worth (NPV/PW)

SOLUTION / DERIVATION:
For mutually exclusive projects, alternatives compete directly and only one can be chosen. When projects meet or exceed the MARR hurdle rate, the decision criterion is to select the alternative that maximizes total net economic worth (highest Net Present Worth / NPV).

TIP FOR UNDERSTANDING: *Do NOT simply pick the highest IRR for mutually exclusive projects, as IRR fails to account for scale/investment size. Use incremental analysis or highest NPW.*

ITEM #064 / 100

QUESTION: When was Republic Act No. 9292 approved by President Gloria Macapagal Arroyo?

-> ANSWER : April 17, 2004

SOLUTION / DERIVATION:
Republic Act No. 9292, titled 'The Electronics Engineering Law of 2004', was officially signed and enacted into law by President Gloria Macapagal-Arroyo on April 17, 2004.

TIP FOR UNDERSTANDING: *RA 9292 repealed Republic Act No. 5734 (The Electronics and Communications Engineering Act of the Philippines).*

ITEM #065 / 100

QUESTION: In workplace safety regulations, the acronym PPE is a critical requirement. What does PPE specifically stand for?

-> ANSWER : Personal Protective Equipment

SOLUTION / DERIVATION:
In occupational safety and health (OSH) terminology, PPE stands for Personal Protective Equipment, encompassing gear (helmets, eye protection, safety footwear, respirators, harnesses) designed to protect workers from occupational hazards.

TIP FOR UNDERSTANDING: *Under OSH hierarchy of controls, PPE is the last line of defense after elimination, substitution, engineering controls, and administrative controls.*

ITEM #066 / 100

QUESTION: In legal terms, what does the term "motu proprio" mean?

-> ANSWER : By one's own initiative or decision without external prompting

SOLUTION / DERIVATION:
'Motu proprio' is a Latin legal term meaning 'on one's own impulse/initiative'. It signifies that a court, administrative board, or legal authority acts on its own official initiative without requiring a formal motion or complaint from an outside party.

TIP FOR UNDERSTANDING: *PRC Professional Regulatory Boards can initiate investigations motu proprio regarding professional malpractice.*

ITEM #067 / 100

QUESTION: Which surface will cool down fastest via radiation, assuming identical temperatures?

-> ANSWER : Black and rough

SOLUTION / DERIVATION:
Stefan-Boltzmann Law: Radiated thermal power P = epsilon * sigma * A * T^4. A matte, rough black surface approaches an ideal blackbody with emissivity epsilon ≈ 1.0, making it both the best possible absorber and the fastest radiator/emitter of thermal energy.

TIP FOR UNDERSTANDING: *Good absorbers are good emitters (Kirchhoff's Law of Thermal Radiation). Dark and rough radiates fastest; polished silver radiates slowest.*

ITEM #068 / 100

QUESTION: Three identical resistors are connected in series to a battery. If one resistor is removed and replaced by a wire, what happens to the current in the circuit?

-> ANSWER : It increases

SOLUTION / DERIVATION:
In a series circuit with 3 identical resistors: R_total = 3R => I = V / (3R). Replacing one resistor with a short-circuit jumper wire leaves two resistors in series: R'_total = 2R. New circuit current: I' = V / (2R) = 1.5 * I. Because total resistance drops from 3R to 2R, circuit current increases by 50%.

TIP FOR UNDERSTANDING: *Reducing circuit series resistance increases current: I = V / R_total.*

ITEM #069 / 100

QUESTION: Which particle-like excitations describe lattice vibrations in a crystal?

-> ANSWER : Phonons

SOLUTION / DERIVATION:
In solid-state physics, a phonon is a quantized quasiparticle representing the collective vibrational modes of atoms propagating through an elastic crystalline lattice (quantized acoustic or optical lattice waves).

TIP FOR UNDERSTANDING: *Photons = Quantized electromagnetic energy; Phonons = Quantized vibrational/sound energy in crystal lattices.*

ITEM #070 / 100

QUESTION: Which architecture uses separate memories for program instructions and data, allowing simultaneous access?

-> ANSWER : Harvard Architecture

SOLUTION / DERIVATION:
The Harvard Architecture employs physically separate storage memories and dedicated bus lines for program code (instructions) and data, allowing the CPU to fetch an instruction and read/write operand data simultaneously in the same clock cycle.

TIP FOR UNDERSTANDING: *Von Neumann architecture shares a single memory bus for code and data (bottleneck); Harvard uses dual separate buses.*

ITEM #071 / 100

QUESTION: The rotational kinetic energy of a rigid body is:

-> ANSWER : (1/2) I omega^2

SOLUTION / DERIVATION:
Rotational kinetic energy of a body rotating about a fixed axis with moment of inertia I and angular velocity omega is: KE_rot = (1/2) I omega^2. This is the direct rotational analogue of translational kinetic energy KE_trans = (1/2) m v^2.

TIP FOR UNDERSTANDING: *Linear: m <-> Rotational: I; Linear: v <-> Rotational: omega; Linear: p = mv <-> Rotational: L = I*omega.*

ITEM #072 / 100

QUESTION: Which statement reflects the interdependency of management functions?

-> ANSWER : Leading relies on a well-structured organization and clear plans

SOLUTION / DERIVATION:
The core management functions (Planning, Organizing, Leading, Controlling - POLC) are interdependent. Effective leadership requires clear strategic plans to communicate direction and a structured organizational hierarchy to execute responsibilities.

TIP FOR UNDERSTANDING: *Planning sets objectives -> Organizing allocates resources -> Leading motivates personnel -> Controlling monitors performance.*

ITEM #073 / 100

QUESTION: What is the primary advantage of polymorphism in object-oriented programming?

-> ANSWER : It allows functions or methods to behave differently based on input (or object type)

SOLUTION / DERIVATION:
Polymorphism (meaning 'many forms') allows an entity (such as a method, function, or operator) to execute different underlying behaviors based on the specific subclass object invoking it at runtime (dynamic method dispatch).

TIP FOR UNDERSTANDING: *Polymorphism enables loose coupling and extensible code: base class pointers can invoke overridden derived class methods uniformly.*

ITEM #074 / 100

QUESTION: What is the primary purpose of the Miller indices (hkl) in crystallography?

-> ANSWER : To specify the orientation of crystal planes

SOLUTION / DERIVATION:
Miller indices (hkl) form a standard crystallographic notation system using reciprocal lattice intercepts to uniquely designate the spatial orientation of parallel planes of atoms within a crystal lattice.

TIP FOR UNDERSTANDING: *Miller indices procedure: (1) Find intercepts on unit cell axes, (2) Take reciprocals, (3) Clear fractions to smallest integers (hkl).*

ITEM #075 / 100

QUESTION: Why is differentiation important in a value proposition?

-> ANSWER : It helps customers understand why your solution is better or unique

SOLUTION / DERIVATION:
In entrepreneurship, a value proposition explains how a company's product solves customer problems uniquely. Differentiation highlights distinctive competitive advantages, enabling prospective buyers to understand why this solution is superior to competing alternatives.

TIP FOR UNDERSTANDING: *Without clear differentiation, products devolve into commoditized price competition.*

ITEM #076 / 100

QUESTION: A startup has completed its business plan and is now assigning responsibilities, allocating resources, and setting timelines. Which phase of management is being performed?

-> ANSWER : Business plan execution (Organizing / Implementation)

SOLUTION / DERIVATION:
Once strategic planning is complete, translating goals into reality by allocating resources, assigning task ownership, and tracking milestone timelines falls under the implementation and execution phase of management.

TIP FOR UNDERSTANDING: *Plan the work, then work the plan: execution operationalizes strategic planning.*

ITEM #077 / 100

QUESTION: What type of crystal imperfection occurs when an atom is displaced from its normal lattice position to an interstitial site within the same crystal?

-> ANSWER : Frenkel defect

SOLUTION / DERIVATION:
A Frenkel defect is a point defect in ionic crystals where an atom or ion (usually a smaller cation) vacates its normal lattice site, creating a vacancy, and relocates into a nearby interstitial position within the crystal.

TIP FOR UNDERSTANDING: *Schottky defect = Pair of cation and anion vacancies; Frenkel defect = Vacancy-interstitial combination without mass loss.*

ITEM #078 / 100

QUESTION: A projectile is launched at 40 m/s at an angle of 30°. What is its horizontal range?

-> ANSWER : 141 m

SOLUTION / DERIVATION:
Horizontal range of a projectile launched over level ground: R = [v_0^2 * sin(2 * theta)] / g Given v_0 = 40 m/s, theta = 30°, g = 9.81 m/s^2: R = [(40)^2 * sin(2 * 30°)] / 9.81 = [1600 * sin(60°)] / 9.81 = (1600 * 0.8660) / 9.81 = 1385.64 / 9.81 ≈ 141.25 m ≈ 141 m.

TIP FOR UNDERSTANDING: *Maximum range occurs at theta = 45° where sin(2*theta) = sin(90°) = 1. Launch angles that sum to 90° (e.g. 30° and 60°) achieve identical horizontal range.*

ITEM #079 / 100

QUESTION: What is the minimum punishment imposed to any person who shall give any false or fraudulent statement to the Board to obtain a Certificate of Registration and/or Professional ID Card as Electronic Engineer?

-> ANSWER : P 100,000 or imprisonment of 6 months

SOLUTION / DERIVATION:
Under Section 35 (Penal Provision) of Republic Act No. 9292: Any person submitting false or fraudulent statements/evidence to the Board shall be punished by a fine of not less than P100,000 nor more than P1,000,000, or imprisonment of not less than 6 months nor more than 6 years, or both. Thus, the statutory minimum punishment is a fine of P100,000 or imprisonment of 6 months.

TIP FOR UNDERSTANDING: *Section 35, RA 9292: Minimum = P100,000 / 6 months; Maximum = P1,000,000 / 6 years.*

ITEM #080 / 100

QUESTION: A company concludes that favorable government policies, low inflation, changing consumer preferences, and rapid digital innovation will significantly affect its future strategy. Which statement best describes this conclusion?

-> ANSWER : The company integrated all four dimensions of PEST analysis into strategic planning

SOLUTION / DERIVATION:
PEST analysis examines macro-environmental influences: - Political: Favorable government policies - Economic: Low inflation - Social: Changing consumer preferences - Technological: Rapid digital innovation. Synthesizing all four factors confirms full integration of PEST dimensions into strategic planning.

TIP FOR UNDERSTANDING: *PEST (or PESTLE: Political, Economic, Social, Technological, Legal, Environmental) assesses external macro trends.*

[SECTION 5 : QUESTIONS 081 - 100]

ITEM #081 / 100

QUESTION: In rechargeable batteries, why does capacity tend to decrease with repeated cycles?

-> ANSWER : Irreversible chemical changes and electrode degradation

SOLUTION / DERIVATION:
Repeated charge-discharge cycles in secondary cells (Li-ion) induce irreversible degradation: growth of the solid electrolyte interphase (SEI layer), active lithium plating, microcracking of cathode materials, and loss of active electrode surface area.

TIP FOR UNDERSTANDING: *Battery capacity fade stems from parasitic chemical reactions and mechanical lattice fatigue during ion intercalation.*

ITEM #082 / 100

QUESTION: Which statement best distinguishes crystalline materials from amorphous materials?

-> ANSWER : Crystalline materials possess long-range atomic order

SOLUTION / DERIVATION:
Crystalline materials feature long-range atomic order where atoms occupy regular, repeating lattice positions extending across macroscopic dimensions. Amorphous solids lack this long-range periodic translational order, exhibiting only short-range nearest-neighbor bonding.

TIP FOR UNDERSTANDING: *X-ray diffraction (XRD) produces sharp Bragg peaks for crystalline structures and broad diffuse halos for amorphous structures.*

ITEM #083 / 100

QUESTION: An RLC circuit has R = 40 Ω, X_L = 60 Ω, and X_C = 20 Ω. Another circuit has the same resistance but X_L = 20 Ω and X_C = 60 Ω. Which statement is true?

-> ANSWER : Both circuits have the same impedance (Z = 56.57 Ω)

SOLUTION / DERIVATION:
Total series circuit impedance magnitude: Z = sqrt(R^2 + (X_L - X_C)^2). Circuit 1: Z_1 = sqrt(40^2 + (60 - 20)^2) = sqrt(1600 + 40^2) = sqrt(3200) = 56.57 Ω. Circuit 2: Z_2 = sqrt(40^2 + (20 - 60)^2) = sqrt(1600 + (-40)^2) = sqrt(3200) = 56.57 Ω. Because the net reactive difference |X_L - X_C| = 40 Ω in both circuits, their total impedance magnitudes are identical.

TIP FOR UNDERSTANDING: *Impedance magnitude depends on the square of net reactance (X_L - X_C)^2; inductive or capacitive dominance changes phase sign, not magnitude.*

ITEM #084 / 100

QUESTION: What term refers to the maximum stress a material can withstand before failure occurs?

-> ANSWER : Ultimate tensile strength

SOLUTION / DERIVATION:
Ultimate Tensile Strength (UTS) is the maximum engineering stress recorded on the engineering stress-strain curve before necking and macroscopic rupture occur: UTS = Maximum Tensile Force / Original Cross-Sectional Area.

TIP FOR UNDERSTANDING: *Yield strength marks the boundary of plastic deformation; Ultimate strength marks the maximum stress capacity.*

ITEM #085 / 100

QUESTION: Why are fossil fuels considered unsustainable?

-> ANSWER : They emit greenhouse gases and are finite in supply, leading to long-term depletion

SOLUTION / DERIVATION:
Fossil fuels are unsustainable because their reserves are geologically finite (irreversible depletion), and their combustion releases massive volumes of greenhouse gases (CO2, NOx, SO2) driving climate destabilization.

TIP FOR UNDERSTANDING: *Sustainability requires meeting current energy needs without compromising the ability of future generations to meet theirs.*

ITEM #086 / 100

QUESTION: Which of the following does not constitute the foundation of ethics?

-> ANSWER : Integrity (or personal bias / convenience; here contextual to question choices)

SOLUTION / DERIVATION:
Ethical frameworks are grounded in universal moral virtues: honesty, justice, fairness, respect for autonomy, and integrity. In standard philosophical ethics, personal bias or subjective expedience does not constitute a moral foundation.

TIP FOR UNDERSTANDING: *Professional ethics requires placing public safety, health, and welfare above self-interest (Code of Ethics for ECEs).*

ITEM #087 / 100

QUESTION: Which of the following statements is true about methane (CH4) as a greenhouse gas?

-> ANSWER : It is more potent than CO2 but lasts for a shorter time

SOLUTION / DERIVATION:
Methane (CH4) has a Global Warming Potential (GWP) roughly 28 to 36 times higher than CO2 over a 100-year horizon (and >80x over 20 years), but its atmospheric lifespan is relatively short (~12 years) compared to centuries for carbon dioxide.

TIP FOR UNDERSTANDING: *Methane: Short atmospheric residence time, but vastly higher radiative thermal forcing per molecule than CO2.*

ITEM #088 / 100

QUESTION: In business ethics, an ethical dilemma arises when a decision involves a conflict between moral principles. Which situation best illustrates an ethical dilemma?

-> ANSWER : Deciding whether to report internal fraud

SOLUTION / DERIVATION:
An ethical dilemma occurs when two compelling moral duties clash, requiring a choice where following one principle forces the violation of another. Deciding whether to blow the whistle on internal corporate fraud pits loyalty to colleagues/employer against honesty, legality, and public welfare.

TIP FOR UNDERSTANDING: *Whistleblowing dilemmas involve conflicting duties: Corporate loyalty vs. Public protection.*

ITEM #089 / 100

QUESTION: Technopreneurs often apply effectual thinking. What does this approach emphasize?

-> ANSWER : Building ventures using resources they already control

SOLUTION / DERIVATION:
Effectual thinking (pioneered by Saras Sarasvathy) is an entrepreneurial mindset focusing on 'Bird in Hand': starting with available means (Who am I? What do I know? Whom do I know?), and co-creating goals, rather than analyzing distant predictive market goals first.

TIP FOR UNDERSTANDING: *Causal thinking = Set a goal and assemble resources; Effectual thinking = Start with existing resources and explore emerging possibilities.*

ITEM #090 / 100

QUESTION: A company identifies increasing demand for electric vehicles and decides to develop charging stations. This decision is an example of:

-> ANSWER : Exploiting an opportunity

SOLUTION / DERIVATION:
In SWOT analysis, an Opportunity is an external positive market trend or environmental shift that a firm can capitalize on. Recognizing growing consumer EV adoption and installing charging stations is a direct commercial exploitation of an emerging market opportunity.

TIP FOR UNDERSTANDING: *SWOT: Strengths & Weaknesses are internal; Opportunities & Threats are external.*

ITEM #091 / 100

QUESTION: Excel Review Center offers review classes to aspiring engineering licensure examinees through face-to-face and online instruction in exchange for tuition fees. Which business model best describes its primary operation?

-> ANSWER : Service-Based Business Model

SOLUTION / DERIVATION:
Excel Review Center delivers educational instruction, review lectures, and mock board exams to students in exchange for fees. Because it creates value through professional knowledge transfer rather than manufacturing physical goods, it operates a service-based business model.

TIP FOR UNDERSTANDING: *Service business models monetize specialized intellectual, technical, or educational labor.*

ITEM #092 / 100

QUESTION: A disc of radius 0.2 m rotates at 10 rad/s. What is the linear speed of a point on its rim?

-> ANSWER : 2.0 m/s

SOLUTION / DERIVATION:
Linear tangential speed relates to angular speed by: v = omega * r. Given radius r = 0.2 m and angular velocity omega = 10 rad/s: v = 10 rad/s * 0.2 m = 2.0 m/s.

TIP FOR UNDERSTANDING: *Linear speed increases linearly with radial distance from the rotation center: v = omega * r.*

ITEM #093 / 100

QUESTION: A species whose impact on the ecosystem is disproportionately large compared to its abundance is known as a:

-> ANSWER : Keystone species

SOLUTION / DERIVATION:
A keystone species is an organism that plays a critical, foundational role in maintaining ecological community structure; without it, the ecosystem would be dramatically altered or collapse entirely (e.g. sea otters regulating sea urchin kelp forests).

TIP FOR UNDERSTANDING: *The impact of a keystone species is disproportionately massive relative to its raw population count.*

ITEM #094 / 100

QUESTION: Which Philippine law is known as the Clean Water Act?

-> ANSWER : Republic Act 9275

SOLUTION / DERIVATION:
Republic Act No. 9275 is the 'Philippine Clean Water Act of 2004', enacted to protect the country's water bodies from land-based pollution sources (industrial, agricultural, and domestic sewage).

TIP FOR UNDERSTANDING: *RA 9275 established Wastewater Discharge Permits and designated Water Quality Management Areas (WQMAs).*

ITEM #095 / 100

QUESTION: Which of the following best describes the primary goal of managing the marketing function?

-> ANSWER : Delivering superior value to customers

SOLUTION / DERIVATION:
Modern marketing management centers on creating, communicating, and delivering superior customer value, establishing enduring customer relationships that generate sustainable long-term profitability for the enterprise.

TIP FOR UNDERSTANDING: *Customer value = Perceived benefits minus total acquisition costs.*

ITEM #096 / 100

QUESTION: An investor provides funding to a startup during its early stage in exchange for an ownership interest. This investor is commonly known as a:

-> ANSWER : Angel investor

SOLUTION / DERIVATION:
An angel investor is an affluent high-net-worth individual who provides capital for early-stage business startups, typically in exchange for convertible debt or equity ownership, filling the gap before venture capital funding.

TIP FOR UNDERSTANDING: *Angel investors invest their personal funds in seed stages; Venture capitalists (VCs) invest pooled institutional funds in growth stages.*

ITEM #097 / 100

QUESTION: Which of the following is an adverse effect of acid rain?

-> ANSWER : All of the choices (corrosion of buildings, soil nutrient depletion, acidification of aquatic ecosystems)

SOLUTION / DERIVATION:
Acid rain (precipitation with pH < 5.6 caused by atmospheric SO2 and NOx emissions) produces widespread ecological damage: it acidifies aquatic ecosystems (killing fish), leaches toxic aluminum into groundwater, strips nutrients from forest soils, and chemically corrodes marble buildings and metal structures.

TIP FOR UNDERSTANDING: *All of the adverse effects cited occur simultaneously from acid precipitation.*

ITEM #098 / 100

QUESTION: A very light object and a very heavy object have the same momentum. Which has more kinetic energy?

-> ANSWER : The light object

SOLUTION / DERIVATION:
Kinetic energy relates to linear momentum p by: KE = p^2 / (2m). If two objects possess the exact same momentum p, kinetic energy is inversely proportional to mass m (KE ∝ 1/m). Therefore, the lighter object (smaller m) carries far greater kinetic energy.

TIP FOR UNDERSTANDING: *KE = p^2 / (2m). Light object (low m) has high velocity, making v^2 dominate over mass in kinetic energy.*

ITEM #099 / 100

QUESTION: Which of the following best describes a thermoplastic polymer?

-> ANSWER : Softens when heated and can be reshaped

SOLUTION / DERIVATION:
Thermoplastic polymers (e.g. polyethylene, PVC, nylon) consist of linear or branched chains held together by weak intermolecular van der Waals forces. They soften and melt upon heating and solidify upon cooling, allowing them to be repeatedly remelted and reshaped.

TIP FOR UNDERSTANDING: *Thermoplastics can be remelted and recycled; Thermosets decompose upon heating and cannot be reshaped due to permanent cross-links.*

ITEM #100 / 100

QUESTION: If the velocity of an object is doubled, what happens to its kinetic energy?

-> ANSWER : It becomes four times

SOLUTION / DERIVATION:
Kinetic energy formula: KE = (1/2) m v^2. If velocity is doubled (v' = 2v): KE' = (1/2) m (2v)^2 = (1/2) m (4 v^2) = 4 * [(1/2) m v^2] = 4 * KE. Kinetic energy quadruples (becomes four times as large).

TIP FOR UNDERSTANDING: *Kinetic energy depends on the square of velocity (KE ∝ v^2). Doubling speed quadruples the kinetic energy and stopping distance.*

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[END OF EXAM SOLUTION MANUAL]

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